

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\112-C1-C2Stack.pvdb

Date of Analysis : Sep 8, 2026 6:27pm

PV Elite 28, January 2026



PV Elite 28 Licensee: SL User
 FileName : 112-C1-C2_(Stack) Page 2 of 224
 Vessel Design Summary: Step: 33 6:34pm Sep 8,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 1920.000 x 1926.000 x 1920.000 mm ID
 Vessel Design Length, Tangent to Tangent 16970.35 mm
 Specified Datum Line Distance 0.00 mm
 Shell Side Design Temperature 160 °C
 Channel Side Design Temperature 150 °C
 Shell Side Design Pressure 1310.000 KPa.
 Channel Side Design Pressure 520.000 KPa.
 Shop Shell Side Test Pressure 2176.523 KPa.
 Field Shell Side Test Pressure 1703.000 KPa.
 Shop Channel Side Test Pressure 936.811 KPa.
 Field Channel Side Test Pressure 676.000 KPa.
 Wind Design Code ASCE/SEI 7-16
 Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-240 304L	...	...	S30403	
Shell	SA-516 70	...	...	K02700	
Flange	SA-965 F304L	...	...	S30403	
Nozzle	SA-182 F304L	...	> 130	S30403	
Nozzle	SA-106 B	...	...	K03006	
Nozzle	SA-266 2	...	...	K03506	
Nozzle	SA-312 TP304	...	...	S30400	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-182 F304	...	> 130	S30400	
Nozzle Flg	SA-105	...	...	K03504	
Rings	SA-516 70	...	...	K02700	
Tubes	SA-213 TP304L	...	...	S30403	
Tubesheet	SA-765 II	...	...	K03047	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-307 A	...	6 <= t <= 100		
Saddles	SA-516 70	...	...	K02700	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
LHS Blind Flang	540.976	51.70	573.000	3.0000	N/A	No
LHS ch Flange	538.820	51.70	1286.000	3.0000	N/A	No
LHS ch Shell	538.830	51.70	1293.000	1.0000	N/A	No
Main Shell	1328.888	51.70	1789.000	1.0000	N/A	No
Main Shell	1328.888	51.70	1550.000	1.0000	N/A	No
Main Shell	1328.634	51.70	1790.000	1.0000	N/A	No
RHS ch Shell	538.634	51.70	1891.000	1.0000	N/A	No
RHS Ch Flg	538.624	51.70	1286.000	3.0000	N/A	No



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Vessel Design Summary:

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RHS Blind Flang	540.829	51.70	573.000	3.0000	N/A	No
-----------------	---------	-------	---------	--------	-----	----

Liquid Level: 2140.00 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000

Stiffener Ring Specifications:

Elevation mm	Selected Type	Ring Description
4426.17	Bar 271.0 x 30.0	Ring R1 Fr40
12044.18	Bar 271.0 x 30.0	Ring R1 Fr60

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Body Flg	123.2	120.0	128.0	120.0	115.1	...	1.00	1.00
Body Flg	335.2	212.0	188.0	180.0	174.1	...	1.00	1.00
Cylinder	1835.2	1500.0	14.0	14.0	6.3	...	0.85	0.85
Cylinder	6121.2	4230.0	16.0	16.0	12.0	...	0.85	0.85
Cylinder	10351.2	4230.0	14.0	14.0	12.0	...	0.85	0.85
Cylinder	14579.2	4228.0	16.0	16.0	12.0	...	0.85	0.85
Cylinder	16635.2	2000.0	20.0	20.0	6.3	...	0.85	0.85
Body Flg	16850.4	212.0	188.0	180.0	174.1	...	1.00	1.00
Body Flg	16970.4	120.0	128.0	120.0	115.1	...	1.00	1.00

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	120	95.885	51.7045	No Calc
20	30	180	174.117	51.7045	No Calc
30	Tsht	14	5.40268	51.7045	399.737
Tsht	40	14	No Calc	51.7045	No Calc
40	Ring	16	6.29255	51.7045	694.317
Ring	50	16	9.39831	51.7045	216.617
50	60	14	9.38814	51.7045	151.628
60	Ring	16	9.39831	51.7045	216.617
Ring	Tsht	16	6.2925	51.7045	694.334
Tsht	70	16	No Calc	51.7045	No Calc
70	80	20	5.98157	51.7045	613.166
80	90	180	174.117	51.7045	No Calc
90	100	120	95.885	51.7045	No Calc

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1499.97	19071.5	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Ring	2535.03	53294.6	No Calc	No Calc
Ring	50	7618	157800	136.354	9271.13
50	60	7618	21151.5	No Calc	No Calc
60	Ring	7618	157800	No Calc	No Calc
Ring	Tsht	2534.97	53294.6	136.354	9271.13
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	2000.03	49662.2	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc



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90	100	No Calc	No Calc	No Calc	No Calc
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Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.957	Nozzle	Passed
N3-Shell Inlet	WRC-297	0.990	Shell	Passed
N4-Shell Outle	WRC-297	0.993	Shell	Passed
N2-Tube Outlet	WRC-297	0.971	Shell	Passed

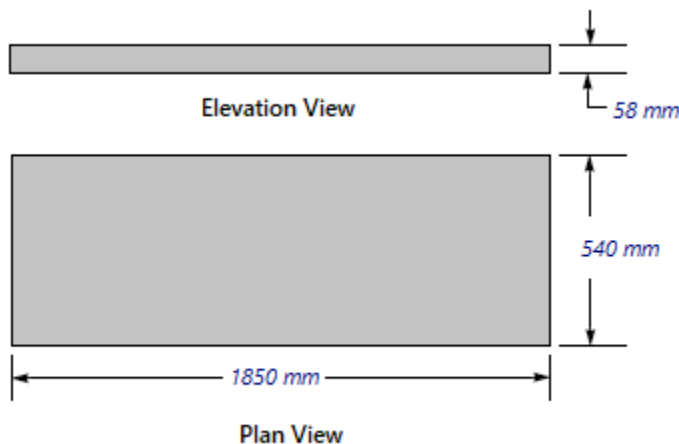
Foundation/Support Summary

Saddle Parameters:

Saddle Width	515.000 mm
Saddle Bearing Angle	166.000 deg.
Centerline Dimension	1580.000 mm
Wear Pad Width	800.000 mm
Wear Pad Thickness	16.000 mm
Wear Pad Bearing Angle	180.000 deg.
Distance from Saddle to Tangent	2535.000 mm

Baseplate Length	1850.000 mm
Baseplate Thickness	58.000 mm
Baseplate Width	540.000 mm
Number of Ribs (including outside ribs)	4
Rib Thickness	16.000 mm
Web Thickness	16.000 mm
Height of Center Web	620.000 mm
Number of Bolts in Baseplate	6

Baseplate Sketch



Maximum Tensile Bolt Load 1384. Kg

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	229740.03 Kg
Maximum Transverse Saddle Shear Load	42116.38 Kg
Maximum Longitudinal Saddle Shear Load	14371.93 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	130147.98 Kg
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Vessel Design Summary:

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Maximum Transverse Saddle Shear Load	1513.48	Kgf
Maximum Longitudinal Saddle Shear Load	226.30	Kgf

Weights:

Fabricated - Bare w/o removable internals	153199.9	kg.
Shop Test - Fabricated + water (full)	241927.9	kg.
Shipping - Fab. + rem. intls.+ shipping app.	153199.9	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	153199.9	kg.
Empty - Fab. + intls. + details + wghts.	157242.5	kg.
Operating - Empty + operating liquid (no ca)	246835.6	kg.
Field Test - Empty weight + test medium (full)	245240.7	kg.

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Nozzle Summary:

Step: 31

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Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	1293	OK	...	OK	5.36	OK	Passed
N3-Shell Inlet	1789	OK	...	OK	9.33	OK	Passed
N4-Shell Outlet	1790	OK	...	OK	9.33	OK	Passed
N2-Tube Outlet	1891	OK	...	OK	5.51	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1789 Nozzle : N3-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 1293 Nozzle : N1-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
30	N1-Tube Inlet	1085.175	0.000	977.200
40	N3-Shell Inlet	3142.405	0.000	915.250
60	N4-Shell Outlet	13575.306	180.000	915.250
70	N2-Tube Outlet	15635.176	180.000	1480.000

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 30 6:34pm Sep 8,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube Inlet	20.000 in	Actual	WNF	20.000	12.700	708.00	26.00	356.21	150
N3-Shell Inlet	20.000 in	80	WNF	20.000	26.187	948.00	16.00	350.10	150
N4-Shell Outlet	20.000 in	80	WNF	20.000	26.187	948.00	16.00	350.10	150
N2-Tube Outlet	30.000 in	Actual	WNF	30.000	14.000	1362.00	30.00	398.84	150

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube Inl	SA-182 F304L	14.000	12.000	20.000	20.000	...
N3-Shell In	SA-106 B	14.000	10.000	12.000	16.000	...
N4-Shell Ou	SA-266 2	14.000	10.000	12.000	16.000	...
N2-Tube Out	SA-312 TP304	14.000	12.000	20.000	25.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube Inlet	1085.175	0.0	300.00	0.00	LHS ch Shell
N3-Shell Inlet	3142.405	0.0	300.00	0.00	Main Shell
N4-Shell Outlet	13575.306	180.0	300.00	0.00	Main Shell
N2-Tube Outlet	15635.175	180.0	300.00	0.00	RHS ch Shell

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Bill of Material:

Step: 29 6:34pm Sep 8,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 128.0mm THK X 2140.0mm OD	SA-965 F304L
2	WELD NECK FLANGE: 188.0mm THK X 2140.0mm OD	SA-965 F304L
1	BLIND FLANGE: 128.0mm THK X 2145.0mm OD	SA-965 F304L
2	INSULATION: 120mm X 100mm THK	
2	INSULATION: 212mm X 100mm THK	
1	INSULATION: 1500mm X 100mm THK	
22	BAFFLES: 1910.0mm DIA X 10.0mm THK	SA-516 70
12	TIE RODS: 16mm DIA X 11680mm	SA-193 B7M
2	TUBESHEET: 1954.0mm X 56.0mm THK	SA-765 II
2895	TUBES: 12800.0mm X 25.4mm DIA X 2.1mm THK	SA-213 TP304L
2	SADDLE: 515.0mm X 166 DEG	SA-516 70
2	INSULATION: 4230mm X 100mm THK	
2	BAR RING STIFFENER: 1958.0mm ID X 2500.0mm OD X 30.0mm	SA-516 70
1	INSULATION: 4228mm X 100mm THK	
1	CLADDING: 4228.000mm X 3.000mm THK	...
1	INSULATION: 2000mm X 100mm THK	
2	GASKET: 1990.0mm OD X 1952.0mm ID	...
120	BODY FLANGE BOLTS: 34.9mm DIA	SA-193 B7
240	NUTS FOR BODY FLANGE BOLTS: 34.9mm DIA	...
1	CYLINDRICAL SEGMENT 14.0mm THK X 3080.0mm X 1500.0mm	SA-240 304L
1	CYLINDRICAL SEGMENT 14.0mm THK X 2951.7mm X 1500.0mm	SA-240 304L
1	CYLINDRICAL SEGMENT 16.0mm THK X 3048.0mm X 4230.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 3002.7mm X 4230.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 3048.0mm X 4230.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 14.0mm THK X 3002.7mm X 4230.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 3089.7mm X 4228.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 16.0mm THK X 2960.9mm X 4228.0mm	SA-516 70
1	CYLINDRICAL SEGMENT 20.0mm THK X 3080.0mm X 2000.0mm	SA-240 304L
1	CYLINDRICAL SEGMENT 20.0mm THK X 2951.7mm X 2000.0mm	SA-240 304L
1	NAMEPLATE	...



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Bill of Material:

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MDMT Summary:

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Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Main Shell [8]	B	-14	-29	-29	0.723	16.000	0.85	No
Main Shell [8]	B	-18	-27	-29	0.834	14.000	0.85	No
Main Shell [8]	B	-14	-29	-29	0.723	16.000	0.85	No
Ring R1 Fr40 [23]	B	-14	-29	-29	0.723	16.000	0.85	No
Ring R1 Fr60 [23]	B	-14	-29	-29	0.723	16.000	0.85	No
LH floating s[24]	B	-14	-29	-29	0.723	16.000	0.85	No
New Saddle [24]	B	-14	-29	-29	0.723	16.000	0.85	No
N3-Shell Inlet[1]	B	-14	-36	-29	0.614	16.000	1.00	No
Nozzle Flg [5]	A	-18	-36					
N4-Shell Outle[1]	B	-14	-35	-29	0.622	16.000	1.00	No
Nozzle Flg [5]	A	-18	-36					
Tubesheet: SS[13]	B	-14	-44	-29	0.521	16.000	1.00	No

Warmest MDMT: -14 -27

LHS ch Shell [16]		-196						
RHS ch Shell [16]		-196						
N1-Tube Inlet [1]	B	-21	-104	-29	0.118	12.700	1.00	No
N2-Tube Outlet[1]	B	-18	-104	-29	0.135	14.000	1.00	No
Tubesheet: CS[14]	B	-18	-104	-29	0.333	14.000	1.00	No
Bolting [21]		-48						
Bolting [21]		-48						

Warmest MDMT: -18 -104

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-29.0	-12.0	Pass
Channel/Tube	-48.0	-12.0	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.



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MDMT Summary:

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Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 6:27pm Sep 8, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



One or more nozzles in this model was specified using the actual thickness basis and a standard flange is also specified. In many cases a matching nominal diameter cannot be determined and the weight of the flange and optional blind flange cannot be determined. When actual thickness basis is specified, you must calculate the weight of the assembly (by hand) and enter it into the 'Overriding Weight' field in the nozzle dialog when there is a standard flange specified. However, for FVC nozzles that have their nominal diameter specified in the FVC dialog, entering the overriding weight for that nozzle is not required.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1

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Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	1310	KPa.
Channel Side Design Pressure	520	KPa.
Shell Side Design Temperature	160.0	°C
Channel Side Design Temperature	150.0	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-12.0	°C
MDMT (CET), Tube Side	-12.0	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

Field Pressure Test:

Type of Pressure Test UG-99(b) Note [35]
 Pressure Test Position Horizontal
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	13100 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.600
Table Value Fv	2.400
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9
Element From Node	10
Element To Node	20
Element Type	Flange
Description	LHS Blind Flang
Distance "FROM" to "TO"	120 mm
Flange Outside Diameter	2140 mm
Element Thickness	120 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	128 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	150 °C
Design External Pressure	51.704 KPa.
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304L
Allowable Stress, Ambient	115 N./mm ²
Allowable Stress, Operating	88.101 N./mm ²
Allowable Stress, Hydrotest	153 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	132 N./mm ²
External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2140 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	10
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	120 mm



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8, 2026

Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element#	2/9
Element From Node	20
Element To Node	30
Element Type	Flange
Description	LHS ch Flange
Distance "FROM" to "TO"	212 mm
Flange Inside Diameter	1920 mm
Element Thickness	180 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	188 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	150 °C
Design External Pressure	51.704 KPa.
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304L
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1920 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	20
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	212 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	3/9
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	LHS ch Shell
Distance "FROM" to "TO"	1500 mm
Inside Diameter	1920 mm
Element Thickness	14 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	14 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	150 °C
Design External Pressure	51.704 KPa.
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304L
Allowable Stress, Ambient	115 N./mm ²
Allowable Stress, Operating	115 N./mm ²
Allowable Stress, Hydrotest	153 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	132 N./mm ²



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1920 mm
Liquid Density	0.0009984 kg./cm³

Element From Node	30
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	1500 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm³

Element From Node	30
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	750 mm
Nozzle Diameter	20 in.
Nozzle Schedule	30
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	184.58 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-182 F304L

Element From Node	30
Detail Type	Weight
Detail ID	Pass partiton
Dist. from "FROM" Node / Offset dist	750 mm
Miscellaneous Weight	750 Kg
Offset from Element Centerline	0 mm

Element From Node	30
Detail Type	For./Mom.
Detail ID	TOP HE FORCES
Dist. from "FROM" Node / Offset dist	1556 mm
Force in X Direction	700 Kg
Force in Y Direction	0 Kg
Force in Z Direction	2900 Kg
Moment about X Axis	0 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	0 Kg-m.
Force/Moment Combination Method	SRSS

Element From Node	30
Detail Type	For./Mom.
Detail ID	TOP HE FORCES
Dist. from "FROM" Node / Offset dist	1556 mm
Force in X Direction	700 Kg
Force in Y Direction	0 Kg
Force in Z Direction	2900 Kg
Moment about X Axis	0 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	0 Kg-m.
Force/Moment Combination Method	SRSS



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

Element From Node	30
Detail Type	For./Mom.
Detail ID	TOP HE FORCES
Dist. from "FROM" Node / Offset dist	1500 mm
Force in X Direction	9800 Kgf
Force in Y Direction	0 Kgf
Force in Z Direction	9800 Kgf
Moment about X Axis	0 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	0 Kg-m.
Force/Moment Combination Method	SRSS

Element From Node	30
Detail Type	For./Mom.
Detail ID	TOP HE FORCES
Dist. from "FROM" Node / Offset dist	1556 mm
Force in X Direction	10700 Kgf
Force in Y Direction	0 Kgf
Force in Z Direction	10700 Kgf
Moment about X Axis	0 Kg-m.
Moment about Y Axis	0 Kg-m.
Moment about Z Axis	0 Kg-m.
Force/Moment Combination Method	SRSS

Element#	4/9
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	4230 mm
Inside Diameter	1926 mm
Element Thickness	16 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	16 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1310 KPa.
Design Temperature Internal Pressure	160 °C
Design External Pressure	51.704 KPa.
Design Temperature External Pressure	160 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	234 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.4 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K02700
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	LH floating sad
Dist. from "FROM" Node / Offset dist	2535 mm
Width of Saddle	515 mm
Height of Saddle at Bottom	1580 mm
Saddle Contact Angle	166.0



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	800	mm
Thickness of Wear Plate	16	mm
Contact Angle, Wear Plate (degrees)	180.0	
Friction coefficient	0.30000001	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1926	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	40	
Detail Type	Insulation	
Detail ID	Ins: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	4230	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	40	
Detail Type	Ring	
Detail ID	Ring R1 Fr40	
Dist. from "FROM" Node / Offset dist	2535	mm
Inside Diameter of Ring	1958	mm
Thickness of Ring	30	mm
Outside Diameter of Ring	2500	mm
Material Name	SA-516 70	
Height of Section Ring	0	mm
Using Custom Stiffener Section	No	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	N3-Shell Inlet	
Dist. from "FROM" Node / Offset dist	1251.2	mm
Nozzle Diameter	20	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	255.96	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-106 B	

Element#	5/9	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	4230	mm
Inside Diameter	1926	mm
Element Thickness	14	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	14	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1310	KPa.
Design Temperature Internal Pressure	160	°C



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

Design External Pressure	51.704	KPa.
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	
Weld is pre-Heated	No	

Element From Node	50	
Detail Type	Liquid	
Detail ID	Liquid: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	1926	mm
Liquid Density	0.0009984	kg./cm ³

Element From Node	50	
Detail Type	Insulation	
Detail ID	Ins: 50	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Insulation	4230	mm
Thickness of Insulation	100	mm
Density	0.000352	kg./cm ³

Element From Node	50	
Detail Type	Weight	
Detail ID	TOP HE EMPTY	
Dist. from "FROM" Node / Offset dist	2241.7	mm
Miscellaneous Weight	75850	Kgf
Offset from Element Centerline	3160	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	TOP HE OPERATING	
Dist. from "FROM" Node / Offset dist	2297.8	mm
Miscellaneous Weight	46800	Kgf
Offset from Element Centerline	3160	mm

Element From Node	50	
Detail Type	Weight	
Detail ID	TOP HE TEST	
Dist. from "FROM" Node / Offset dist	2296.6	mm
Miscellaneous Weight	46000	Kgf
Offset from Element Centerline	3160	mm

Element#	6/9	
Element From Node	60	
Element To Node	70	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	4228	mm
Inside Diameter	1926	mm
Element Thickness	16	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	16	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1310	KPa.
Design Temperature Internal Pressure	160	°C
Design External Pressure	51.704	KPa.
Design Temperature External Pressure	160	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Efficiency, Longitudinal Seam	0.85	
Efficiency, Circumferential Seam	0.85	



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

Weld is pre-Heated	No
Element From Node	60
Detail Type	Saddle
Detail ID	New Saddle
Dist. from "FROM" Node / Offset dist	1693 mm
Width of Saddle	515 mm
Height of Saddle at Bottom	1580 mm
Saddle Contact Angle	166.0
Height of Composite Ring Stiffener	0 mm
Width of Wear Plate	800 mm
Thickness of Wear Plate	16 mm
Contact Angle, Wear Plate (degrees)	180.0
Friction coefficient	0.0
Moment Factor	3.0
Dimension E at base (optional)	0 mm
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 40
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1900 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	60
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	4228 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³
Element From Node	60
Detail Type	Lining
Detail ID	SS lining
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Lining	4228 mm
Thickness of Lining	3 mm
Density	0.008 kg./cm ³
Element From Node	60
Detail Type	Ring
Detail ID	Ring R1 Fr60
Dist. from "FROM" Node / Offset dist	1693 mm
Inside Diameter of Ring	1958 mm
Thickness of Ring	30 mm
Outside Diameter of Ring	2500 mm
Material Name	SA-516 70
Height of Section Ring	0 mm
Using Custom Stiffener Section	No
Element From Node	60
Detail Type	Nozzle
Detail ID	N4-Shell Outlet
Dist. from "FROM" Node / Offset dist	3224.1 mm
Nozzle Diameter	20 in.
Nozzle Schedule	80
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	255.35 Kg
Grade of Attached Flange	GR 1.1



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

Nozzle Matl

SA-266 2

Element#	7/9
Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	RHS ch Shell
Distance "FROM" to "TO"	2000 mm
Inside Diameter	1920 mm
Element Thickness	20 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	20 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	150 °C
Design External Pressure	51.704 KPa.
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-240 304L
Allowable Stress, Ambient	115 N./mm ²
Allowable Stress, Operating	115 N./mm ²
Allowable Stress, Hydrotest	153 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	132 N./mm ²
External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.85
Weld is pre-Heated	No
Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1900 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	70
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	2000 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³
Element From Node	70
Detail Type	Nozzle
Detail ID	N2-Tube Outlet
Dist. from "FROM" Node / Offset dist	1000 mm
Nozzle Diameter	30 in.
Nozzle Schedule	30
Nozzle Class	150
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	540.57 Kg
Grade of Attached Flange	GR 2.1
Nozzle Matl	SA-312 TP304



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8,2026

Element From Node	80
Element To Node	90
Element Type	Flange
Description	RHS Ch Flg
Distance "FROM" to "TO"	212 mm
Flange Inside Diameter	1920 mm
Element Thickness	180 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	188 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	150 °C
Design External Pressure	51.704 KPa.
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304L
Allowable Stress, Ambient	115 N./mm ²
Allowable Stress, Operating	88.101 N./mm ²
Allowable Stress, Hydrotest	153 N./mm ²
Material Density	0.008027 kg./cm ³
P Number Thickness	0 mm
Yield Stress, Operating	132 N./mm ²
External Pressure Chart Name	HA-3
UNS Number	S30403
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No

Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1900 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	80
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	212 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

Element#	9/9
Element From Node	90
Element To Node	100
Element Type	Flange
Description	RHS Blind Flang
Distance "FROM" to "TO"	120 mm
Flange Outside Diameter	2145 mm
Element Thickness	120 mm
Internal Corrosion Allowance	3 mm
Nominal Thickness	128 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	520 KPa.
Design Temperature Internal Pressure	150 °C
Design External Pressure	51.704 KPa.
Design Temperature External Pressure	150 °C
Effective Diameter Multiplier	1.2
Material Name	SA-965 F304L
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



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FileName : 112-C1-C2_(Stack)

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Input Echo:

Step: 1 6:27pm Sep 8, 2026

Element From Node	90
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	2125 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	90
Detail Type	Insulation
Detail ID	Insulation
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Insulation	120 mm
Thickness of Insulation	100 mm
Density	0.000352 kg./cm ³

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XY Coordinate Calculations:

Step: 2 6:27pm

Sep 8,2026

XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
LHS Blind Flang		123.175	...	120	...
LHS ch Flange		335.175	...	212	...
LHS ch Shell		1835.17	...	1500	...
Main Shell		6121.17	...	4230	...
Main Shell		10351.2	...	4230	...
Main Shell		14579.2	...	4228	...
RHS ch Shell		16635.2	...	2000	...
RHS Ch Flg		16850.4	...	212	...
RHS Blind Flang		16970.4	...	120	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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FileName : 112-C1-C2_(Stack)

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Flg Calc [Int P]: LHS Blind Fl

Flng: 9

6:27pm

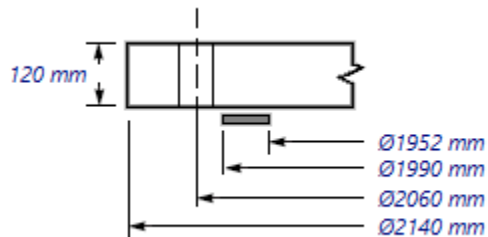
Sep 8, 2026

Flange Input Data Values

Description: LHS Blind Fl :

LHS Blind Flang

Flange Type	TEMA Channel Cover		
Design Pressure	P	540.98	KPa.
Design Temperature		150	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2140.000	mm
Flange Thickness	t	120.0000	mm
Flange Material	SA-965 F304L		
Flange Material UNS number	S30403		
Flange Allowable Stress At Temperature	Sfo	88.10	N./mm ²
Flange Allowable Stress At Ambient	Sfa	115.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	6471.681	mm
Youngs Modulus for Blind Flange Matl.	Sy	186083.27	N./mm ²
Allowed Channel Cover Deflection	dfAllow	2.4075	mm
Diameter of Bolt Circle	C	2060.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	60		
Flange Face Outside Diameter	Fod	2001.000	mm
Flange Face Inside Diameter	Fid	1920.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1990.000	mm
Gasket Inside Diameter	Gi	1952.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	19.000 mm
Basic Gasket Width,	$bo = N/2$	9.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1974.466 mm



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Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1974.4663^2 \cdot 540.976$$

$$= 168913.438 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.7669 \cdot \pi \cdot 1974.4663 \cdot 3.0 \cdot 540.98$$

$$= 15946.832 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(168913 + 15947 + 0, 0)$$

$$= 184860.266 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 7.7669 \cdot \pi \cdot 1974.466 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 338730.156 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg / Sa)$$

$$= \max((184860 + 0 + 4 \cdot 0 / 1974.47) / 172, 338730 / 172)$$

$$= 193.135 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	193.135	447.096	
Radial Distance between Bolts and Edge:	34.925	40.000	
Circ. Spacing between the Bolts:	77.800	107.812	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 447.096 \cdot 172.0 / (68.95 \cdot \pi (1990.0 + 1952.0))$$

$$= 9.006 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 447.0959$$

$$= 784138.12 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 184860.27 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2060.0 - 1974.4663) / 2$$

$$= 42.7669 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 184860 \cdot 42.7669 \cdot 1.0 / (117.0^2 \cdot 1974.4663) + 0.3 \cdot 1.0 \cdot 540.98 \cdot 1974.4663^2 / 117.0^2$$

$$= 51.67 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:



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$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 784138 \cdot 42.7669 \cdot 1.0 / (117.0^2 \cdot 1974.4663)$$

$$= 23.12 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 184860 / 447.0959$$

$$= 40.55 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (338730 / 447.0959)$$

$$= 74.30 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	51.67	88.10	23.12	115.00
Bolting	40.55	172.00	74.30	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 2.2895 mm

Allowed deflection at center of Cover 2.4075 mm

Reqd. Blind Flange Thickness at Center 92.604 mm

Reqd. Blind Flange Thickness at Gasket 55.459 mm

Min. Required Flange Thickness per TEMA 115.056 mm

Estimated M.A.W.P. (Operating) 594 KPa.

Estimated Finished Weight of Flange at given Thk. 3464.7 kg.

Estimated Unfinished Weight of Forging at given Thk 3464.7 kg.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	49.12	115.00	21.98	115.00
Bolting	40.55	172.00	74.30	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 113.259 mm

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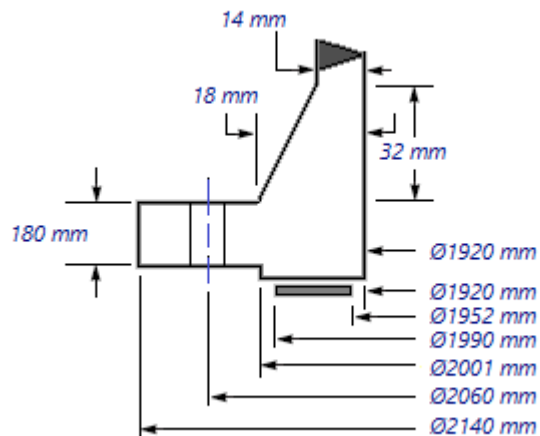
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Flange Input Data Values

Description: LHS Ch Flg :

LHS ch Flange

Flange Type	Integral Weld Neck		
Design Pressure	P	538.82	KPa.
Design Temperature		150	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1920.000	mm
Flange Outside Diameter	A	2140.000	mm
Flange Thickness	t	180.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	g1	18.0000	mm
Length of Hub	h	32.0000	mm
Flange Material		SA-965 F304L	
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	88.10	N./mm ²
Flange Allowable Stress At Ambient	Sfa	115.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2060.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		60	
Flange Face Outside Diameter	Fod	2001.000	mm
Flange Face Inside Diameter	Fid	1920.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1990.000	mm
Gasket Inside Diameter	Gi	1952.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (538.82 \cdot (1920.0/2 + 3.0)) / (115.0 \cdot 1.0 - 0.6 \cdot 538.82) + Ca$$

$$= 7.5251 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.0 \cdot 1.0 \cdot 11.0) / (963.0 + 0.6 \cdot 11.0)$$

$$= 1304.562 \text{ KPa}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.0 \cdot 1.0 \cdot 14.0) / (960.0 + 0.6 \cdot 14.0)$$

$$= 1662.410 \text{ KPa}$$

Corroded Flange Thickness, $t_c = T - c_i$	177.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1926.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	15.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	52.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width, $b_o = N/2$	9.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.767 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1974.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1974.4663^2 \cdot 538.82$$

$$= 168240.109 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.7669 \cdot \pi \cdot 1974.4663 \cdot 3.0 \cdot 538.82$$

$$= 15883.265 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1926.0^2 \cdot 538.8199$$

$$= 160082.062 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 168240 - 160082$$

$$= 8158.039 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o,1 + W_o,2 + H'p, 0)$$

$$= \max(168240 + 15883 + 0, 0)$$

$$= 184123.375 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 7.7669 \cdot \pi \cdot 1974.466 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 338730.156 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((184123 + 0 + 4 \cdot 0/1974.47)/172, 338730/172)$$

$$= 193.135 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 34.925 + 6 \cdot 177.0/(3.0 + 0.5)$
 $= 373.279 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 2060.0 \cdot \sin(\pi/60)$
 $= 107.812 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{Bs/(2a + t)}, 1)$
 $= \max(\sqrt{107.812/(2 \cdot 34.925 + 177.0)}, 1)$
 $= 1.0000$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	193.135	447.096	
Radial Distance between Hub and Bolts:	47.625	52.000	
Radial Distance between Bolts and Edge:	34.925	40.000	
Circ. Spacing between the Bolts:	77.800	107.812	373.279

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 447.096 \cdot 172.0/(68.95 \cdot \pi(1990.0 + 1952.0))$
 $= 9.006 \text{ mm}$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
 $= Sa \cdot Ab$
 $= 172.0 \cdot 447.0959$
 $= 784138.12 \text{ Kgf}$

Gasket Load for the Operating Condition [HG]:
 $= Wo - H$
 $= 184123 - 168240$
 $= 15883.27 \text{ Kgf}$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2060.0 - 1974.4663) / 2$
 $= 42.7669 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5(C - B)/2 + hg$
 $= 0.5(2060.0 - 1926.0)/2 + 42.7669$
 $= 54.8834 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2060.0 - 1926.0 - 15.0) / 2$
 $= 59.5000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	160082.	59.5000	1.0000	9525.
Face Pressure, Mt	8158.	54.8834	1.0000	448.
Gasket Load, Mg	15883.	42.7669	1.0000	679.
Gasket Seating, Matm	784138.	42.7669	1.0000	33536.

Total Moment for Operation, Mo 10652. Kg-m.
Total Moment for Gasket seating, Mg 33536. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 145.554 mm
Hub Ratio, h/h0 = HL / H0 0.220
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.364

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.900
Factor V per 4.16.5 0.428
Factor f per 4.16.5 1.190

Stress factor from Table 4.16.4 [K]:

= A / B
= 2140.0/1926.0
= 1.111

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 * (K² * log10(K))) / (K² - 1)) / (K - 1)
= (0.66845 + (5.7169 * (1.111² * log10(1.111))) / (1.111² - 1)) / (1.111 - 1)
= 18.407

Stress factor from Table 4.16.4 [T]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / ((1.0472 + 1.9448 * K²) * (K - 1))
= (1.111² * (1 + 8.55246 * log10(1.111)) - 1) / ((1.0472 + 1.9448 * 1.111²) * (1.111 - 1))
= 1.873

Stress factor from Table 4.16.4 [U]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / (1.36136 * (K² - 1) * (K - 1))
= (1.111² * (1 + 8.55246 * log10(1.111)) - 1) / (1.36136 * (1.111² - 1) * (1.111 - 1))
= 20.228

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.111² + 1) / (1.111² - 1)
= 9.526

Stress factor from Table 4.16.4 for integral flanges [d]:

= U * g0² * h0 / V
= 20.228 * 11.0² * 145.554 / 0.428
= 832961.625 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.9 / 145.554
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:

= (t * e + 1) / (T + t³/d)
= (177.0 * 0.006 + 1) / (1.873 + 177.0³/832962)
= 7.776

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f * Mop / Bcor) / (L * g1²)
= (1.1897 * 10652 / 1926.0) / (7.7755 * 15.0²)
= 36.88 N/mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.1897 \cdot 33536 / 1926.0) / (7.7755 \cdot 15.0^2)$$

$$= 116.11 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.4597 \cdot 10652 / 1926.0) / (7.7755 \cdot 177.0^2)$$

$$= 0.55 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.4597 \cdot 33536 / 1926.0) / (7.7755 \cdot 177.0^2)$$

$$= 1.72 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (18.4072 \cdot 10652 / (177.0^2 \cdot 1926.0)) - 9.5263 \cdot 0.5$$

$$= 26.65 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (18.4072 \cdot 33536 / (177.0^2 \cdot 1926.0)) - 9.5263 \cdot 2$$

$$= 83.90 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (37 + \max(0.5, 27)) / 2$$

$$= 31.77 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (116 + \max(2, 84)) / 2$$

$$= 100.01 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 184123 / 447.0959$$

$$= 40.39 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (338730 / 447.0959)$$

$$= 74.30 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	36.88	132.15	116.11	172.50
Radial Flange	0.55	88.10	1.72	115.00
Tangential Flange	26.65	88.10	83.90	115.00
Maximum Average	31.77	88.10	100.01	115.00
Bolting	40.39	172.00	74.30	172.00

Minimum Required Flange Thickness [Rigidity] 174.117 mm

Estimated M.A.W.P. (Operating) 1305 KPa.

Estimated Finished Weight of Flange at given Thk. 1038.6 kg.

Estimated Unfinished Weight of Forging at given Thk 1193.8 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	283.07	301.72	0.938	283.07	339.99	0.833
Tangential Flange	St	209.21	226.29	0.925	209.21	255.00	0.820
Tangential Flange	Sto	61.09	150.86	0.405	61.09	170.00	0.359
Combined	Sh/f + St	452.57	452.57	1.000	452.57	509.99	0.887
Combined	Sr + St	213.50	452.57	0.472	213.50	509.99	0.419
Combined	Sh + Sto	344.16	452.57	0.760	344.16	509.99	0.675
Radial Flange	Sr	4.30	301.72	0.014	4.30	339.99	0.013
Maximum Stress Ratio				1.000			0.887

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.3079/1.1244)

= 1.1631

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)

= max(0.25; 0.2593/1.0331) if h/ho < 0.35, else max (0.25; 0.232/0.7944)

= 0.2510

In this case, the hub ratio h/ho, 0.2198 was < 0.35.

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	428.9 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	7.4516 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	170.0 N./mm ²
Flange Material Operating Yield Stress	Syo	132.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.375 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1990.0^2 - 1952.0^2)$$

$$= 1176.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 207.0 \cdot 1176.494 / (60 \cdot 7.452)$$

$$= 544.7 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(544.703; 507.5)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(507.5; 290.0)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(507.5; 428.87)$$

$$= 428.9 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 428.87 \geq 140.0 (1176.494 / (60 \cdot 7.452))$$

$$:: 428.9 \geq 368.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 428.87 \geq (97.0 \cdot 1176.494 + \pi/4 \cdot 538.82 \cdot 1952.0^2) / (0.7 \cdot 7.452 \cdot 60)$$

$$:: 428.9 \geq 416.2 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 428.87 \leq 276.0 (1176.494 / (60 \cdot 7.452))$$

$$:: 428.9 \leq 726.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 428.87 \leq 428.87 (1.0/0.375)$$



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:: 428.9 ≤ 1143.4 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 428.87 \cdot 0.2 \cdot 7.452 \cdot 34.925$$

$$= 227.6 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 103 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 33535.8 / 1.0 \cdot 9806.637 \cdot 0.428 / (7.113 \cdot 195129 \cdot 11.0^2 \cdot 145.554 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 10652.1 / 1.0 \cdot 9806.637 \cdot 0.428 / (7.113 \cdot 186083 \cdot 11.0^2 \cdot 145.554 \cdot 0.3)$$

$$= 0.333 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 33535.8 / 1.0 \cdot 9806.637 \cdot 0.428 / (7.776 \cdot 195129 \cdot 11.0^2 \cdot 145.554 \cdot 0.3)$$

$$= 0.914 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 10652.1 / 1.0 \cdot 9806.637 \cdot 0.428 / (7.776 \cdot 186083 \cdot 11.0^2 \cdot 145.554 \cdot 0.3)$$

$$= 0.304 \text{ (should be } \leq 1)$$

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	36.82	172.50	113.28	172.50
Radial Flange	0.76	115.00	2.34	115.00
Tangential Flange	23.69	115.00	72.89	115.00
Maximum Average	30.26	115.00	93.09	115.00
Bolting	40.39	172.00	74.30	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 163.119 mm

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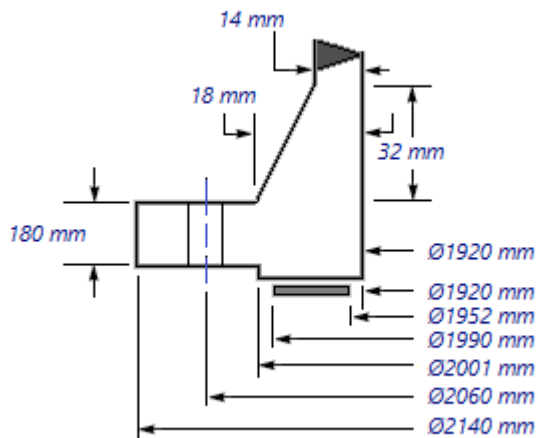
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Flange Input Data Values

Description: RHS Ch Flg :

RHS Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	538.62	KPa.
Design Temperature		150	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	1920.000	mm
Flange Outside Diameter	A	2140.000	mm
Flange Thickness	t	180.0000	mm
Thickness of Hub at Small End	go	14.0000	mm
Thickness of Hub at Large End	g1	18.0000	mm
Length of Hub	h	32.0000	mm
Flange Material		SA-965 F304L	
Flange Material UNS number		S30403	
Flange Allowable Stress At Temperature	Sfo	88.10	N./mm ²
Flange Allowable Stress At Ambient	Sfa	115.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	2060.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		60	
Flange Face Outside Diameter	Fod	2001.000	mm
Flange Face Inside Diameter	Fid	1920.000	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	1990.000	mm
Gasket Inside Diameter	Gi	1952.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm





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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (538.62 \cdot (1920.0/2 + 3.0)) / (115.0 \cdot 1.0 - 0.6 \cdot 538.62) + Ca$$

$$= 7.5234 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.0 \cdot 1.0 \cdot 11.0) / (963.0 + 0.6 \cdot 11.0)$$

$$= 1304.562 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (115.0 \cdot 1.0 \cdot 14.0) / (960.0 + 0.6 \cdot 14.0)$$

$$= 1662.410 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	177.000 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	1926.000 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	15.000 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	11.000 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	52.000 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	19.000 mm
Basic Gasket Width, $b_o = N/2$	9.500 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	7.767 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	1974.466 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1974.4663^2 \cdot 538.624$$

$$= 168178.906 \text{ Kg}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.7669 \cdot \pi \cdot 1974.4663 \cdot 3.0 \cdot 538.62$$

$$= 15877.486 \text{ Kg}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 1926.0^2 \cdot 538.6239$$

$$= 160023.828 \text{ Kg}$$

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 168179 - 160024$$

$$= 8155.076 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_o, 1 + W_o, 2 + H'p, 0)$$

$$= \max(168179 + 15877 + 0, 0)$$

$$= 184056.391 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 7.7669 \cdot \pi \cdot 1974.466 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 338730.156 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G)/S_b, W_g/S_a)$$

$$= \max((184056 + 0 + 4 \cdot 0/1974.47)/172, 338730/172)$$

$$= 193.135 \text{ cm}^2$$



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ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 34.925 + 6 \cdot 177.0/(3.0 + 0.5)$
 $= 373.279 \text{ mm}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 2060.0 \cdot \sin(\pi/60)$
 $= 107.812 \text{ mm}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{Bs/(2a + t)}, 1)$
 $= \max(\sqrt{107.812/(2 \cdot 34.925 + 177.0)}, 1)$
 $= 1.0000$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	193.135	447.096	
Radial Distance between Hub and Bolts:	47.625	52.000	
Radial Distance between Bolts and Edge:	34.925	40.000	
Circ. Spacing between the Bolts:	77.800	107.812	373.279

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 447.096 \cdot 172.0/(68.95 \cdot \pi(1990.0 + 1952.0))$
 $= 9.006 \text{ mm}$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 19 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:
 $= Sa \cdot Ab$
 $= 172.0 \cdot 447.0959$
 $= 784138.12 \text{ Kgf}$

Gasket Load for the Operating Condition [HG]:
 $= Wo - H$
 $= 184056 - 168179$
 $= 15877.49 \text{ Kgf}$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:
 $= (C - G) / 2$
 $= (2060.0 - 1974.4663) / 2$
 $= 42.7669 \text{ mm}$

Distance to Face Pressure Reaction [ht]:
 $= 0.5((C - B) / 2 + hg)$
 $= 0.5((2060.0 - 1926.0) / 2 + 42.7669)$
 $= 54.8834 \text{ mm}$

Distance to End Pressure Reaction [hd]:
 $= (C - B - g1) / 2$
 $= (2060.0 - 1926.0 - 15.0) / 2$
 $= 59.5000 \text{ mm}$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
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End Pressure, Md	160024.	59.5000	1.0000	9522.
Face Pressure, Mt	8155.	54.8834	1.0000	448.
Gasket Load, Mg	15877.	42.7669	1.0000	679.
Gasket Seating, Matm	784138.	42.7669	1.0000	33536.

Total Moment for Operation, Mo 10648. Kg-m.
Total Moment for Gasket seating, Mg 33536. Kg-m.

Effective Hub Length, ho = sqrt(Bcor*goCor) 145.554 mm
Hub Ratio, h/h0 = HL / H0 0.220
Thickness Ratio, g1/g0 = (g1Cor/goCor) 1.364

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.900
Factor V per 4.16.5 0.428
Factor f per 4.16.5 1.190

Stress factor from Table 4.16.4 [K]:

= A / B
= 2140.0/1926.0
= 1.111

Stress factor from Table 4.16.4 [Y]:

= (0.66845 + (5.7169 * K² * log10(K))) / (K² - 1) * (K - 1)
= (0.66845 + (5.7169 * 1.111² * log10(1.111))) / (1.111² - 1) * (1.111 - 1)
= 18.407

Stress factor from Table 4.16.4 [T]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / ((1.0472 + 1.9448 * K²) * (K - 1))
= (1.111² * (1 + 8.55246 * log10(1.111)) - 1) / ((1.0472 + 1.9448 * 1.111²) * (1.111 - 1))
= 1.873

Stress factor from Table 4.16.4 [U]:

= (K² * (1 + 8.55246 * log10(K)) - 1) / (1.36136 * (K² - 1) * (K - 1))
= (1.111² * (1 + 8.55246 * log10(1.111)) - 1) / (1.36136 * (1.111² - 1) * (1.111 - 1))
= 20.228

Stress factor from Table 4.16.4 [Z]:

= (K² + 1) / (K² - 1)
= (1.111² + 1) / (1.111² - 1)
= 9.526

Stress factor from Table 4.16.4 for integral flanges [d]:

= U * g0² * h0 / V
= 20.228 * 11.0² * 145.554 / 0.428
= 832961.625 mm³

Stress factor from Table 4.16.4 for integral flanges [e]:

= F / h0
= 0.9 / 145.554
= 0.006 1/mm

Stress factor from Table 4.16.4 [L]:

= (t * e + 1) / T + t³ / d
= (177.0 * 0.006 + 1) / 1.873 + 177.0³ / 832962
= 7.776

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

= (f * Mop / Bcor) / (L * g1²)
= (1.1897 * 10648 / 1926.0) / (7.7755 * 15.0²)
= 36.87 N/mm²

Longitudinal Hub Stress, Seating [SHa]:



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$$= (f \cdot \text{Matm} / \text{Bcor}) / (L \cdot g^{12})$$

$$= (1.1897 \cdot 33536 / 1926.0) / (7.7755 \cdot 15.0^2)$$

$$= 116.11 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot \text{Mop} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.4597 \cdot 10648 / 1926.0) / (7.7755 \cdot 177.0^2)$$

$$= 0.55 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot \text{Matm} / \text{Bcor}) / (L \cdot t^2)$$

$$= (2.4597 \cdot 33536 / 1926.0) / (7.7755 \cdot 177.0^2)$$

$$= 1.72 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot \text{Mo} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRo}$$

$$= (18.4072 \cdot 10648 / (177.0^2 \cdot 1926.0)) - 9.5263 \cdot 0.5$$

$$= 26.64 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot \text{Matm} / (t^2 \cdot \text{Bcor})) - Z \cdot \text{SRa}$$

$$= (18.4072 \cdot 33536 / (177.0^2 \cdot 1926.0)) - 9.5263 \cdot 2$$

$$= 83.90 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (\text{SHo} + \max(\text{SRo}, \text{STo})) / 2$$

$$= (37 + \max(0.5, 27)) / 2$$

$$= 31.75 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (\text{SHa} + \max(\text{SRa}, \text{STa})) / 2$$

$$= (116 + \max(2, 84)) / 2$$

$$= 100.01 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= \text{Wm1} / \text{Ab}$$

$$= 184056 / 447.0959$$

$$= 40.37 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (\text{Wm2} / \text{Ab})$$

$$= (338730 / 447.0959)$$

$$= 74.30 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	36.87	132.15	116.11	172.50
Radial Flange	0.55	88.10	1.72	115.00
Tangential Flange	26.64	88.10	83.90	115.00
Maximum Average	31.75	88.10	100.01	115.00
Bolting	40.37	172.00	74.30	172.00

Minimum Required Flange Thickness [Rigidity] 174.117 mm

Estimated M.A.W.P. (Operating) 1305 KPa.

Estimated Finished Weight of Flange at given Thk. 1038.6 kg.

Estimated Unfinished Weight of Forging at given Thk 1193.8 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):



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		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	283.07	301.72	0.938	283.07	339.99	0.833
Tangential Flange	St	209.21	226.29	0.925	209.21	255.00	0.820
Tangential Flange	Sto	61.09	150.86	0.405	61.09	170.00	0.359
Combined	Sh/f + St	452.57	452.57	1.000	452.57	509.99	0.887
Combined	Sr + St	213.50	452.57	0.472	213.50	509.99	0.419
Combined	Sh + Sto	344.16	452.57	0.760	344.16	509.99	0.675
Radial Flange	Sr	4.30	301.72	0.014	4.30	339.99	0.013
Maximum Stress Ratio				1.000			0.887

Coefficients for f and f_2

	f	f_2	f_3
a	-0.71375912	-0.01447638	-0.06456312
b	-0.12846279	-0.27814745	-0.08232239
c	1.08037907	0.01035395	0.13691677
d	0.99766848	1.37984158	-0.77879888
e	1.21823466	-0.34346764	0.14909019
f	0.01024612	0.02536192	0.00660062
g	0.4262313	-0.0014508	0.0616252
h	1.4204976	2.25543162	0.74639834
i	-0.70278181	-0.60889597	-0.079595
j	-0.02483937	-0.15735462	0.09851673
k	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g_1}{g_0} \right) + e \left(\frac{h}{h_0} \right) + g \left(\frac{g_1}{g_0} \right)^2 + i \left(\frac{h}{h_0} \right)^2 + k \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}{\left[1 + b \left(\frac{g_1}{g_0} \right) + d \left(\frac{h}{h_0} \right) + f \left(\frac{g_1}{g_0} \right)^2 + h \left(\frac{h}{h_0} \right)^2 + j \left(\frac{g_1}{g_0} \right) \left(\frac{h}{h_0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 1.3079/1.1244)

= 1.1631

Hub Stress Correction Factor [f2]:

= max(0.25; fn2) if $h/h_0 < 0.35$, else max (0.25; fn3)

= max(0.25; 0.2593/1.0331) if $h/h_0 < 0.35$, else max (0.25; 0.232/0.7944)

= 0.2510

In this case, the hub ratio h/h_0 , 0.2198 was < 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.



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Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	428.9 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	7.4516 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	170.0 N./mm ²
Flange Material Operating Yield Stress	Syo	132.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.375 deg

Gasket Contact Area [Ag]:

$$= \pi/4 (GO.D.^2 - GI.D.^2)$$

$$= \pi/4 (1990.0^2 - 1952.0^2)$$

$$= 1176.5 \text{ cm}^2$$

Target Bolt Stress [Sbsel]: (Eq. O-1)

$$= SgT \cdot Ag / (nb \cdot Abs)$$

$$= 207.0 \cdot 1176.494 / (60 \cdot 7.452)$$

$$= 544.7 \text{ N./mm}^2$$

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

$$= \min(Sbsel; Sbmax)$$

$$= \min(544.703; 507.5)$$

$$= 507.5 \text{ N./mm}^2$$

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

$$= \max(Sbsel; Sbmin)$$

$$= \max(507.5; 290.0)$$

$$= 507.5 \text{ N./mm}^2$$

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

$$= \min(Sbsel; Sfmax)$$

$$= \min(507.5; 428.87)$$

$$= 428.9 \text{ N./mm}^2$$

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

$$:: Sbsel \geq Sgmin-S (Ag / (nb \cdot Abs))$$

$$:: 428.87 \geq 140.0 (1176.494 / (60 \cdot 7.452))$$

$$:: 428.9 \geq 368.4 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Operating Stress is maintained. (Eq. O-8)

$$:: Sbsel \geq (Sgmin-O \cdot Ag + \pi/4 \cdot Pmax \cdot GI.D.^2) / (\phi G \cdot Abs \cdot nb)$$

$$:: 428.87 \geq (97.0 \cdot 1176.494 + \pi/4 \cdot 538.624 \cdot 1952.0^2) / (0.7 \cdot 7.452 \cdot 60)$$

$$:: 428.9 \geq 416.1 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: Sbsel \leq Sgmax (Ag / (nb \cdot Abs))$$

$$:: 428.87 \leq 276.0 (1176.494 / (60 \cdot 7.452))$$

$$:: 428.9 \leq 726.3 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: Sbsel \leq Sfmax (thetagmax / thetafmax)$$

$$:: 428.87 \leq 428.87 (1.0/0.375)$$



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 Flg Calc [Int P]: RHS Ch Flg Flng: 11 6:27pm Sep 8,2026

:: 428.9 ≤ 1143.4 N./mm² Check Passes

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= Sbsel \cdot K \cdot Abs \cdot \phi b$$

$$= 428.87 \cdot 0.2 \cdot 7.452 \cdot 34.925$$

$$= 227.6 \text{ Kg-m.}$$

Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 103 N./mm²
 Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since Sb >= SbB8, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 33535.8 / 1.0 \cdot 9806.637 \cdot 0.428 / (7.113 \cdot 195129 \cdot 11.0^2 \cdot 145.554 \cdot 0.3)$$

$$= 1.000 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 10648.2 / 1.0 \cdot 9806.637 \cdot 0.428 / (7.113 \cdot 186083 \cdot 11.0^2 \cdot 145.554 \cdot 0.3)$$

$$= 0.333 \text{ (should be } \leq 1)$$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:

$$= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 33535.8 / 1.0 \cdot 9806.637 \cdot 0.428 / (7.776 \cdot 195129 \cdot 11.0^2 \cdot 145.554 \cdot 0.3)$$

$$= 0.914 \text{ (should be } \leq 1)$$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:

$$= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$$

$$= 52.14 \cdot 10648.2 / 1.0 \cdot 9806.637 \cdot 0.428 / (7.776 \cdot 186083 \cdot 11.0^2 \cdot 145.554 \cdot 0.3)$$

$$= 0.304 \text{ (should be } \leq 1)$$

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Longitudinal Hub	36.81	172.50	113.28	172.50
Radial Flange	0.76	115.00	2.34	115.00
Tangential Flange	23.68	115.00	72.89	115.00
Maximum Average	30.24	115.00	93.09	115.00
Bolting	40.37	172.00	74.30	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 163.119 mm

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Flg Calc [Int P]: RHS Blind Fl

Flng: 12

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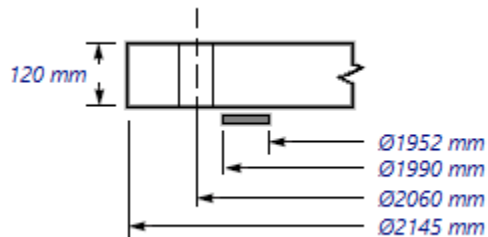
Sep 8, 2026

Flange Input Data Values

Description: RHS Blind Fl :

RHS Blind Flang

Flange Type	TEMA Channel Cover		
Design Pressure	P	540.83	KPa.
Design Temperature		150	°C
Internal Corrosion Allowance	ci	3.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	2145.000	mm
Flange Thickness	t	120.0000	mm
Flange Material	SA-965 F304L		
Flange Material UNS number	S30403		
Flange Allowable Stress At Temperature	Sfo	88.10	N./mm ²
Flange Allowable Stress At Ambient	Sfa	115.00	N./mm ²
Bolt Material	SA-193 B7		
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	6471.681	mm
Youngs Modulus for Blind Flange Matl.	Sy	186083.27	N./mm ²
Allowed Channel Cover Deflection	dfAllow	2.4075	mm
Diameter of Bolt Circle	C	2060.000	mm
Nominal Bolt Diameter	a	34.9250	mm
Type of Threads	TEMA Thread Series		
Number of Bolts	60		
Flange Face Outside Diameter	Fod	2001.000	mm
Flange Face Inside Diameter	Fid	1920.000	mm
Flange Facing Sketch	Sketch 1a		
Gasket Outside Diameter	Go	1990.000	mm
Gasket Inside Diameter	Gi	1952.000	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm



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Gasket Contact Width,	$N = (Go - Gi)/2$	19.000 mm
Basic Gasket Width,	$bo = N/2$	9.500 mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	7.767 mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	1974.466 mm



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 Flg Calc [Int P]: RHS Blind Fl Flng: 12 6:27pm Sep 8,2026

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 1974.4663^2 \cdot 540.829$$

$$= 168867.531 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 7.7669 \cdot \pi \cdot 1974.4663 \cdot 3.0 \cdot 540.83$$

$$= 15942.498 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(168868 + 15942 + 0, 0)$$

$$= 184810.031 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$$

$$= 68.95 \cdot 7.7669 \cdot \pi \cdot 1974.466 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 338730.156 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$$

$$= \max((184810 + 0 + 4 \cdot 0/1974.47)/172, 338730/172)$$

$$= 193.135 \text{ cm}^2$$

Bolting Information for TEMA Imperial Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	193.135	447.096	
Radial Distance between Bolts and Edge:	34.925	42.500	
Circ. Spacing between the Bolts:	77.800	107.812	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 447.096 \cdot 172.0 / (68.95 \cdot \pi (1990.0 + 1952.0))$$

$$= 9.006 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 447.0959$$

$$= 784138.12 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 184810.03 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (2060.0 - 1974.4663) / 2$$

$$= 42.7669 \text{ mm}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot Wm1 \cdot hG \cdot Bsc / (t^2 \cdot G) + C \cdot Z \cdot Peq \cdot G^2 / t^2$$

$$= 1.9 \cdot 184810 \cdot 42.7669 \cdot 1.0 / (117.0^2 \cdot 1974.4663) + 0.3 \cdot 1.0 \cdot 540.83 \cdot 1974.4663^2 / 117.0^2$$

$$= 51.66 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:



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 Flg Calc [Int P]: RHS Blind Fl Flng: 12 6:27pm Sep 8,2026

$$= 1.9 \cdot W \cdot hG \cdot Bsc / (t^2 \cdot G)$$

$$= 1.9 \cdot 784138 \cdot 42.7669 \cdot 1.0 / (117.0^2 \cdot 1974.4663)$$

$$= 23.12 \text{ N./mm}^2$$

Bolt Stress, Operating [BSO]:

$$= Wm1 / Ab$$

$$= 184810 / 447.0959$$

$$= 40.54 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm2 / Ab)$$

$$= (338730 / 447.0959)$$

$$= 74.30 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	51.66	88.10	23.12	115.00
Bolting	40.54	172.00	74.30	172.00

TEMA Channel Cover Deflection Calculations:

Actual deflection at center per TEMA formula 2.2892 mm
 Allowed deflection at center of Cover 2.4075 mm

Reqd. Blind Flange Thickness at Center 92.592 mm
 Reqd. Blind Flange Thickness at Gasket 55.459 mm
 Min. Required Flange Thickness per TEMA 115.051 mm

Estimated M.A.W.P. (Operating) 594 KPa.
 Estimated Finished Weight of Flange at given Thk. 3480.9 kg.
 Estimated Unfinished Weight of Forging at given Thk 3480.9 kg.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	49.11	115.00	21.98	115.00
Bolting	40.54	172.00	74.30	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 113.259 mm

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Internal Pressure Calculations: Step: 7 6:27pm Sep 8,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
LHS Blind Flang		540.98	128	3	2140	88.101	1
LHS ch Flange		538.82	188	3	1920	88.101	2
LHS ch Shell		538.83	14	1	1920	97.748	3
Main Shell		1328.9	16	1	1926	117.3	4
Main Shell		1328.9	14	1	1926	117.3	5
Main Shell		1328.6	16	1	1926	117.3	6
RHS ch Shell		538.63	20	1	1920	97.748	7
RHS Ch Flg		538.62	188	3	1920	88.101	8
RHS Blind Flang		540.83	128	3	2145	88.101	9

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
LHS Blind Flang		520	573.131	736.797	120	115.056	1
LHS ch Flange		520	1285.74	1662.41	180	174.117	2
LHS ch Shell		520	1292.74	1413.05	14	6.31533	3
Main Shell		1310	1789.3	1929.53	16	11.9967	4
Main Shell		1310	1550.14	1690.42	14	11.9967	5
Main Shell		1310	1789.55	1929.53	16	11.9946	6
RHS ch Shell		520	1891.19	2011.16	20	6.31339	7
RHS Ch Flg		520	1285.94	1662.41	180	174.117	8
RHS Blind Flang		520	573.278	736.797	120	115.051	9

Warning: Shell Side MAWP governed by a standard Flange!

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1540 KPa.
 Shell Side MAPnc = 1690 KPa.
 Channel Side MAWP = 573 KPa.
 Channel Side MAPnc = 737 KPa.

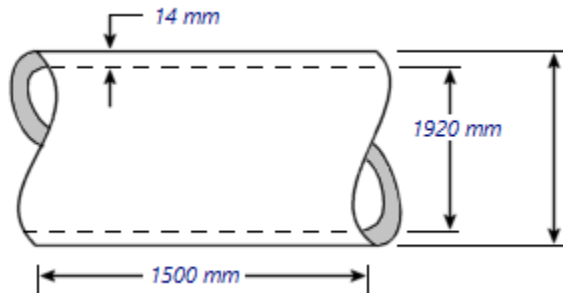
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: LHS ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	538.830 KPa.	Int. Design Temperature	150 °C
Material	SA-240 304L	Ext. Design Temperature	150 °C
Length	1500.000 mm	External Pressure Chart	HA-3
		UNS Number	S30403
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	115.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	115.0 N./mm ²
Inside Diameter Di	1920.000 mm	Actual Stress	47.2 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	91797.4 cm ²	SG of Contents	1.000
Element Volume	4343716.0 cm ³	Weight of Contents	4341.1 Kg
Empty Weight	1024.2 Kg	Operating Weight	5365.3 Kg
		Comp. All. Ope. Stress	60.1 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (538.83 \cdot 961.0) / (115.0 \cdot 0.85 - 0.6 \cdot 538.83) \\
 &= 5.3153 + 1.0000 = 6.3153 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 18.830 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (115.0 \cdot 0.85 \cdot 13.0) / (961.0 + 0.6 \cdot 13.0) \\
 &= 1312 - 19 = 1293 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (115.0 \cdot 0.85 \cdot 14.0) / (960.0 + 0.6 \cdot 14.0) \\
 &= 1413 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (538.83 \cdot (961.0 + 0.6 \cdot 13.0)) / (0.85 \cdot 13.0) \\
 &= 47.244 \text{ N./mm}^2
 \end{aligned}$$



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Element: LHS ch Shell

Step: 8

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% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 0.724 %

Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: LHS ch Shell

Cylindrical Shell From 30 to Tsht Ext. Chart: HA-3 at 150 °C

Elastic Modulus from Chart: HA-3 at 150 °C: 183300992 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	1948.00	1499.97	149.85	0.7700	0.0009603	44.93

$$MAEP = (4*B)/(3*(Do/t)) = (4*44.9269)/(3*149.8461) = 399.7372 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.403	1948.00	1499.97	442.46	0.7700	0.0001872	17.16

$$MAEP = (4*B)/(3*(Do/t)) = (4*17.1594)/(3*442.458) = 51.7062 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	1948.00	19071.50	149.85	9.7903	0.0000634	5.82

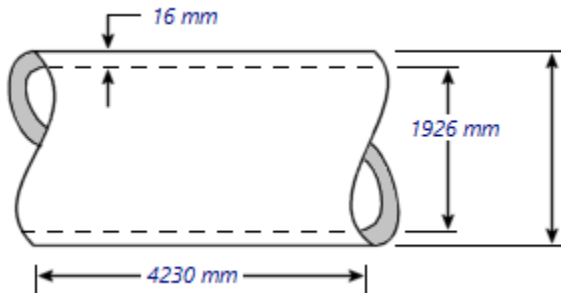
$$MAEP = (4*B)/(3*(Do/t)) = (4*5.8153)/(3*149.8461) = 51.7416 \text{ KPa.}$$

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1328.888 KPa.	Int. Design Temperature	160 °C
Material	SA-516 70	Ext. Design Temperature	160 °C
Length	4230.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1926.000 mm	Actual Stress	101.4 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	260197.4 cm ²	SG of Contents	1.000
Element Volume	6119781.5 cm ³	Weight of Contents	6116.0 Kg
Empty Weight	3200.3 Kg	Operating Weight	9316.3 Kg
		Comp. All. Ope. Stress	120.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1328.888 \cdot 964.0) / (138.0 \cdot 0.85 - 0.6 \cdot 1328.888) \\
 &= 10.9967 + 1.0000 = 11.9967 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 18.889 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 15.0) / (964.0 + 0.6 \cdot 15.0) \\
 &= 1808 - 19 = 1789 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 16.0) / (963.0 + 0.6 \cdot 16.0) \\
 &= 1930 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1328.888 \cdot (964.0 + 0.6 \cdot 15.0)) / (0.85 \cdot 15.0) \\
 &= 101.418 \text{ N./mm}^2
 \end{aligned}$$



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Element: Main Shell

Step: 9

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% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.824 %

Minimum Design Metal Temperature Results:

Govern. thk, $t_g = 16.0$, $t_r = 12.758$, $c = 1.0$ mm, $E^* = 0.85$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.723$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1540.046 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.723	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	-29 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 40 to Ring R1 Fr40 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 198566240 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1958.00	2535.03	130.53	1.2947	0.0006846	67.98

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 67.9776) / (3 \cdot 130.5333) = 694.3171 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.293	1958.00	2535.03	369.95	1.2947	0.0001445	14.35

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 14.3476) / (3 \cdot 369.9541) = 51.7064 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1958.00	53294.61	130.53	27.2189	0.0000656	6.51

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 6.5136) / (3 \cdot 130.5333) = 66.5296 \text{ KPa.}$$

Cylindrical Shell From Ring R1 Fr40 to 50 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 198566240 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1958.00	7618.00	130.53	3.8907	0.0002136	21.21



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Element: Main Shell

Step: 9

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$$MAEP = (4*B)/(3*(Do/t)) = (4*21.208)/(3*130.5333) = 216.6165 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.398	1958.00	7618.00	233.14	3.8907	0.0000911	9.04

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.0419)/(3*233.1421) = 51.7074 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1958.00	157800.47	130.53	50.0000	0.0000651	6.46

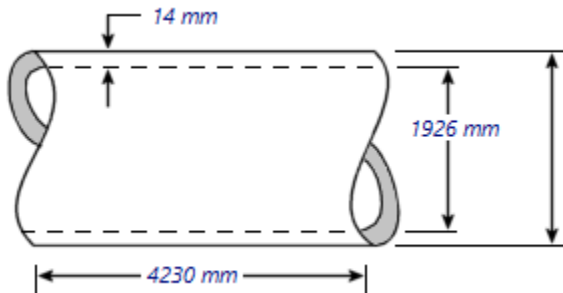
$$MAEP = (4*B)/(3*(Do/t)) = (4*6.4638)/(3*130.5333) = 66.0211 \text{ KPa.}$$

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1328.888 KPa.	Int. Design Temperature	160 °C
Material	SA-516 70	Ext. Design Temperature	160 °C
Length	4230.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1926.000 mm	Actual Stress	116.9 N./mm ²
Spec. Min. Thk t,e, or sa	14.0000 mm	Spec. Nominal Thk.	14.000 mm
Surface Area	259665.9 cm ²	SG of Contents	1.000
Element Volume	6119782.5 cm ³	Weight of Contents	6116.0 Kg
Empty Weight	2797.4 Kg	Operating Weight	8913.4 Kg
		Comp. All. Ope. Stress	116.5 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1328.888 \cdot 964.0) / (138.0 \cdot 0.85 - 0.6 \cdot 1328.888) \\
 &= 10.9967 + 1.0000 = 11.9967 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 18.889 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 13.0) / (964.0 + 0.6 \cdot 13.0) \\
 &= 1569 - 19 = 1550 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 14.0) / (963.0 + 0.6 \cdot 14.0) \\
 &= 1690 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1328.888 \cdot (964.0 + 0.6 \cdot 13.0)) / (0.85 \cdot 13.0) \\
 &= 116.877 \text{ N./mm}^2
 \end{aligned}$$



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 Element: Main Shell Step: 10 6:34pm Sep 8,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.722 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 14.0$, $t_r = 12.758$, $c = 1.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.834$, Temp. Reduction = 9 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-27 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1540.046 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	14.000 mm	Basic MDMT	-18 °C
Stress/Thk. Ratio	0.834	Temperature Difference	9 °C
UCS-66(c) red.	No	Computed Min. Temp.	-27 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 198566240 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	1954.00	7618.00	150.31	3.8987	0.0001722	17.09

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 17.0941) / (3 \cdot 150.3077) = 151.6279 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.388	1954.00	7618.00	232.95	3.8987	0.0000910	9.03

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.0343) / (3 \cdot 232.948) = 51.7068 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
13.000	1954.00	21151.54	150.31	10.8247	0.0000588	5.84

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 5.8359) / (3 \cdot 150.3077) = 51.7659 \text{ KPa.}$$



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Element: Main Shell

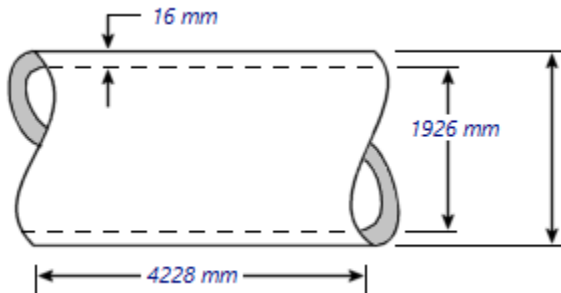
Step: 11

6:34pm

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	1328.634 KPa.	Int. Design Temperature	160 °C
Material	SA-516 70	Ext. Design Temperature	160 °C
Length	4228.000 mm	External Pressure Chart	CS-2
		UNS Number	K02700
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	138.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	138.0 N./mm ²
Inside Diameter Di	1926.000 mm	Actual Stress	101.4 N./mm ²
Spec. Min. Thk t,e, or sa	16.0000 mm	Spec. Nominal Thk.	16.000 mm
Surface Area	260074.4 cm ²	SG of Contents	1.000
Element Volume	6116888.0 cm ³	Weight of Contents	6113.2 Kg
Empty Weight	3198.8 Kg	Operating Weight	9311.9 Kg
		Comp. All. Ope. Stress	120.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (1328.634 \cdot 964.0) / (138.0 \cdot 0.85 - 0.6 \cdot 1328.634) \\
 &= 10.9946 + 1.0000 = 11.9946 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 18.634 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 15.0) / (964.0 + 0.6 \cdot 15.0) \\
 &= 1808 - 19 = 1790 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (138.0 \cdot 0.85 \cdot 16.0) / (963.0 + 0.6 \cdot 16.0) \\
 &= 1930 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (1328.634 \cdot (964.0 + 0.6 \cdot 15.0)) / (0.85 \cdot 15.0) \\
 &= 101.399 \text{ N./mm}^2
 \end{aligned}$$



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 Element: Main Shell Step: 11 6:34pm Sep 8,2026

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 0.824 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 16.0$, $t_r = 12.758$, $c = 1.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.723$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1540.046 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-12 °C
Governing Thk.	16.000 mm	Basic MDMT	-14 °C
Stress/Thk. Ratio	0.723	Temperature Difference	15 °C
UCS-66(c) red.	No	Computed Min. Temp.	-29 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	85 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 60 to Ring R1 Fr60 Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 198566240 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1958.00	7618.00	130.53	3.8907	0.0002136	21.21

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 21.208) / (3 \cdot 130.5333) = 216.6165 \text{ KPa.}$$

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
8.398	1958.00	7618.00	233.14	3.8907	0.0000911	9.04

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 9.0419) / (3 \cdot 233.1421) = 51.7074 \text{ KPa.}$$

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1958.00	157800.47	130.53	50.0000	0.0000651	6.46

$$MAEP = (4 \cdot B) / (3 \cdot (Do/t)) = (4 \cdot 6.4638) / (3 \cdot 130.5333) = 66.0211 \text{ KPa.}$$

Cylindrical Shell From Ring R1 Fr60 to Tsht Ext. Chart: CS-2 at 160 °C

Elastic Modulus from Chart: CS-2 at 160 °C: 198566240 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1958.00	2534.97	130.53	1.2947	0.0006847	67.98



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Element: Main Shell

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$$MAEP = (4*B)/(3*(Do/t)) = (4*67.9792)/(3*130.5333) = 694.3339 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
5.292	1958.00	2534.97	369.96	1.2947	0.0001445	14.35

$$MAEP = (4*B)/(3*(Do/t)) = (4*14.3477)/(3*369.9576) = 51.7063 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
15.000	1958.00	53294.57	130.53	27.2189	0.0000656	6.51

$$MAEP = (4*B)/(3*(Do/t)) = (4*6.5136)/(3*130.5333) = 66.5296 \text{ KPa.}$$

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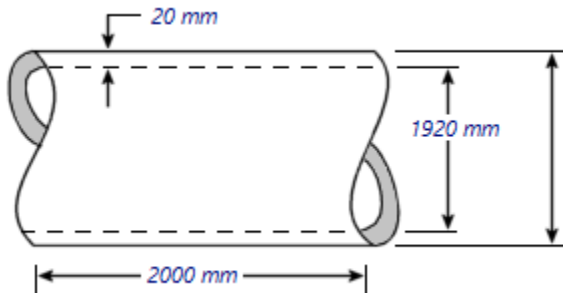
Element: RHS ch Shell

Step: 12 6:34pm

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Internal Pressure Calculation Results:

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Internal Pressure Results for: RHS ch Shell

Cylindrical Shell Design Information:			
Design Pressure P	538.634 KPa.	Int. Design Temperature	150 °C
Material	SA-240 304L	Ext. Design Temperature	150 °C
Length	2000.000 mm	External Pressure Chart	HA-3
		UNS Number	S30403
Int. Corr. All. c	1.0000 mm	Allowable Stress (ope) S	115.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	115.0 N./mm ²
Inside Diameter Di	1920.000 mm	Actual Stress	32.4 N./mm ²
Spec. Min. Thk t,e, or sa	20.0000 mm	Spec. Nominal Thk.	20.000 mm
Surface Area	123150.4 cm ²	SG of Contents	1.000
Element Volume	5791621.0 cm ³	Weight of Contents	5777.7 Kg
Empty Weight	1957.0 Kg	Operating Weight	7734.6 Kg
		Comp. All. Ope. Stress	120.4 N./mm ²

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v 85 %

Required Thickness due to Internal Pressure [tr]:

$$\begin{aligned}
 &= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)} \\
 &= (538.634 \cdot 961.0) / (115.0 \cdot 0.85 - 0.6 \cdot 538.634) \\
 &= 5.3134 + 1.0000 = 6.3134 \text{ mm}
 \end{aligned}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 18.634 KPa.

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (115.0 \cdot 0.85 \cdot 19.0) / (961.0 + 0.6 \cdot 19.0) \\
 &= 1910 - 19 = 1891 \text{ KPa.}
 \end{aligned}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$\begin{aligned}
 &= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)} \\
 &= (115.0 \cdot 0.85 \cdot 20.0) / (960.0 + 0.6 \cdot 20.0) \\
 &= 2011 \text{ KPa.}
 \end{aligned}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$\begin{aligned}
 &= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t) \\
 &= (538.634 \cdot (961.0 + 0.6 \cdot 19.0)) / (0.85 \cdot 19.0) \\
 &= 32.433 \text{ N./mm}^2
 \end{aligned}$$



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 Element: RHS ch Shell Step: 12 6:34pm Sep 8,2026

% Elongation per Table UG-79-1 (50*tnom/Rf)*(1-Rf/Ro) 1.031 %
 Note: Please Check Requirements of Table UHA-44 for Elongation limits.

SA-240 304L, Min Metal Temp without impact per UHA-51: -196 °C

EXTERNAL PRESSURE RESULTS for: RHS ch Shell

Cylindrical Shell From 70 to 80 Ext. Chart: HA-3 at 150 °C

Elastic Modulus from Chart: HA-3 at 150 °C: 183300992 KPa.

Results for Maximum Allowable Ext. Pressure					MAEP	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	1960.00	2000.03	103.16	1.0204	0.0012575	47.44

$$MAEP = (4*B)/(3*(Do/t)) = (4*47.4424)/(3*103.1579) = 613.1661 \text{ KPa.}$$

Results for Required Thickness					Tca	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
4.982	1960.00	2000.03	393.45	1.0204	0.0001665	15.26

$$MAEP = (4*B)/(3*(Do/t)) = (4*15.2587)/(3*393.4506) = 51.706 \text{ KPa.}$$

Results for Maximum Stiffened Length					Slen	
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
19.000	1960.00	49662.19	103.16	25.3378	0.0001043	9.56

$$MAEP = (4*B)/(3*(Do/t)) = (4*9.5579)/(3*103.1579) = 123.5309 \text{ KPa.}$$



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 Shop Pressure Test Results: Step: 13 6:34pm Sep 8,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2002.060 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1703.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2176.523 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1694.051 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1873.300 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1873.300 KPa.
App 27-4	= M.A.W.P.	1540.046 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	745.071 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	936.811 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	630.444 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	573.131 KPa.

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS ch Shell	83.8	153.0	0.548	955.63
Main Shell	167.5	234.0	0.716	2195.40
Main Shell	193.1	234.0	0.825	2195.40
Main Shell	167.5	234.0	0.716	2195.40
RHS ch Shell	57.5	153.0	0.376	955.63

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	115.00	88.10	1.305
N1-Tube Inlet	Pad	138.00	138.00	1.000
N3-Shell Inlet	Nozzle	118.00	118.00	1.000
N3-Shell Inlet	Pad	138.00	138.00	1.000
N4-Shell Outlet	Nozzle	138.00	138.00	1.000
N4-Shell Outlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	117.00	111.00	1.054
N2-Tube Outlet	Pad	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1 Fr40	138.00	138.00	1.000
Ring R1 Fr60	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):



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Shop Pressure Test Results:

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Description	Ambient	Operating	Ratio
LHS Blind Flang	115.00	88.10	1.305
LHS ch Flange	115.00	88.10	1.305
LHS ch Shell	115.00	115.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS ch Shell	115.00	115.00	1.000
RHS Ch Flg	115.00	88.10	1.305
RHS Blind Flang	115.00	88.10	1.305

Minimum

1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	115.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	24.64	149.50	0.165
N3-Shell Inlet	21.26	216.00	0.098
N4-Shell Outlet	21.26	179.40	0.119
N2-Tube Outlet	32.72	152.10	0.215

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 14 6:34pm Sep 8,2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	2002.060 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1703.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2176.523 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1694.051 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1873.300 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1873.300 KPa.
App 27-4	= M.A.W.P.	1540.046 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	745.071 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	676.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	936.811 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	630.444 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	743.600 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	743.600 KPa.
App 27-4	= M.A.W.P.	573.131 KPa.

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 1310.0 • 1.0

= 1703.000 KPa.

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 520.0 • 1.0

= 676.000 KPa.

Horizontal Test performed per: UG-99b (Note 35)

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
LHS ch Shell	60.9	153.0	0.398	694.82
Main Shell	131.4	234.0	0.562	1721.88
Main Shell	151.4	234.0	0.647	1721.88
Main Shell	131.4	234.0	0.562	1721.88
RHS ch Shell	41.8	153.0	0.273	694.82

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	115.00	88.10	1.305
N1-Tube Inlet	Pad	138.00	138.00	1.000
N3-Shell Inlet	Nozzle	118.00	118.00	1.000
N3-Shell Inlet	Pad	138.00	138.00	1.000
N4-Shell Outlet	Nozzle	138.00	138.00	1.000
N4-Shell Outlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	117.00	111.00	1.054
N2-Tube Outlet	Pad	138.00	138.00	1.000

Minimum

1.000



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Field Pressure Test Results:

Step: 14

6:34pm

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Stress ratios for Stiffening Ring Materials (N./mm²):

Description	Ambient	Operating	Ratio
Ring R1 Fr40	138.00	138.00	1.000
Ring R1 Fr60	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
LHS Blind Flang	115.00	88.10	1.305
LHS ch Flange	115.00	88.10	1.305
LHS ch Shell	115.00	115.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
Main Shell	138.00	138.00	1.000
RHS ch Shell	115.00	115.00	1.000
RHS Ch Flg	115.00	88.10	1.305
RHS Blind Flang	115.00	88.10	1.305

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	115.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	17.92	149.50	0.120
N3-Shell Inlet	16.68	216.00	0.077
N4-Shell Outlet	16.68	179.40	0.093
N2-Tube Outlet	23.79	152.10	0.156

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 6:34p Sep 8,2026

Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		1703.00 KPa.
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1926.000 mm
Shell Circumferential Joint Efficiency	Esw	0.850
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS ch Shell

Channel Type:		Cylinder
Channel Test Pressure		676.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1920.000 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-240 304L

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		S30403
Channel Allowable Stress at Temperature	Sc	115.00 N./mm ²
Channel Allowable Stress at Ambient		115.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2895
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	12800.00 mm
Straight Tube Length (bet. inner tubsht faces)	L	12688.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	170.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	937.029 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	25256.1 cm ²



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ASME TS Calc: Pressure Test

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Perimeter of Tube Layout Cp 6023.4 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03047
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 56.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 1954.000 mm

Area of the Untubed Lanes AL 305.8 cm²

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration l 560.000 mm
Tube End condition corresponding to Span (l) k 1.00

Ignore Radial Thermal Exp. effects (4.18.8/4.18.9) Yes
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Shell Side Min. Design Pressure Psd,min -52.0000 KPa.
Channel Side Min. Design Pressure Ptd,min -52.0000 KPa.

Mean Shell Metal Temp. along Shell len. Tsm 21.11 °C
Mean Tube Metal Temp. along Tube length Ttm 21.11 °C
Junction Stress Reduction option Perform Plastic Calculation

Tubesheet Integral with Both
Tubesheet Extended as Flange No

Is Exchanger in Creep range (skip EP, Use 3S for Sps) No



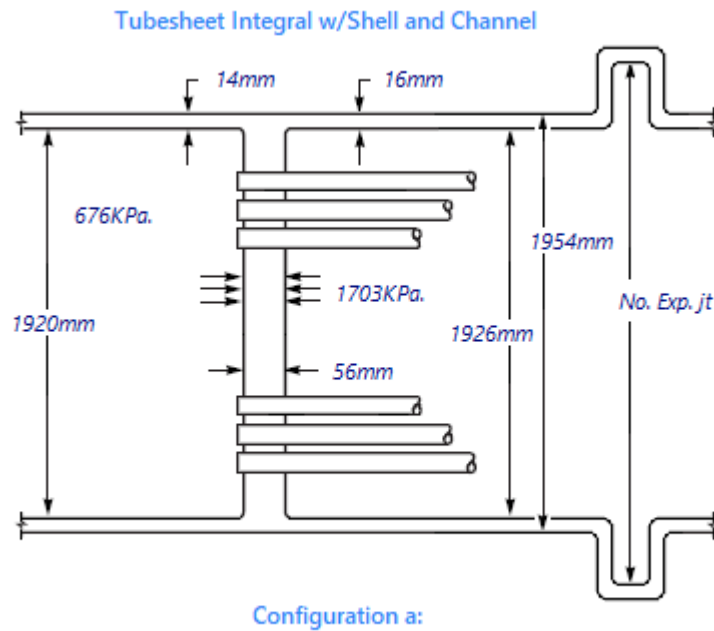
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ASME TS Calc: Pressure Test

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ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties:

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000115191 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115191 /°C
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Austenitic Group 3 Steels Group 1
Channel - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Metal Temp. along Len	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:



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ASME TS Calc: Pressure Test

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The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (728.0 \cdot (25.4/2 - 0)) / (115.0 \cdot 1.0 + 0.4 \cdot 728.0)$$

$$= 0.0802 + 0.0000 = 0.0802 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 21.11 °C
Elastic Modulus for Material 193048800.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	12688.00	12.04	50.0000	0.0075908	87.65
EMAWP = (4*B)/(3*(D/T)) = (4 *87.6487)/(3 *12.0379) = 9707.5039 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6164	25.40	12688.00	41.21	50.0000	0.0006478	54.25
EMAWP = (4*B)/(3*(D/T)) = (4 *54.2487)/(3 *41.2084) = 1755.16 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6164 mm
Allowable Internal Pressure at Corroded thickness	20464.87 KPa.
Required Internal Design Pressure	728.00 KPa.
Allowable External Pressure at Corroded thickness	9707.50 KPa.
Required External Design Pressure	1755.00 KPa.
Required Thickness due to Shell Side pressure	0.6164 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [ρ]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/56.0$$

$$= 0.0000 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (StT/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (0.19512E+09/.20270E+09) \cdot (114/137) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 937.029 + 25.4$$

$$= 1899.458 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$



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Equivalent outer tube limit radius [ao]:

$$= D_o / 2$$

$$= 1899.458 / 2$$

$$= 949.7290 \text{ mm}$$

Effective tube pitch [p*]:

$$= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)}$$

$$= 31.75 / \sqrt{1 - 4 \cdot \min(305.75 \cdot 100.0 ; 4 \cdot 1899.458 \cdot 31.75) / (\pi \cdot 1899.458^2)}$$

$$= 31.9227 \text{ mm}$$

Effective ligament efficiency for bending [μ^*]:

$$= (p^* - d^*) / p^*$$

$$= (31.9227 - 25.4) / 31.9227$$

$$= 0.2043$$

Ratio [ps]:

$$= a_s / a_o$$

$$= 963.0 / 949.729$$

$$= 1.0140$$

Ratio [pc]:

$$= a_c / a_o$$

$$= 960.0 / 949.729$$

$$= 1.0108$$

Compute parameter [xt]:

$$= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot a_o))^2$$

$$= 1 - 2895 / ((25.4 - 2 \cdot 2.11) / (2 \cdot 949.729))^2$$

$$= 0.6401$$

Determine parameter [xs]:

$$= 1 - Nt / (dt / (2 \cdot a_o))^2$$

$$= 1 - 2895 / (25.4 / (2 \cdot 949.729))^2$$

$$= 0.4823$$

Determine the value [h'g]:

$$= \max((h_g - CATC); 0) \text{ (For pressure only cases)}$$

$$= \max((0 - 0); 0)$$

$$= 0.000 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$= \pi \cdot t_s (D_s + t_s) E_s / L$$

$$= \pi \cdot 16.0 (1926.0 + 16.0) \cdot 20270E+09 / 12688.0$$

$$= 159029.9375 \text{ Kgf/mm}$$

Axial Tube Stiffness [Kt]:

$$= \pi \cdot t_t (D_t - t_t) E_t / L$$

$$= \pi \cdot 2.11 (25.4 - 2.11) \cdot 19512E+09 / 12688.0$$

$$= 242.1033 \text{ Kgf/mm}$$

Stiffness Factor [Ks,t]:

$$= K_s / (Nt \cdot K_t)$$

$$= 159029.938 / (2895 \cdot 242.103)$$

$$= 0.22690$$

Rigidity Ratio [J]:

$$= 1 / (1 + K_s / K_j)$$

$$= 1 / (1 + 159029.938 / 0)$$

$$= 1.0000 (= 1 \text{ if No Exp. Jt.})$$

Shell coefficient [β_s]:

$$= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s)$$

$$= ((12(1 - 0.3^2))^{1/4}) / ((1926.0 + 16.0) 16.0)$$



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= 0.0103 1/mm

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.01 \cdot 0.20270E+09 \cdot 16.0^3 / (6(1 - 0.3^2))$$

$$= 1568175872.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [ls]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1926.0 + 16.0) \cdot 0.15682E+1 / 56.0^3 \cdot (1 + 56.0 \cdot 0.01 + (56.0 \cdot 0.01)^2 / 2)$$

$$= 181486528.0000 \text{ KPa}$$

Shell coefficient [ds]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1926.0^2 / (4 \cdot 0.20270E+09 \cdot 16.0) \cdot (1 - 1926.0 \cdot 0.3 / (2(1926.0 + 16.0)))$$

$$= 0.24339E+00 \text{ mm/N./mm}^2$$

Channel coefficient [bc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.31^2))^{1/4}) / \sqrt{(1920.0 + 14.0)14.0}$$

$$= 0.0110 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.011 \cdot 0.19512E+09 \cdot 14.0^3 / (6(1 - 0.31^2))$$

$$= 1088782592.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [lc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1920.0 + 14.0 / \cos(0)) \cdot 0.10888E+1 / 56.0^3 \cdot (1.0 + 56.0 \cdot 0.011 \cdot (1.0 + 1.0) / 2 + (56.0 \cdot 0.011)^2 / 2 \cdot 1.0)$$

$$= 130096760.0 \text{ KPa}$$

Channel coefficient [dc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1920.0^2 / (4 \cdot 0.19512E+09 \cdot 14.0) \cdot (1 - 1920.0 \cdot 0.31 / (2(1920.0 + 14.0)))$$

$$= 0.28545E+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$h/p = 1.763780$; $\mu^* = 0.204328$; $E^* = 33998984 \text{ KPa}$.

$$E^*/E = 0.167730$$
 ; $\nu^* = 0.441119$

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.441^2)2895 \cdot 0.19512E+09 \cdot 2.11(25.4 - 2.11)949.729^2) / (33998984 \cdot 12688.0 \cdot 56.0^3))^{1/4}$$

$$= 8.9403$$

Values from Table 4.18.3

$$Z_d = 0.002056$$
 ; $Z_v = 0.012502$; $Z_m = 0.158936$
 $Z_a = 0.365837E+04$; $Z_w = 0.012502$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1954.0 / 1899.458$$



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ASME TS Calc: Pressure Test

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= 1.0287

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.44) / 33998984 \cdot (0.18149E+09 + 0.13010E+09 + 0.20270E+09 \cdot \ln(1.03))$$

= 5.2162

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.4411) 5.2162$$

= 7.5171

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.014 - 1 - 7.5171 \cdot 0.0125) / (1 + 7.5171 \cdot 0.1589)$$

= -0.036452

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.00206 + -0.03645 \cdot 0.0125) / 2 \cdot 8.94027^4$$

= 5.1108

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.0125 + -0.03645 \cdot 0.1589) / 2 \cdot 8.94027^4$$

= 21.4273

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.0125 + (1.014 - 1) 0.1589) 8.94027^4 / (1 + 7.5171 \cdot 0.1589)$$

= 42.8546

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

= 0 (config a)

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.014 \cdot 0.15682E+1 \cdot 0.0103 \cdot 0.243389 (1 + 56.0 \cdot 0.0103)$$

= 6296.3740 mm²

Calculate parameter [ω^*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$$

$$= 949.729^2 (1.014^2 - 1) \cdot (1.014 - 1) / 4 - 6296.374$$

= -6207.6987 mm²

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.011 \cdot 0.10888E+1 [0.285448 (1.0 + 0.011 \cdot 1.0 \cdot 56.0) - 0 (1.0/56.0 + 1.0 \cdot 0.011/2)]$$

= 5604.9805 mm²

Parameter [ω^*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega_c$$

$$= 949.729^2 [(1.01081^2 + 1) \cdot (1.01081 - 1) / 4 - (1.01397 - 1) / 2] - 5604.98047$$

= -6976.5479 mm²

Pressure [P^*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D



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Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 1703.0 \{ 0.482 + 2(1 - 0.482) 0.31 + [2/0.227 \cdot (1926.0/1899.458)^2] 0.3 - [(1.014^2 - 1)/(1.0 \cdot 0.227)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 0.227)] [(0^2 - (1926.0)^2)/1899.458^2] \}$$

$$= 5786.8594 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.64 + 2(1 - 0.64) 0.31 + 1/(1.0 \cdot 0.227)] 676.0$$

$$= 3562.8560 \text{ KPa.}$$

Pressure [Pγ]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 2895 \cdot 242.103 \cdot 0 / (3.142 \cdot 949.729^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 42.855 \cdot 0 / (2 \cdot \pi \cdot 949.729^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(42.855/949.729^2) \cdot (-6207.699 \cdot 1703.0 - 6976.548 \cdot 676.0)$$

$$= 278.2061 \text{ KPa.}$$

Calculate the pressure [Pω]:

$$= (U/ao^2) \cdot (\omega*s \cdot P*s - \omega*c \cdot P*c)$$

$$= (42.855/949.729^2) \cdot (6296.374 \cdot 0 - 5604.9805 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + P\gamma + P\omega + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.227 / (1 + 1.0 \cdot 0.227 (5.111 + (1.014 - 1)21.427)) \cdot (5786.859 - 3562.856 + 0 + 0 + 0 + 278.206)$$

$$= 254.8734 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega*s \cdot P*s - \omega*c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-6207.699 \cdot 1703.0 - 6976.548 \cdot 676.0) - (6296.374 \cdot 0 - 5604.98 \cdot 0)) \cdot 0 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 7.51715 \cdot 0.15894)$$

$$= -272.0701 \text{ Kg}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.036 + 2 \cdot -272.07 / (254.873 \cdot 949.729^2)$$

$$= -0.05966$$

Fm Value from Table 4.18.3 = 0.029831

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$



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ASME TS Calc: Pressure Test

Case: 1 6:34p Sep 8, 2026

$$= (1.5 \cdot 0.0298 / 0.2043) \cdot (2 \cdot 949.729 / (56.0 - 0))^2 \cdot 254.87$$

$$= 64.2198 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot \beta T \cdot S_y = 1.5 \cdot 0.95 \cdot 250.0 = 356.24 \text{ N./mm}^2, 4.18.233$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0143 / 0.2043) \cdot (2 \cdot 949.729 / (13.334 - 0))^2 \cdot 167.21$$

$$= 356.2376 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 13.3335 + 0 + 0$$

$$= 13.3335 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ_{limit}]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot (0.8 \cdot \beta T \cdot S_y)$$

$$= (2 \cdot 0.2 \cdot 56.0 / 949.729) \cdot 189.997$$

$$= 4.4812 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ_{limit}

Could not converge on a minimum required tubesheet thk. to satisfy the Shear Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

$$\text{Tubesheet thickness} = 7.6185 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 35.5461 \text{ N./mm}^2$$

Required Tubesheet Thickness for Given Loadings (Including CA) [H reqd]:

$$= \max(H_{reqB}, H_{reqS})$$

$$= \max(13.3335, 7.6185)$$

$$= 13.3335 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.2671, F_{tmax} = 4.1839$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((1703.0 \cdot 0.4823 - 676.0 \cdot 0.6401) - (254.873) \cdot -0.267) /$$

$$(0.6401 - 0.4823)$$

$$= 2.8964 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((1703.0 \cdot 0.4823 - 676.0 \cdot 0.6401) - (254.873) \cdot 4.184) /$$

$$(0.6401 - 0.4823)$$

$$= -4.2965 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 4.297 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:



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ASME TS Calc: Pressure Test

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$$= \beta T \cdot S_{y,t} = 161.4971 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q_3 \cdot Z_w)X_{a^4}); 1.25)$$

$$= \max((3.25 - 0.25(0.002 + -0.06 \cdot 0.013)8.94^4); 1.25)$$

$$= 1.2500 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$$

$$= (\sqrt{ (25.4^2 + (25.4 - 2 \cdot 2.11)^2) }) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / S_{y,t}) }$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.19512E+09 / 169) }$$

$$= 166.8131$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1.0 \cdot 560.0 / 8.268$$

$$= 67.7312$$

Buckling Allowable Stress [Stb]:

$$= (S_{y,t} / F_s) \cdot (1 - F_t / (2 \cdot C_t))$$

$$= (169 / 1.25) \cdot (1 - 67.7312 / (2 \cdot 166.813))$$

$$= 161.497 \text{ N./mm}^2 \text{ (Never greater than } S_{ot} \text{)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 4.3 \cdot 1.5438$$

$$= 67.64 \text{ Kg}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 161.497$$

$$= 2542.365 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 161.5$$

$$= 2487.9636 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 161.497 / 161.497$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 161.5$$

$$= 1609.8589 \text{ Kg}$$



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Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1.0 \cdot 1.0} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 2542.36 is >= tube strength 2542.36 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= \beta T \cdot Sy,s = 0.95 \cdot 260.0 = 370.49 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= ao^2 / ((Ds + ts) \cdot ts) \cdot [Pe + (ps^2 - 1)(Ps - Pt)] + as^2 \cdot Pt / ((Ds + ts) \cdot ts)$$

$$= 949.729^2 / ((1926.0 + 16.0) \cdot 16.0) \cdot [254.87 + (1.014^2 - 1) \cdot (1703.0 - 676.0)] + 963.0^2 \cdot 676.0 / ((1926.0 + 16.0) \cdot 16.0)$$

$$= 28.4151 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 16.0 mm, the minimum Shell length adjacent to the tubesheet should be 315.981 mm

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	64.2 <=	356.2 N./mm ²	Pass
Tubesheet Shear Stress	9.5 <=	190.0 N./mm ²	Pass
Maximum Tube Stress	4.3 <=	161.5 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-4.3 <=	-161.5 N./mm ²	Pass
Maximum Force on any one Tube	67.6 <=	2542.4 Kgf	Pass
Axial Membrane Stress in Shell	28.4 <=	370.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	13.334	56.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 3 Load Cases:

--Reqd. Thk. + CA			---- Tubesheet		Stresses		Case	Pass/
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	20.235	...	84	356	15	190	Ps+Pt-Th	D1
D2uc	31.234	...	150	356	25	190	Ps+Pt-Th	D2
D3uc	13.334	...	64	356	10	190	Ps+Pt-Th	D3
Max:	31.2338	... mm		0.421		0.132	(Str. Ratio)	

Load Case Definitions:



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ASME TS Calc: Pressure Test

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[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:

With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:

Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

---- Shell Stresses ----					:	---- Shell Band Stress ----					:	Pass
Case#	Ten	Allwd	Cmp	Allwd	:	Ten	Allwd	Cmp	Allwd	:	Fail	
D1uc	8	370	...	...	:	...	...	...	...	:		Ok
D2uc	19	370	...	...	:	...	...	...	...	:		Ok
D3uc	28	370	...	...	:	...	...	...	...	:		Ok
Max RATIO	0.077		...	...	:	...	...	...	...	:		

Tube, Shell and Channel Stress Summary:

---- Tube Stresses ----					Tube Loads		Shell Stress		Channel Stress		Pass
Case#	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	Fail
D1uc	9	161	-4	-161	141	2542	...	370	...	242	Ok
D2uc	14	161	-14	-161	213	2542	...	370	...	242	Ok
D3uc	4	161	-4	-161	68	2542	...	370	...	242	Ok
Max RATIO	0.084		0.084		0.084		...	...	...	...	

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	31.234	56.000	Ok
Tube thickness :	0.616	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (16.000) adj. to tubesheet: 315.981 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Extends by : +.000 mm
Due to Pressure Shell Compresses by : -0.166 mm
Due to Pressure + Thermal Shell Compresses by : -0.166 mm

Summary of Exchanger Pressure and Thermal Displacements: (mm)



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ASME TS Calc: Pressure Test

Case: 1 6:34p Sep 8, 2026

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0325	...	...	...	...	...
2	-0.1656	...	...	...	...	...
3	-0.1360	...	...	...	...	...
4	...	...	...	...	...	...
5	...	...	...	...	...	...
6	...	...	...	...	...	...
7	...	...	...	...	...	...
8	...	...	...	...	...	...

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 External Pressure Calculations: Step: 15 6:34pm Sep 8,2026

External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

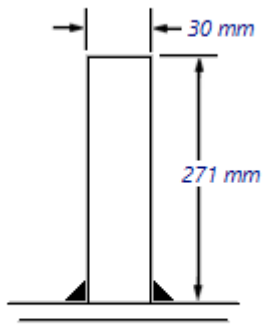
MDMT Calculations for: Ring R1 Fr40, Shell to Ring Junction, Curve: B

Govrn. thk, $t_g = 16.0$, $t_r = 12.758$, $c = 1.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.723$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -29 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Stiffening Ring Calculations for: Ring R1 Fr40, SA-516 70 , Bar Ring: 271.0 x 30.0 mm

Material treatment options:



Effective Length of Shell: 189 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	28.277	7.5000	212.079
Ring :	81.300	150.5000	12235.653
Total:	109.577		12447.732

Centroid of Ring plus Shell: 114 mm

	Inertia	Distance	A*Dist ²
Shell:	5.302	106.0978	318309.125
Ring :	4975.625	-36.9022	110711.914
Total:	4980.926		429021.031

Available Moment of Inertia, Ring + Shell: 0.927E+04 cm⁴

Required Stress in Ring plus Shell Breq 4.57 N./mm²
 Required Strain in Ring plus Shell Areq 0.0000460

Required Moment of Inertia, Ring plus Shell:

= $(D_o^2 * L_s(t + A_s/L_s)A) / 10.9$
 = $(1958.0^2 * 5076.5156(15.0 + 81.3/5076.5156)0.000046) / 10.9$
 = 136 cm⁴



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External Pressure Calculations: Step: 15 6:34pm Sep 8, 2026

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size	Wleg	8.000	mm
Stiffening Ring Attachment Style		CONTINUOUS	
Location of Stiffening Ring		EXTERNAL	
Radial Pressure Load	Pext*Slen	26.77	Kgf/mm
The Radial Shear Load	V	524.09	Kgf
The First Moment of the Area (Ring + Shell)	Q	300014.66	mm ³
Weld Shear Flow due to Rad. Shear Load	VQ/I	1.70	Kgf/mm
The Weld Allowable Stress	0.55*S	75.90	N./mm ²
Minimum Weld Leg Size Min(6mm, t, tw)	Wldmin	6.00	mm
The Weld Allowable Load	2* Wleg*0.55*S	123.83	Kgf/mm
The Combined Weld Load	SRSS of VQ/I and Pext*Slen	26.82	Kgf/mm
MAEP of Stiffening Ring Weld	Weld MAEP	0.24	N./mm ²

MDMT Calculations for: Ring R1 Fr60, Shell to Ring Junction, Curve: B

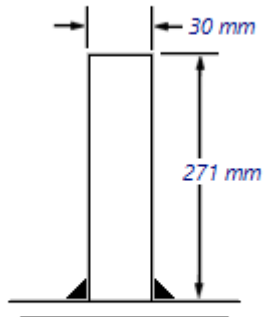
Govern. thk, tg = 16.0, tr = 12.758, c = 1.0 mm, E* = 0.85

Thickness Ratio = $tr * E^* / (tg - c) = 0.723$, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Stiffening Ring Calculations for: Ring R1 Fr60, SA-516 70 , Bar Ring: 271.0 x 30.0 mm

Material treatment options:



Effective Length of Shell: 189 mm

	Area (cm ²)	Distance (mm)	Area*Dist
Shell:	28.277	7.5000	212.079
Ring :	81.300	150.5000	12235.653
Total:	109.577		12447.732

Centroid of Ring plus Shell: 114 mm

	Inertia	Distance	A*Dist ²
Shell:	5.302	106.0978	318309.125
Ring :	4975.625	-36.9022	110711.914
Total:	4980.926		429021.031

Available Moment of Inertia, Ring + Shell: 0.927E+04 cm⁴



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 External Pressure Calculations: Step: 15 6:34pm Sep 8,2026

Required Stress in Ring plus Shell Breq 4.57 N./mm²
 Required Strain in Ring plus Shell Areq 0.0000460

Required Moment of Inertia, Ring plus Shell:

$$= (D_o^2 \cdot L_s(t + A_s/L_s)A) / 10.9$$

$$= (1958.0^2 \cdot 5076.4863(15.0 + 81.3/5076.4863)0.000046) / 10.9$$

$$= 136 \text{ cm}^4$$

Results for Stiffening Ring Weld Calculations per UG-30:

Given Stiffening Ring Fillet Weld Size Wleg 8.000 mm
 Stiffening Ring Attachment Style CONTINUOUS
 Location of Stiffening Ring EXTERNAL
 Radial Pressure Load Pext*Slen 26.77 Kg/mm
 The Radial Shear Load V 524.08 Kg
 The First Moment of the Area (Ring + Shell) Q 300014.66 mm³
 Weld Shear Flow due to Rad. Shear Load VQ/I 1.70 Kg/mm
 The Weld Allowable Stress 0.55*S 75.90 N./mm²
 Minimum Weld Leg Size Min(6mm, t, tw) Wldmin 6.00 mm
 The Weld Allowable Load 2* WLeg*0.55*S 123.83 Kg/mm
 The Combined Weld Load SRSS of VQ/I and Pext*Slen 26.82 Kg/mm
 MAEP of Stiffening Ring Weld Weld MAEP 0.24 N./mm²

External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	...	117	No Calc	No Calc
20	30	No Calc	...	177	No Calc	No Calc
30	Tsht	1499.97	1948	13	0.00096034	44.9269
Tsht	40	No Calc	...	13	No Calc	No Calc
40	Ring	2535.03	1958	15	0.00068464	67.9776
Ring	50	7618	1958	15	0.0002136	21.208
50	60	7618	1954	13	0.00017217	17.0941
60	Ring	7618	1958	15	0.0002136	21.208
Ring	Tsht	2534.97	1958	15	0.00068466	67.9792
Tsht	70	No Calc	...	15	No Calc	No Calc
70	80	2000.03	1960	19	0.0012575	47.4424
80	90	No Calc	...	177	No Calc	No Calc
90	100	No Calc	...	117	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	120	95.885	51.7045	No Calc
20	30	180	174.117	51.7045	No Calc
30	Tsht	14	5.40268	51.7045	399.737
Tsht	40	14	No Calc	51.7045	No Calc
40	Ring	16	6.29255	51.7045	694.317
Ring	50	16	9.39831	51.7045	216.617
50	60	14	9.38814	51.7045	151.628
60	Ring	16	9.39831	51.7045	216.617
Ring	Tsht	16	6.2925	51.7045	694.334
Tsht	70	16	No Calc	51.7045	No Calc
70	80	20	5.98157	51.7045	613.166
80	90	180	174.117	51.7045	No Calc
90	100	120	95.885	51.7045	No Calc
Ring R1 Fr40					239



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External Pressure Calculations: Step: 15 6:34pm Sep 8,2026

Ring R1 Fr60

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Minimum

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External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1499.97	19071.5	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	Ring	2535.03	53294.6	No Calc	No Calc
Ring	50	7618	157800	136.354	9271.13
50	60	7618	21151.5	No Calc	No Calc
60	Ring	7618	157800	No Calc	No Calc
Ring	Tsht	2534.97	53294.6	136.354	9271.13
Tsht	70	No Calc	No Calc	No Calc	No Calc
70	80	2000.03	49662.2	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

Step: 16

6:34pm

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3695.7	...	3695.7	...	...
20	30	1083.68	640486	1051.67	641066	...
30	40	1024.23	4343716	951.565	4352770	...
40	50	3200.3	6119781	3001.82	6145395	...
50	60	2797.38	6119782	2598.89	6145396	...
60	70	3198.78	6116888	3000.4	6142489	...
70	80	1956.97	5791621	1860.07	5803693	...
80	90	1083.68	640486	1051.67	641066	...
90	100	3712.99	...	3712.99	...	...
Total		21753	29772762.00	20924	29871876.00	0



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	...	60	...	...	Liquid: 10
10	Insl	148.368	60	...	...	Insulation
20	Liqd	640.095	106	...	...	Liquid: 20
20	Insl	29.9549	106	...	...	Insulation
30	Liqd	4341.07	750	...	...	Liquid: 30
30	Insl	340.112	750	...	...	Insulation
30	Nozl	184.576	750	1214	...	N1-Tube Inlet
30	Wght	750	750	...	...	Pass partiton
30	Forc	...	1556	...	...	TOP HE FORCES
30	Forc	...	1556	...	...	TOP HE FORCES
30	Forc	...	1500	...	...	TOP HE FORCES
30	Forc	...	1556	...	...	TOP HE FORCES
40	Sadl	1207.52	2535	1255.5	...	LH floating sad
40	Liqd	6116.05	2115	...	...	Liquid: 40
40	Insl	963.799	2115	...	...	Ins: 40
40	Ring	441.246	2535	...	...	Ring R1 Fr40
40	Nozl	255.961	1251.23	1190.81	...	N3-Shell Inlet
50	Liqd	6116.05	2115	...	...	Liquid: 50
50	Insl	961.926	2115	...	...	Ins: 50
50	Wght	75850	2241.74	3159.51	-55.7066	TOP HE EMPTY
50	Wght	46800	2297.81	3159.51	-55.7066	TOP HE OPERATING
50	Wght	46000	2296.64	3159.51	-55.7066	TOP HE TEST
60	Sadl	1207.52	1693	1255.5	...	New Saddle
60	Liqd	6113.16	2114	13	...	Liquid: 40
60	Insl	963.343	2114	...	...	Insulation
60	Lini	613.744	2114	...	...	SS lining
60	Ring	441.246	1693	...	...	Ring R1 Fr60
60	Nozl	255.347	3224.13	-1190.81	...	N4-Shell Outlet
70	Liqd	5777.67	1000	10	...	Liquid: 50
70	Insl	456.14	1000	...	...	Insulation
70	Nozl	540.568	1000	-1341	...	N2-Tube Outlet
80	Liqd	638.944	106	10	...	Liquid: 60
80	Insl	29.9549	106	...	...	Insulation
90	Liqd	...	60	9.99997	...	Liquid: 70
90	Insl	149.033	60	...	...	Insulation



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Element and Detail Weights:

Step: 16

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30 FTsh	664.855	1528	...	...	Fixed Tubesheet
30 Tube	45923	7900	...	...	Tubes
30 RTsh	664.855	14272	...	...	Tubesheet
40 Baf1	111.14	560	...	...	Baffles
40 Baf1	111.14	1120	...	...	Baffles
40 Baf1	111.14	1680	...	...	Baffles
40 Baf1	111.14	2240	...	...	Baffles
40 Baf1	111.14	2800	...	...	Baffles
40 Baf1	111.14	3360	...	...	Baffles
40 Baf1	111.14	3920	...	...	Baffles
40 Baf1	111.14	4480	...	...	Baffles
40 Baf1	111.14	5040	...	...	Baffles
40 Baf1	111.14	5600	...	...	Baffles
40 Baf1	111.14	6160	...	...	Baffles
40 Baf1	111.14	6720	...	...	Baffles
40 Baf1	111.14	7280	...	...	Baffles
40 Baf1	111.14	7840	...	...	Baffles
40 Baf1	111.14	8400	...	...	Baffles
40 Baf1	111.14	8960	...	...	Baffles
40 Baf1	111.14	9520	...	...	Baffles
40 Baf1	111.14	10080	...	...	Baffles
40 Baf1	111.14	10640	...	...	Baffles
40 Baf1	111.14	11200	...	...	Baffles
40 Baf1	111.14	11760	...	...	Baffles
40 Baf1	111.777	12320	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	2415.0
Liquid	29743.0
Insulation	4042.6
Cladding & Overlay	613.7
Stiffeners	882.5
Nozzles	1236.5
Weights	169400.0
Exchanger Components	49698.4
Liquid in Tubes	13050.0

Sum of the Detail Weights 271081.8 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	21753.7	21753.7	21753.7	21753.7	21753.7	21753.7
Saddles	2415.0	2415.0	2415.0	2415.0	2415.0	2415.0
Cladding	613.7	613.7	613.7	613.7	613.7	613.7
Stif. Rings	882.5	882.5	882.5	882.5	882.5	882.5
Nozzles	1236.5	1236.5	1236.5	1236.5	1236.5	1236.5
Wld Weights	76600.0	76600.0	76600.0	76600.0	76600.0	76600.0
Exchanger	49698.4	49698.4	49698.4	49698.4	49698.4	49698.4
Insulation	...	...	...	...	4042.6	4042.6 *
Test Weights	...	46000.0	...	...	...	...
Ope Weights	...	...	...	...	...	46800.0
Ope. Liquid	...	...	...	...	...	29743.0
Tube Ope Lqd	...	...	...	...	...	13050.0
Test Liquid	...	29678.0	...	...	...	...
Tube Tst Lqd	...	13050.1	...	...	...	...
Totals	153199.9	241927.9	153199.9	153199.9	157242.5	246835.6

Field Installation Options:

* Insulation installed after lifting.



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Element and Detail Weights:

Step: 16

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Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	153199.9 kg.
Shop Test Wt.	- Fabricated weight + water (full)	241927.9 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	153199.9 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	153199.9 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	157242.5 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	246835.6 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	246006.6 kg.
Field Test Wt.	- Empty weight + test medium (full)	245240.7 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	13058043.0 cm ³
Weight of Ope Liq in tubes :	13050.0 kg.
Weight of Water in tubes :	13050.1 kg.

Field Test weight is computed in the corroded condition.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	44573.5	...
20	30	21620.8	13270.1
30	40	91797.4	90477.9
40	50	260197	255945
50	60	259666	255945
60	70	260074	255824
70	80	123150	120637
80	90	21620.8	13270.1
90	100	44761.9	...

Total	1127462.6	1005369.1 cm ²
Total	1213.6	1082.2 ft ²

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Nozzle Flange MAWP:

Step: 17

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Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating Ope.	Flange Rating Ambient	Des. Temp	Class	Grade/Group	Equiv. Press	UG-44	Max Pressure 50%	DNV
N1-Tube Inlet	1420.0	1910.0	150	150	GR 2.1	1657.6	1420.0	1420.0	472.4
N3-Shell Inlet	1540.0	1960.0	160	150	GR 1.1	1657.6	1540.0	1540.0	652.4
N4-Shell Outle	1540.0	1960.0	160	150	GR 1.1	1657.6	1540.0	1540.0	652.4
N2-Tube Outlet	1420.0	1910.0	150	150	GR 2.1	1269.2	749.2	1420.0	860.8

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 1540.046 KPa.

Lowest Flange Pressure Rating was (Amb)[ShellSide]: 1960.006 KPa.

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 749.209 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 1910.006 KPa.

Selected Method for Derating ASME B16.5 Flange MAWP: ASME UG-44(b)

The UG-44(b) Method is based on the paper PVP 2013-97814. PV Elite uses the maximum loads from each load category to determine ME and FE. In many cases, the computed maximum allowable pressure will be greater than the flange rating. In these cases, the minimum of the rating from the table and the UG-44(b) method will be used. SA-193 B8 Cl. 2 bolts or ones with higher allowable stresses at the specified bolt size shall be used.

The axial force FE must put the nozzle in tension (-P for 107/297; +Fr for PD 5500 AnnexG) to be considered.

Nozzle Description	Gasket Dimension G mm	Fm Value	Resolved Moment ME Kg-m.	Axial Force FE Kgf	MAWP PD KPa.	Maximum Pressure Pmax KPa.	Pass Fail
N1-Tube Inlet	551.688	1.20	5046	3824	573.1	1420.0	Pass
N3-Shell Inlet	551.688	1.20	5046	3824	1540.0	1540.0	Pass
N4-Shell Outlet	551.688	1.20	5046	3824	1540.0	1540.0	Pass
N2-Tube Outlet	787.400	0.42	11353	5354	573.1	749.2	Pass

Evaluating Flange on Nozzle [N1-Tube Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(5046) + 4 \cdot \text{abs}(3824) \cdot 551.688 \leq \pi \cdot 551.688^3 [(1420.034 - 573.131) + 1.200 \cdot 1420.034]$$

$$= 0.89171E+05 \leq 0.13723E+06 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N1-Tube Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1420.034, 1.200 \cdot 1420.034 + 1420.034 - 89170.9 / (\pi \cdot 551.688^3))$$

$$= \min(1420.034, 1466.446)$$

$$= 1420.034 \text{ KPa.}$$

Evaluating Flange on Nozzle [N3-Shell Inlet]:

$$= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [((PR - PD) + FM \cdot PR)]$$

$$= 16(5046) + 4 \cdot \text{abs}(3824) \cdot 551.688 \leq \pi \cdot 551.688^3 [(1540.046 - 1540.046) + 1.200 \cdot 1540.046]$$

$$= 0.89171E+05 \leq 0.99415E+05 \text{ Check Passes}$$

Maximum Pressure for Flange on Nozzle N3-Shell Inlet [Pmax]:

$$= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3))$$

$$= \min(1540.046, 1.200 \cdot 1540.046 + 1540.046 - 89170.9 / (\pi \cdot 551.688^3))$$

$$= \min(1540.046, 1730.473)$$

$$= 1540.046 \text{ KPa.}$$

Evaluating Flange on Nozzle [N4-Shell Outlet]:



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Nozzle Flange MAWP:

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$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(5046) + 4 \cdot \text{abs}(3824) 551.688 \leq \pi \cdot 551.688^3 [(1540.046 - 1540.046) + 1.200 \cdot 1540.046] \\
 &= 0.89171E+05 \leq 0.99415E+05 \quad \text{Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N4-Shell Outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(1540.046, 1.200 \cdot 1540.046 + 1540.046 - 89170.9 / (\pi \cdot 551.688^3)) \\
 &= \min(1540.046, 1730.473) \\
 &= 1540.046 \text{ KPa.}
 \end{aligned}$$

Evaluating Flange on Nozzle [N2-Tube Outlet]:

$$\begin{aligned}
 &= 16 \cdot ME + 4 \cdot FE \cdot G \leq \pi \cdot G^3 [(PR - PD) + FM \cdot PR] \\
 &= 16(11353) + 4 \cdot \text{abs}(5354) 787.400 \leq \pi \cdot 787.400^3 [(1420.034 - 573.131) + 0.421 \cdot 1420.034] \\
 &= 0.19851E+06 \leq 0.22605E+06 \quad \text{Check Passes}
 \end{aligned}$$

Maximum Pressure for Flange on Nozzle N2-Tube Outlet [Pmax]:

$$\begin{aligned}
 &= \min(PR, FM \cdot PR + PR - (16 \cdot ME + 4 \cdot FE \cdot G) / (\pi \cdot G^3)) \\
 &= \min(1420.034, 0.421 \cdot 1420.034 + 1420.034 - 198509.2 / (\pi \cdot 787.400^3)) \\
 &= \min(1420.034, 749.209) \\
 &= 749.209 \text{ KPa.}
 \end{aligned}$$

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Large Diameter Flange Pressure Ratings are per ASME B16.47 2025 Edition

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Wind Load Calculation:

Step: 18

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	13100 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:

$$= \min(0.85, 0.925((1 + 1.7 \cdot gQ \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot gV \cdot I_{zbar})))$$

$$= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.208 \cdot 0.932) / (1 + 1.7 \cdot 3.4 \cdot 0.208)))$$

$$= \min(0.85, 0.89)$$

$$= 0.850$$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient [Cf]	0.605
Structure Height to Diameter ratio	7.919

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of [α] and [Zg]:

Exposure Category: C from Table 26.11-1
 $\alpha = 9.5$: $Z_g = 274320.0$ mm

Effective Height [z]:

$$= \text{Centroid Height} + \text{Vessel Base Elevation}$$

$$= 1580.0 + 13100.0 = 14680.0 \text{ mm}$$

$$= 48.163 \text{ ft. Imperial Units}$$

Velocity Pressure coefficient evaluated at height z [Kz]:

$$\text{Because } z (48.163 \text{ ft.}) > 15 \text{ ft.}$$

$$= 2.01 (z / Z_g)^{2/\alpha}$$

$$= 2.01 (48.163 / 900.0)^{2/9.5}$$

$$= 1.085$$

Type of Hill: No Hill



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Wind Load Calculation:

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Wind Directionality Factor [Kd]:

= 1, per Table 26.6-1

As there is No Hill Present: [Kzt]:

K1 = 0, K2 = 0, K3 = 0

Topographical Factor [Kzt]:

= (1 + K1 • K2 • K3)²

= (1 + 0.0 • 0.0 • 0.0)²

= 1.0

Velocity Pressure evaluated at height z, Imperial Units [qz]:

= max(16, 0.00256 • Kz • Kzt • Kd • Ke • V(mph)²)

= max(16, 0.00256 • 1.085 • 1.0 • 1.0 • 1.0 • 60.399²)

= 16.0 psf [78.12] Kgs/m²

Force on the first element [F]:

= qz • G • Cf • WindArea

= 16.0 • 0.85 • 0.605 • 3.627

= 29.8 lbs. [13.5] Kgf

Element	Hgt (z) mm	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
LHS Blind Flang	14680.0	0.000	0.000	0.000	1.085	1.000	78.120
LHS ch Flange	14680.0	0.000	0.000	0.000	1.085	1.000	78.120
LHS ch Shell	14680.0	0.000	0.000	0.000	1.085	1.000	78.120
Main Shell	14680.0	0.000	0.000	0.000	1.085	1.000	78.120
Main Shell	14680.0	0.000	0.000	0.000	1.085	1.000	78.120
Main Shell	14680.0	0.000	0.000	0.000	1.085	1.000	78.120
RHS ch Shell	14680.0	0.000	0.000	0.000	1.085	1.000	78.120
RHS Ch Flg	14680.0	0.000	0.000	0.000	1.085	1.000	78.120
RHS Blind Flang	14680.0	0.000	0.000	0.000	1.085	1.000	78.120

Wind Loads on Masses/Equipment/Piping

ID	Wind Area cm ²	Elevation mm	Pressure Kgs/m ²	Force Kgf
Pass partiton	0.00	14680.00	78.12	0.00
TOP HE EMPTY	480042.00	17839.51	78.12	3750.10
TOP HE OPERAT	-1.01	17839.51	78.12	-0.01
TOP HE TEST	-1.01	17839.51	78.12	-0.01

Wind Load Calculation:

From	To	Wind Height mm	Wind Diameter mm	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	14680	2808	3369.6	78.12	14.8931
20	30	14680	2544	5393.28	78.12	23.8374
30	40	14680	2577.6	38664	78.12	170.889
40	50	14680	2589.6	109540	78.12	484.149
50	60	14680	2584.8	109337	78.12	4608.34
60	70	14680	2589.6	109488	78.12	483.92
70	80	14680	2592	51840	78.12	229.124
80	90	14680	2544	5393.28	78.12	23.8374
90	100	14680	2814	3376.8	78.12	14.9249

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.



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Wind Load Calculation:

Step: 18

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Be sure the wind speed is in accordance with the specified wind design code.

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 Earthquake Load Calculation: Step: 19 6:34pm Sep 8,2026

Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-16
Seismic Load Reduction Scale Factor		1.000
Importance Factor		1.250
Table Value Fa		1.600
Table Value Fv		2.400
Max. Mapped Res. Acceleration	[Ss]	0.147
Max. Eff. Ground Acceleration	[S]	0.065
Force Modification Factor R		3.000
Site Class		D
Component Elevation Ratio	z/h	0.000
Amplification Factor	Ap	0.000
Force Factor		0.000
Consider Vertical Acceleration		Yes
Minimum Acceleration Multiplier		0.000
User Value of Sds (used if > 0)		0.000
User Value of Sd1 (used if > 0)		0.000
Moment Reduction Factor Tau		1.000

Seismic Analysis Results:

$$\begin{aligned} SMS &= Fa * Ss = 1.6 * 0.147 = 0.235 \\ SM1 &= Fv * S1 = 2.4 * 0.065 = 0.156 \\ SDS &= 2/3 * Sms = 2/3 * 0.235 = 0.157 \\ SD1 &= 2/3 * Sm1 = 2/3 * 0.156 = 0.104 \end{aligned}$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_n^x$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(8.70240.75)$
 = 0.101 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = (1/Natural Frequency) = (1/33.0)
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of ($1.7 \cdot 0.101$, 0.03) per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.157 \cdot 244421 \cdot 1.25$
 = 14371.930 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.157 \cdot 244421$
 = 7665.0 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	1070	22220	1424.22



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Earthquake Load Calculation:

Step: 19

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20	30	960	22220	1277.8
30	40	960	22220	1277.8
40	50	963	22220	1281.79
40	50	963	22220	1281.79
50	60	963	22220	1281.79
60	0	963	22220	1281.79
60	70	963	22220	1281.79
70	80	960	22220	1277.8
80	90	960	22220	1277.8
90	100	1072.5	22220	1427.54

The Earthquake Weight does not include the weight of the saddles.

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Center of Gravity Calculation:

Step: 20

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Shop/Field Installation Options :

Insulation is installed in the Field after being lifted.

Lining is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	8235.176 mm
Center of Gravity of the Liquid	8639.587 mm
Center of Gravity of the Insulation	8492.103 mm
Center of Gravity of the Lining	12465.176 mm
Center of Gravity of the Stiffening Rings	8235.176 mm
Center of Gravity of the Nozzles	10451.614 mm
Center of Gravity of the Added Weights (Operating)	8339.947 mm
Center of Gravity of the Added Weights (Empty)	8291.658 mm
Center of Gravity of the Tubesheet(s)	8235.176 mm
Center of Gravity of the Tubes	8235.175 mm
Center of Gravity of the Baffles	8332.707 mm
Center of Gravity of Bare Shell New and Cold	8681.661 mm
Center of Gravity of Bare Shell Corroded	8689.145 mm
Vessel CG in the Operating Condition	8402.123 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	8363.208 mm
Vessel CG in the Test Condition	8397.058 mm

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Saddle Calcs: Road Transport:

Step: 21

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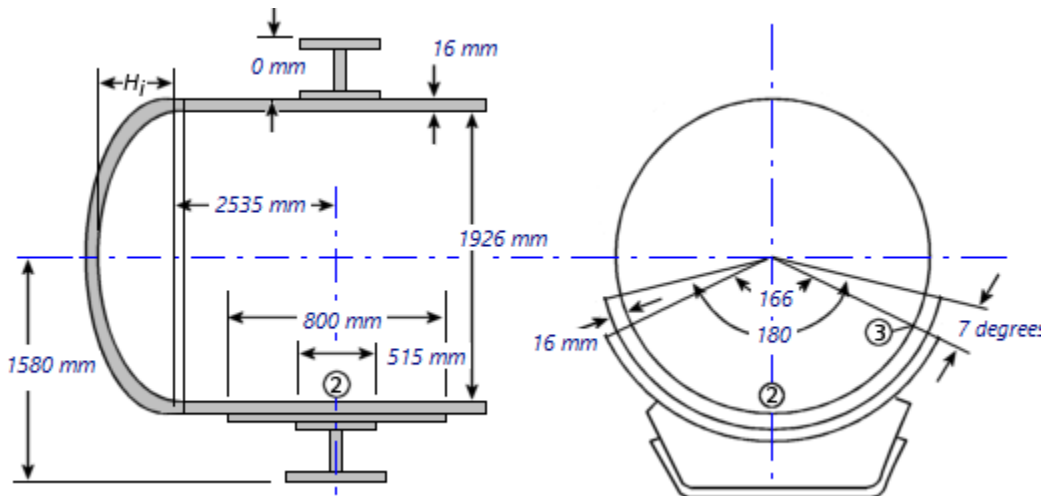
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Road Transport

Input and Calculated Values:

Vessel Mean Radius	Rm	971.00	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	12688.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2535.00	mm
Saddle Width	b1 or b	515.00	mm
Saddle Bearing Angle	delta or theta	166.00	degrees
Wear Plate Width	b2 or b1	800.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		88.10	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1580.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		83670.32	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr



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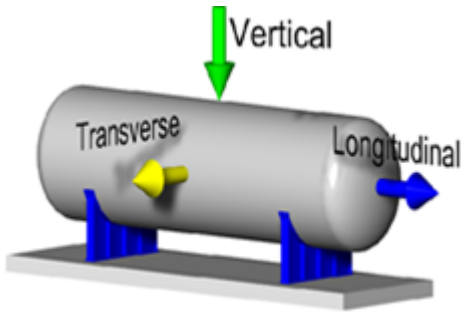
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Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-13.34	-207.36
Long. Stress at Top of Midspan	13.34	138.00
Long. Stress at Bottom of Midspan	13.34	138.00
Long. Stress at Top of Saddles	8.13	138.00
Long. Stress at Bottom of Saddles	-8.13	-234.00
Long. Stress at Bottom of Saddles	8.13	138.00
Tangential Shear in Shell	10.09	110.40
Circ. Compressive Stress in Shell	4.50	138.00
Stiffener Circ. Stress at Shell	0.47	138.00
Stiffener Circ. Stress at Tip	52.27	234.00

Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [F_{tw}]:

$$= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf$$

$$= 469.74 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0$$

$$= 3990.1 \text{ lbf}$$

$$= 1809.905 \text{ Kg}$$

Longitudinal Force due to Wind during Transport [F_{lw}]:

$$= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult}$$

$$= \pi \cdot 7.08^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2$$

$$= 401.3 \text{ lbf}$$

$$= 182.0 \text{ Kg}$$

Transverse Force due to Lateral Acceleration [F_t]:

$$= g_z \cdot \text{Shipping Weight} \cdot g_w + F_{tw}$$

$$= 0.1 \cdot 150784.81 \cdot 1.0 + 1809.9$$

$$= 16888.4 \text{ Kg}$$

Longitudinal Force due to Axial Acceleration [F_l]:

$$= g_x \cdot \text{Shipping Weight} \cdot g_w + F_{lw}$$

$$= 0.1 \cdot 150784.81 \cdot 1.0 + 182.03$$

$$= 15260.5 \text{ Kg}$$

Saddle Reaction Force due to Transverse Load [F_{wt}]:

$$= F_t / \text{Num of Saddles} \cdot B / E$$

$$= 16888.4/2 \cdot 1580.0/1850.0$$

$$= 7211.8 \text{ Kg}$$

Saddle Reaction Force due to Longitudinal Load [F_{wl}]:

$$= F_l \cdot B / L_s$$

$$= 15261 \cdot 1580.0/7618.0015$$

$$= 3165.1 \text{ Kg}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [F_{rv}]:

$$= \text{Saddle Load} \cdot g_y$$



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$$= 72817.6 \cdot 0.05$$

$$= 3640.882 \text{ Kgf}$$

Load Combination Results due to Transport Loading [Q]:

$$= \text{Saddle Load} + \text{Frv} + \max(\text{Fwl}, \text{Fwt})$$

$$= 72818 + 3641 + \max(3165, 7212)$$

$$= 83670.3 \text{ Kgf}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	84877.84	Kgf
Transverse Shear Load	8444.19	Kgf
Longitudinal Shear Load	15260.51	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1850.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1951	K2 = 0.6681	K3 = 0.3500	K4 = 0.2504
K5 = 0.6435	K6 = 0.0233	K7 = 0.0233	K8 = 0.2756
K9 = 0.1812	K10 = 0.0235	K1* = 0.3314	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -83670 \cdot 2535.0 [1 - (1 - 2535.0/12688.0 + (971.0^2 - 0.0^2)/(2 \cdot 2535.0 \cdot 12688.0))/(1 + (4 \cdot 0.0)/(3 \cdot 12688.0))]$$

$$= -39269.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 83670 \cdot 12688/4 ((1 + 2(971^2 - 0^2)/(12688^2))/(1 + (4 \cdot 0)/(3 \cdot 12688)) - 4 \cdot 2535/12688)$$

$$= 56407.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 971.0/(2 \cdot 14.0) - 56407.9/(\pi \cdot 971.0^2 \cdot 14.0)$$

$$= -13.34 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 971.0/(2 \cdot 14.0) + 56407.9/(\pi \cdot 971.0^2 \cdot 14.0)$$

$$= 13.34 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m/(2t) - M1/(\pi \cdot R_m^2 \cdot t)$$

$$= 0.0 \cdot 971.0/(2 \cdot 16.0) - 39269.4/(\pi \cdot 971.0^2 \cdot 16.0)$$

$$= 8.13 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m/(2t) + M1/(\pi \cdot R_m^2 t)$$

$$= 0.0 \cdot 971.0/(2 \cdot 16.0) + 39269.4/(\pi \cdot 971.0^2 \cdot 16.0)$$

$$= -8.13 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 83670(12688.0 - 2 \cdot 2535.0)/(12688.0 + (4 \cdot 0.0/3))$$

$$= 50236.5 \text{ Kgf}$$



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Shear Stress in the shell with a single ring (4.15.13) $[\tau_1]$:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 50236.48 / (\pi \cdot 971.0 \cdot 16.0)$$

$$= 10.09 \text{ N./mm}^2$$

Decay Length (4.15.20) $[x_1, x_2]$:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(971.0 \cdot 16.0)}$$

$$= 97.222 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) $[\sigma_6^*]$:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.643 \cdot 83670 \cdot 0.1/117.21$$

$$= -4.50 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) $[\sigma_8^*]$:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.2756 \cdot 83670/117.21 + 0.0233 \cdot 83670 \cdot 971.0 \cdot 107.53/10112.53$$

$$= 0.47 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) $[\sigma_9^*]$:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.2756 \cdot 83670/117.21 - 0.0233 \cdot 83670 \cdot 971.0 \cdot 179.47/10112.53$$

$$= -52.27 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	58.0000	mm
Baseplate Width	Bpwid	540.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	620.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell $[ShellInertia]$:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + WearPlateWidth) \cdot t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(963.0 \cdot 16.0)} + 800.0) \cdot 16.0^3) / 12$$

$$= 33.916 \text{ cm}^4$$

Resolved Inertia for the Shell $[I, shell]$:

$$= ShellInertia + A_{Shell} (C_1 - Y)^2$$

$$= 33.916 + 158.983 (311.791 - 8.0)^2$$

$$= 146757.578 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	993.6	16.0	8.0	159.0	127186.1	0.147E+06
Wearplate	800.0	16.0	24.0	128.0	307200.0	0.106E+06
Web	16.0	527.0	295.5	84.3	2491656.5	0.197E+05
BasePlate	540.0	58.0	588.0	313.2	18416160.0	0.240E+06
Totals	...	...	...	684.5	21342202.0	0.512E+06

Distance to Centroid $[C_1]$:

$$= AY / A$$

$$= 21342202/684.503$$

$$= 311.791 \text{ mm}$$



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Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 166.0/2$$

$$= 97.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(97.0) - 0.5 \cdot \sin(97.0)^2) / (\pi - 1.693 + \sin(97.0) \cos(97.0))$$

$$= 0.2904$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 83670.32$$

$$= 24298.0879 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) =$$

$$4.5343 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} =$$

$$155.9972 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1580.0 - 963.0 \cdot \sin(166.0/2) / 1.4486$$

$$= 920.186 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d =$$

$$22359.2031 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) =$$

$$13.3434 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} =$$

$$173.3302 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(83670 + 1208) \cdot 540.0) / (4 \cdot 1850.0 \cdot 173.33)}$$

$$= 32.424 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 166.0/2) (963.0 + 16.0 + 16.0)$$

$$= 1975.167 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1975.1669 / (4 - 1)$$

$$= 658.389 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 658.3889 \cdot 540.0/2$$

$$= 1777.650 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 84877.844 / (1850.0 \cdot 540.0)$$

$$= 8.496 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1777.7 \cdot 8.5$$

$$= 15103.414 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 79.84 + 52.671$$

$$= 132.511 \text{ cm}^2$$



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Compressive Stress [Sc]:

= P/A_r

= 15103.4/132.5111

= 11.178 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	515.0	...	82.4	...	0.182E+05

Rib dimension [D]:

= Saddle Width - Web Thickness

= 515.0 - 16.0

= 499.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/132.511

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 515.0/2

= 257.500 mm

Radius of Gyration [r]:

= $\sqrt{(Total\ Inertia / Total\ Area)}$

= $\sqrt{(18212.1/132.511)}$

= 117.234 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - $\cos(\text{saddle bearing angle} / 2) \times (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$

= 1580.0 - $\cos(166.0/2) \times (963.0 + 16.0 + 16.0) - 58.0$

= 1400.740 mm

Length of Outer Rib, step 2, [Lw]:

= $2 \times \sin(\text{saddle bearing angle} / 2) \times (\text{radius} + \text{shlthk} + \text{wpdthk})$

= $2 \times \sin(166.0/2) \times (963.0 + 16.0 + 16.0)$

= 1975.167 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 1850.0 - 50.8

= 1799.200 mm

Length of Outer Rib, step 4, [Dd]:

= $(Lw - Bd) / 2$

= $(1975.167 - 1799.2) / 2$

= 87.983 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{1400.74^2 + 87.983^2}$

= 1403.500 mm

Intermediate Term [Cc]:

= $\sqrt{(2 \times \pi^2 \times \text{Elastic Modulus} / \text{Yield Stress})}$

= $\sqrt{(2 \times \pi^2 \times 0.19994E+09/260.0)}$

= 123.211

Slenderness ratio [KL/r]:



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$$= KL/r$$

$$= 1 \cdot 1403.5/117.234$$

$$= 11.972$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 15260.5 / (2 \cdot 1850.0) \cdot 658.389 \cdot 1403.5 / 2$$

$$= 1905.641 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.9718 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.97)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (11.97) / (8 \cdot 123.21) - (11.97^3) / (8 \cdot 123.21^3))$$

$$= 151.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 11.18/151.95 + (1905.64 \cdot 257.5 / 1.8212E+09) / 173.33$$

$$= 0.226$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	499.0	0.0	79.8	0.0	0.182E+05
Web	658.4	16.0	0.0	105.3	0.0	22.5
Totals	...	...	...	185.2	...	0.182E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/185.182$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 515.0/2$$

$$= 257.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{ (Ro + Wpdthk)^2 - (Pitch/2)^2 } - Bpthk$$

$$= 1580.0 - \sqrt{ (995.0 + 16.0)^2 - (658.389/2)^2 } - 58.0$$

$$= 583.035 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 18234.0/185.182 }$$

$$= 99.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 583.035/99.23$$

$$= 5.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 15260.513 / (2 \cdot 1850.0)$$

$$= 4.124 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 4.124 \cdot 658.389 \cdot 583.035$$

$$= 1583.262 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:



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= Bending Moment / Section Modulus

= 1583.262/708118

= 21.926 N./mm²

Compressive Allowable, $KL/r < C_c$ (5.8756 < 123.2105) per AISC E2-1 [Sca]:

= (1-(Kl/r)²/(2• C_c ²)) F_y /(5/3+3•(Kl/r)/(8• C_c)-(Kl/r)³)/(8• C_c ³)

= (1-(5.88)²/(2 • 123.21²))260/

(5/3+3•(5.88)/(8• 123.21)-(5.88³)/(8•123.21³)

= 154.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c/S_{ca} + (M_{base,c} • C_1/I)/S_{ba}$

= 16.0/154.17 + (1583.26 • 257.5/18234.025)/173.33

= 0.230

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	971.00	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	12688.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2535.00	mm
Saddle Width	b1 or b	515.00	mm
Saddle Bearing Angle	delta or theta	166.00	degrees
Wear Plate Width	b2 or b1	800.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		88.10	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1580.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q for this Transportation Case		89077.34	Kgf
Shipping Longitudinal Acceleration	gx	0.10	g
Shipping Transverse Acceleration	gz	0.10	g
Shipping Vertical Acceleration	gy	0.05	g
Vertical Acceleration Acting with Transverse	gw	1.00	g
Internal Pressure during Transport		0.00	KPa.
Transportation Wind Load Scalar Factor	wlsf	1.00	
Transportation Wind Velocity	wv	92.7	Km/hr

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		-14.20	-207.36
Long. Stress at Top of Midspan		14.20	138.00
Long. Stress at Bottom of Midspan		14.20	138.00
Long. Stress at Top of Saddles		8.65	138.00
Long. Stress at Bottom of Saddles		-8.65	-234.00
Long. Stress at Bottom of Saddles		8.65	138.00
Tangential Shear in Shell		10.75	110.40
Circ. Compressive Stress in Shell		4.80	138.00
Stiffener Circ. Stress at Shell		0.50	138.00
Stiffener Circ. Stress at Tip		55.64	234.00



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Intermediate Results: Saddle Reaction Q due to Road Transport Load:

Transverse Force due to Wind during Transport [Ftw]:

$$\begin{aligned}
 &= \text{Transverse Area} \cdot 0.00256 \cdot wv^2 \cdot wlsf \\
 &= 469.74 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \\
 &= 3990.1 \text{ lbf} \\
 &= 1809.905 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Wind during Transport [Flw]:

$$\begin{aligned}
 &= \pi \cdot D^2/4 \cdot 0.00256 \cdot wv^2 \cdot wlsf \cdot \text{Wind Dia. Mult} \\
 &= \pi \cdot 7.08^2/4 \cdot 0.00256 \cdot 57.6^2 \cdot 1.0 \cdot 1.2 \\
 &= 401.3 \text{ lbf} \\
 &= 182.0 \text{ Kgf}
 \end{aligned}$$

Transverse Force due to Lateral Acceleration [Ft]:

$$\begin{aligned}
 &= gz \cdot \text{Shipping Weight} \cdot gw + Ftw \\
 &= 0.1 \cdot 150784.81 \cdot 1.0 + 1809.9 \\
 &= 16888.4 \text{ Kgf}
 \end{aligned}$$

Longitudinal Force due to Axial Acceleration [Fl]:

$$\begin{aligned}
 &= gx \cdot \text{Shipping Weight} \cdot gw + Flw \\
 &= 0.1 \cdot 150784.81 \cdot 1.0 + 182.03 \\
 &= 15260.5 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Transverse Load [Fwt]:

$$\begin{aligned}
 &= Ft/\text{Num of Saddles} \cdot B / E \\
 &= 16888.4/2 \cdot 1580.0/1850.0 \\
 &= 7211.8 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Longitudinal Load [Fwl]:

$$\begin{aligned}
 &= Fl \cdot B / Ls \\
 &= 15261 \cdot 1580.0/7618.0015 \\
 &= 3165.1 \text{ Kgf}
 \end{aligned}$$

Saddle Reaction Force due to Vertical Transportation Acceleration [Frv]:

$$\begin{aligned}
 &= \text{Saddle Load} \cdot gy \\
 &= 77967.2 \cdot 0.05 \\
 &= 3898.359 \text{ Kgf}
 \end{aligned}$$

Load Combination Results due to Transport Loading [Q]:

$$\begin{aligned}
 &= \text{Saddle Load} + Frv + \max(Fwl, Fwt) \\
 &= 77967 + 3898 + \max(3165, 7212) \\
 &= 89077.3 \text{ Kgf}
 \end{aligned}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	90284.85	Kgf
Transverse Shear Load	8444.19	Kgf
Longitudinal Shear Load	15260.51	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$\begin{aligned}
 &= \text{Baseplate Length} \\
 &= 1850.000 \text{ mm}
 \end{aligned}$$

The Computed K values from Table 4.15.1:

K1 = 0.1951	K2 = 0.6681	K3 = 0.3500	K4 = 0.2504
K5 = 0.6435	K6 = 0.0233	K7 = 0.0233	K8 = 0.2756
K9 = 0.1812	K10 = 0.0235	K1* = 0.3314	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.



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Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(1 - \frac{a}{L} + \frac{(Rm^2 - hm^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4hm)}{3L} \right) \right]$$

$$= -89077 \cdot 2535.0 \left[1 - \left(1 - \frac{2535.0}{12688.0} + \frac{(971.0^2 - 0.0^2)}{(2 \cdot 2535.0 \cdot 12688.0)} \right) / \left(1 + \frac{(4 \cdot 0.0)}{(3 \cdot 12688.0)} \right) \right]$$

$$= -41807.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(Rm^2 - hm^2)/(L^2))}{(1 + (4hm^2)/(3L))} - 4a/L \right)$$

$$= 89077 \cdot 12688 / 4 \left(\frac{(1 + 2(971^2 - 0^2)/(12688^2))}{(1 + (4 \cdot 0)/(3 \cdot 12688))} - 4 \cdot 2535 / 12688 \right)$$

$$= 60053.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 t)$$

$$= 0.0 \cdot 971.0 / (2 \cdot 14.0) - 60053.1 / (\pi \cdot 971.0^2 \cdot 14.0)$$

$$= -14.20 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 971.0 / (2 \cdot 14.0) + 60053.1 / (\pi \cdot 971.0^2 \cdot 14.0)$$

$$= 14.20 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot Rm / (2t) - M1 / (\pi \cdot Rm^2 \cdot t)$$

$$= 0.0 \cdot 971.0 / (2 \cdot 16.0) - 41807.1 / (\pi \cdot 971.0^2 \cdot 16.0)$$

$$= 8.65 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot Rm / (2t) + M1 / (\pi \cdot Rm^2 t)$$

$$= 0.0 \cdot 971.0 / (2 \cdot 16.0) + 41807.1 / (\pi \cdot 971.0^2 \cdot 16.0)$$

$$= -8.65 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 89077(12688.0 - 2 \cdot 2535.0) / (12688.0 + (4 \cdot 0.0 / 3))$$

$$= 53482.9 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot Rm \cdot t)$$

$$= 53482.91 / (\pi \cdot 971.0 \cdot 16.0)$$

$$= 10.75 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(971.0 \cdot 16.0)}$$

$$= 97.222 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K5 \cdot Q \cdot k/A$$

$$= -0.643 \cdot 89077 \cdot 0.1 / 117.21$$

$$= -4.80 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K8 \cdot Q/A + K6 \cdot Q \cdot Rm \cdot c1/l$$

$$= -0.2756 \cdot 89077 / 117.21 + 0.0233 \cdot 89077 \cdot 971.0 \cdot 107.53 / 10112.53$$

$$= 0.50 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K8 \cdot Q/A - K6 \cdot Q \cdot Rm \cdot c2/l$$

$$= -0.2756 \cdot 89077 / 117.21 - 0.0233 \cdot 89077 \cdot 971.0 \cdot 179.47 / 10112.53$$

$$= -55.64 \text{ N./mm}^2$$



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Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	58.0000	mm
Baseplate Width	Bpwid	540.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	620.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(963.0 \cdot 16.0)} + 800.0)^3) / 12$$

$$= 33.916 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}}(C1 - Y)^2$$

$$= 33.916 + 158.983(311.791 - 8.0)^2$$

$$= 146757.578 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	993.6	16.0	8.0	159.0	127186.1	0.147E+06
Wearplate	800.0	16.0	24.0	128.0	307200.0	0.106E+06
Web	16.0	527.0	295.5	84.3	2491656.5	0.197E+05
BasePlate	540.0	58.0	588.0	313.2	18416160.0	0.240E+06
Totals	...	...	...	684.5	21342202.0	0.512E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 21342202 / 684.503$$

$$= 311.791 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 166.0 / 2$$

$$= 97.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(97.0) - 0.5 \cdot \sin(97.0)^2) / (\pi - 1.693 + \sin(97.0) \cos(97.0))$$

$$= 0.2904$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 89077.336$$

$$= 25868.2988 \text{ Kg}$$

Tension Stress, St = (Fh/As) = 4.8273 N./mm²
 Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\text{theta} / 2) / (\text{theta} / 2 \text{ in radians})$$

$$= 1580.0 - 963.0 \cdot \sin(166.0 / 2) / 1.4486$$

$$= 920.186 \text{ mm}$$

Bending Moment, M = Fh * d = 23804.1191 Kg-m.



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Bending Stress, $S_b = (M \cdot C_1 / I) = 14.2056 \text{ N./mm}^2$
 Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$
 $= \sqrt{(3(89077 + 1208) \cdot 540.0) / (4 \cdot 1850.0 \cdot 173.33)}$
 $= 33.441 \text{ mm}$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 \cdot \cos(90 - 166.0/2) (963.0 + 16.0 + 16.0)$
 $= 1975.167 \text{ mm}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nribs} - 1)$
 $= 1975.1669 / (4 - 1)$
 $= 658.389 \text{ mm}$

Baseplate Pressure Area [Ap]:
 $= e \cdot \text{Bpwid} / 2$
 $= 658.3889 \cdot 540.0/2$
 $= 1777.650 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$
 $= 90284.852 / (1850.0 \cdot 540.0)$
 $= 9.038 \text{ Kg/cm}^2$

Axial Load [P]:
 $= A_p \cdot B_p$
 $= 1777.7 \cdot 9.04$
 $= 16065.553 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 79.84 + 52.671$
 $= 132.511 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / A_r$
 $= 16065.6 / 132.5111$
 $= 11.890 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	515.0	...	82.4	...	0.182E+05

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 515.0 - 16.0$
 $= 499.000 \text{ mm}$

Distance to Centroid from Datum [ytot]:
 $= AY / A$
 $= 0.0 / 132.511$
 $= 0.000 \text{ mm}$

Distance to Centroid [C1]:
 $= \text{Saddle Width} / 2$



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$$= 515.0/2$$

$$= 257.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{18212.1/132.511}$$

$$= 117.234 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1580.0 - \cos(166.0/2) (963.0 + 16.0 + 16.0) - 58.0$$

$$= 1400.740 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(166.0/2) (963.0 + 16.0 + 16.0)$$

$$= 1975.167 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1850.0 - 50.8$$

$$= 1799.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (1975.167 - 1799.2) / 2$$

$$= 87.983 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{\text{Lv}^2 + \text{Dd}^2}$$

$$= \sqrt{1400.74^2 + 87.983^2}$$

$$= 1403.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= \text{KL} / r$$

$$= 1 \cdot 1403.5 / 117.234$$

$$= 11.972$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 15260.5 / (2 \cdot 1850.0) \cdot 658.389 \cdot 1403.5 / 2$$

$$= 1905.641 \text{ Kg-m.}$$

Compressive Allowable, $\text{KL}/r < \text{Cc}$ ($11.9718 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/r)^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/r) / (8 \cdot \text{Cc}) - (\text{KL}/r)^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (11.97)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (11.97) / (8 \cdot 123.21) - (11.97^3) / (8 \cdot 123.21^3))$$

$$= 151.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc}/\text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 11.89 / 151.95 + (1905.64 \cdot 257.5 / 18212\text{E}+09) / 173.33$$

$$= 0.231$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	499.0	0.0	79.8	0.0	0.182E+05



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Saddle Calcs: Road Transport:

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Web	658.4	16.0	0.0	105.3	0.0	22.5
Totals	...	...	...	185.2	...	0.182E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/185.182

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 515.0/2

= 257.500 mm

Length of Inner Rib [L]:

= $\text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$

= 1580.0 - $\sqrt{(995.0 + 16.0)^2 - (658.389/2)^2} - 58.0$

= 583.035 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{18234.0/185.182}$

= 99.230 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 583.035/99.23

= 5.876

Unit Force [Force,u]:

= $FI / (2 \cdot \text{Baseplate Length})$

= 15260.513/(2 • 1850.0)

= 4.124 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 4.124 • 658.389 • 583.035

= 1583.262 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 1583.262/708118

= 21.926 N./mm²

Compressive Allowable, $KL/r < C_c$ (5.8756 < 123.2105) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$

= $(1 - (5.88)^2 / (2 \cdot 123.21^2)) 260 /$

$(5/3 + 3 \cdot (5.88) / (8 \cdot 123.21) - (5.88)^3 / (8 \cdot 123.21^3))$

= 154.2 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$

= 17.02/154.17 + (1583.26 • 257.5/18234.025)/173.33

= 0.237

Review notes about nozzle loadings and supports in the Warnings report.

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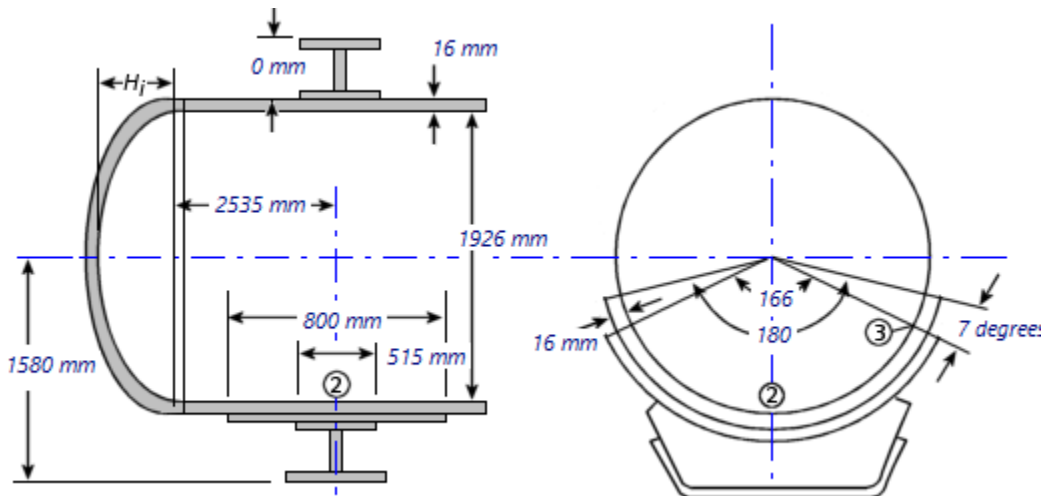
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	971.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	12688.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2535.00	mm
Saddle Width	b1 or b	515.00	mm
Saddle Bearing Angle	delta or theta	166.00	degrees
Wear Plate Width	b2 or b1	800.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		88.10	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1580.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case	228532.52	Kgf
Pressure used in Analysis	P 1319.439	KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	10.10	138.00
Long. Stress at Bottom of Midspan	88.50	138.00
Long. Stress at Top of Saddles	66.38	138.00
Long. Stress at Bottom of Saddles	19.08	138.00
Tangential Shear in Shell	29.39	110.40
Circ. Compressive Stress in Shell	12.64	138.00
Stiffener Circ. Stress at Shell	2.80	172.50



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Stiffener Circ. Stress at Tip	146.05	172.50
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(6053.9/2 + 34930) \cdot 1580.0/1850.0$$

$$= 97253.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(452.6, 11400) \cdot 1580.0/7618.0015$$

$$= 2177.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(\text{Fl,eq}; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(14371.93, 11400) \cdot 1580.0/7618.0015$$

$$= 2980.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$$

$$= 3.0(14372/2 + 34930) \cdot 1580.0/1850.0$$

$$= 107909.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 120624 + \max(2178, 97253, 2981, 107909)$$

$$= 228532.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(2.158 \cdot 1.2)^2)$$

$$= 4438.388 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	229740.03	Kgf
Transverse Shear Load	42116.38	Kgf
Longitudinal Shear Load	14371.93	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1850.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1951	K2 = 0.6681	K3 = 0.3500	K4 = 0.2504
K5 = 0.6435	K6 = 0.0233	K7 = 0.0233	K8 = 0.2756
K9 = 0.1812	K10 = 0.0235	K1* = 0.3314	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -228533 \cdot 2535.0 [1 - (1 - 2535.0/12688.0 + (971.5^2 - 0.0^2)/(2 \cdot 2535.0 \cdot 12688.0))/(1 + (4 \cdot 0.0)/(3 \cdot 12688.0))]$$

$$= -107249.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 228533 \cdot 12688/4((1 + 2(971^2 - 0^2)/(12688^2))/(1 + (4 \cdot 0)/$$



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8, 2026

$$(3 \cdot 12688) - 4 \cdot 2535 / 12688 \\ = 154078.0 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1319.439 \cdot 971.5 / (2 \cdot 13.0) - 154078.0 / (\pi \cdot 971.5^2 \cdot 13.0) \\ = 10.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\ = 1319.439 \cdot 971.5 / (2 \cdot 13.0) + 154078.0 / (\pi \cdot 971.5^2 \cdot 13.0) \\ = 88.50 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t) \\ = 1319.439 \cdot 971.5 / (2 \cdot 15.0) - 107249.6 / (\pi \cdot 971.5^2 \cdot 15.0) \\ = 66.38 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t) \\ = 1319.439 \cdot 971.5 / (2 \cdot 15.0) + 107249.6 / (\pi \cdot 971.5^2 \cdot 15.0) \\ = 19.08 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3)) \\ = 228533(12688.0 - 2 \cdot 2535.0) / (12688.0 + (4 \cdot 0.0 / 3)) \\ = 137213.2 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t) \\ = 137213.16 / (\pi \cdot 971.5 \cdot 15.0) \\ = 29.39 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{R_m \cdot t} \\ = 0.78 \cdot \sqrt{971.5 \cdot 15.0} \\ = 94.159 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k / A \\ = -0.643 \cdot 228533 \cdot 0.1 / 114.05 \\ = -12.64 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q / A + K_6 \cdot Q \cdot R_m \cdot c_1 / I \\ = -0.2756 \cdot 228533 / 114.05 + 0.0233 \cdot 228533 \cdot 971.5 \cdot 109.44 / 9755.48 \\ = 2.80 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q / A - K_6 \cdot Q \cdot R_m \cdot c_2 / I \\ = -0.2756 \cdot 228533 / 114.05 - 0.0233 \cdot 228533 \cdot 971.5 \cdot 176.56 / 9755.48 \\ = -146.05 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 7618.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot L_s (\text{Design Temperature} - \text{Ambient Temperature}) \\ = 0.000012 \cdot 7618.001 (160.0 - 21.1) \\ = 13.216 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length BpLen 1850.0000 mm



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8,2026

Baseplate Thickness	Bpthk	58.0000	mm
Baseplate Width	Bpwid	540.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	620.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t) + \text{WearPlateWidth}})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(964.0 \cdot 15.0) + 800.0})^3) / 12$$

$$= 27.776 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 27.776 + 148.138(315.792 - 7.5)^2$$

$$= 140823.938 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	987.6	15.0	7.5	148.1	111103.8	0.141E+06
Wearplate	800.0	16.0	23.0	128.0	294400.0	0.110E+06
Web	16.0	527.0	294.5	84.3	2483224.2	0.199E+05
BasePlate	540.0	58.0	587.0	313.2	18384840.0	0.231E+06
Totals	...	...	...	673.7	21273568.0	0.502E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 21273568 / 673.658$$

$$= 315.792 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 166.0 / 2$$

$$= 97.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(97.0) - 0.5 \cdot \sin(97.0)^2) / (\pi - 1.693 + \sin(97.0) \cos(97.0))$$

$$= 0.2904$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 228532.516$$

$$= 66366.4609 \text{ Kg}$$

$$\text{Tension Stress, } St = (Fh / As) = 12.3848 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1580.0 - 964.0 \cdot \sin(166.0/2) / 1.4486$$

$$= 919.501 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 61025.2227 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 37.6670 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2$$



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8, 2026

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(228533 + 1208) \cdot 540.0}{4 \cdot 1850.0 \cdot 153.599}}$$

$$= 56.667 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) \cdot (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 166.0/2) \cdot (963.0 + 16.0 + 16.0)$$

$$= 1975.167 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1975.1669 / (4 - 1)$$

$$= 658.389 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 658.3889 \cdot 540.0 / 2$$

$$= 1777.650 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 229740.016 / (1850.0 \cdot 540.0)$$

$$= 22.997 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1777.7 \cdot 23.0$$

$$= 40880.617 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 79.84 + 52.671$$

$$= 132.511 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 40880.6 / 132.5111$$

$$= 30.255 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	515.0	...	82.4	...	0.182E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 515.0 - 16.0$$

$$= 499.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 132.511$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 515.0 / 2$$

$$= 257.500 \text{ mm}$$



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8,2026

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{18212.1/132.511}$$

$$= 117.234 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1580.0 - \cos(166.0/2) (963.0 + 16.0 + 16.0) - 58.0$$

$$= 1400.740 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(166.0/2) (963.0 + 16.0 + 16.0)$$

$$= 1975.167 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1850.0 - 50.8$$

$$= 1799.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (\text{Lw} - \text{Bd}) / 2$$

$$= (1975.167 - 1799.2) / 2$$

$$= 87.983 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{\text{Lv}^2 + \text{Dd}^2}$$

$$= \sqrt{1400.74^2 + 87.983^2}$$

$$= 1403.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 230.4)}$$

$$= 130.885$$

Slenderness ratio [KL/r]:

$$= \text{KL} / \text{r}$$

$$= 1 \cdot 1403.5 / 117.234$$

$$= 11.972$$

Bending Moment [Rm]:

$$= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 14371.9 / (2 \cdot 1850.0) \cdot 658.389 \cdot 1403.5 / 2$$

$$= 1794.680 \text{ Kg-m.}$$

Compressive Allowable, $\text{KL}/\text{r} < \text{Cc}$ ($11.9718 < 130.8854$) per AISC E2-1 [Sca]:

$$= (1 - (\text{KL}/\text{r})^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL}/\text{r}) / (8 \cdot \text{Cc}) - (\text{KL}/\text{r})^3 / (8 \cdot \text{Cc}^3))$$

$$= (1 - (11.97)^2 / (2 \cdot 130.89^2)) 230 /$$

$$(5/3 + 3 \cdot (11.97) / (8 \cdot 130.89) - (11.97^3) / (8 \cdot 130.89^3))$$

$$= 134.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= \text{Sc} / \text{Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba}$$

$$= 30.25 / 134.89 + (1794.68 \cdot 257.5 / 18212\text{E}+09) / 153.6$$

$$= 0.386$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	499.0	0.0	79.8	0.0	0.182E+05
Web	658.4	16.0	0.0	105.3	0.0	22.5
Totals	...	...	...	185.2	...	0.182E+05



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8,2026

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/185.182$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 515.0/2$$

$$= 257.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(\text{Ro} + \text{Wpdthk})^2 - (\text{Pitch}/2)^2} - \text{Bpthk}$$

$$= 1580.0 - \sqrt{(995.0 + 16.0)^2 - (658.389/2)^2} - 58.0$$

$$= 583.035 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{18234.0/185.182}$$

$$= 99.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 583.035/99.23$$

$$= 5.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 14371.931/(2 \cdot 1850.0)$$

$$= 3.884 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.884 \cdot 658.389 \cdot 583.035$$

$$= 1491.073 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1491.073/708118$$

$$= 20.650 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (5.8756 < 130.8854) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.88)^2 / (2 \cdot 130.89^2)) 230 / (5/3 + 3 \cdot (5.88) / (8 \cdot 130.89) - (5.88)^3 / (8 \cdot 130.89^3))$$

$$= 136.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / Sba$$

$$= 43.31/136.72 + (1491.07 \cdot 257.5/18234.025) / 153.6$$

$$= 0.451$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	121831.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2980.8 Kg



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8, 2026

Maximum Transverse Force	Ft	42116.4 Kgf
Maximum Longitudinal Force	F1	14371.9 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:

$= F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 14371.9 / (9.797 \cdot 6)$
 $= 24.0 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:

$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
 $= 42116.4 / (9.797 \cdot 6)$
 $= 70.3 \text{ N./mm}^2$. Must be less than 82.8 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$= 0.0$ (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

$= F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
 $= 14371.93 / (82.8 \cdot 6.0)$
 $= 2.8370 \text{ cm}^2$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$= B \cdot Ft + \text{Sum of X Moments}$
 $= 1580.0 \cdot 42116.38 + 0.0$
 $= 66545.20 \text{ Kg-m}$.

Eccentricity (e):

$= Rmom / QO$
 $= 66545.2 / 121831.03$
 $= 546.20 \text{ mm} > Bplen / 6$ --> Uplift in Transverse direction

$f = Bplen / 2 - Edgedis$

$= 1850.0 / 2 - 50.8$
 $= 874.20 \text{ mm}$

Modular Ratio Of Steel/Concrete (n1):

$= ES / EC$
 $= 203402.5 / 21526.32$
 $= 9.45$

$K1 = 3 (e - 0.5 \cdot Bplen)$

$= 3 (546.2 - 0.5 \cdot 1850.0)$
 $= -1136.41 \text{ mm}$

$K2 = 6 \cdot n1 \cdot At / Bpwid \cdot (f + e)$

$= 6 \cdot 9.45 \cdot 29.39 / 540.0 \cdot (874.2 + 546.2)$
 $= 438282.84 \text{ mm}^2$

$K3 = -K2 \cdot (0.5 \cdot Bplen + f)$

$= -438282.84 \cdot (0.5 \cdot 1850.0 + 874.2)$
 $= -788558466.54 \text{ mm}^3$

Iteratively Solving for the Effective Bearing Length:

$Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 = 0$
 $Y^3 + -1136.41 \cdot Y^2 + 438282.84 \cdot Y + -0.8E+09 = 0$
 $Y = 1276.84 \text{ mm}$

$Num = (Bplen / 2 - Y / 3 - e)$

$= (1850.0 / 2 - 1276.84 / 3 - 546.2)$
 $= -46.81$



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8,2026

$$\begin{aligned}\text{Denom} &= (\text{Bplen} / 2 - Y / 3 + f) \\ &= (1850.0/2 - 1276.84/3 + 874.2) \\ &= 1373.59\end{aligned}$$

$$\begin{aligned}\text{Total Bolt Tension Force [Tforce]:} \\ &= - \text{QO} \cdot \text{Num} / \text{Denom} \\ &= - 121831.03 \cdot -46.81/1373.59 \\ &= 4151.82 \text{ Kgf}\end{aligned}$$

$$\begin{aligned}\text{Bolt Area Required due to Transverse Load [Bltareart]:} \\ &= \text{Tforce} / (\text{Stba} \cdot \text{Nbt}) \\ &= 4151.82 / (138.0 \cdot 3.0) \\ &= 0.9835 \text{ cm}^2\end{aligned}$$

$$\begin{aligned}\text{Required Area of a Single Bolt [Bltarear]:} \\ &= \max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}] \\ &= \max[0.0, 2.837, 0.9835] \\ &= 2.8370 \text{ cm}^2\end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

$$\begin{aligned}\text{Bearing Pressure (fc)} \\ &= 2(\text{QO} + \text{Tforce}) / (Y \cdot \text{Bpwid}) \\ &= 2(121831.03 + 4151.82) / (1276.84 \cdot 540.0) \\ &= 3583.57 \text{ KPa.}\end{aligned}$$

$$\begin{aligned}\text{Distance from Baseplate Edge to the Web [ADIST]:} \\ &= (\text{Bplen} - \text{WebIngh}) / 2 \\ &= (1850.0 - 1799.2) / 2 \\ &= 25.4000 \text{ mm}\end{aligned}$$

$$\begin{aligned}\text{Overturning Moment due To Bolt Tension [Mt]:} \\ &= \text{Tforce} \cdot \text{Adist} \\ &= 4151.82 \cdot 25.4 \\ &= 105.46 \text{ Kg-m.}\end{aligned}$$

$$\begin{aligned}\text{Equivalent Bearing Pressure (f1):} \\ &= \text{fc} \cdot (Y - \text{Adist}) / Y \\ &= 3583.57 \cdot (1276.84 - 25.4) / 1276.84 \\ &= 3512.29 \text{ KPa.}\end{aligned}$$

$$\begin{aligned}\text{Overturning Moment due to Bearing Pressure [Mc]:} \\ &= (\text{Adist}^2 \cdot \text{Bpwid} / 6) \cdot (\text{f1} + 2 \cdot \text{fc}) \\ &= (25.4^2 \cdot 540.0 / 6) \cdot (3512.29 + 2 \cdot 3583.57) \\ &= 63.24 \text{ Kg-m.}\end{aligned}$$

$$\begin{aligned}\text{Baseplate Required Thickness [Treq]:} \\ &= \sqrt{(6 \cdot \max(\text{Mt}, \text{Mc}) / (\text{Bpwid} \cdot \text{Sba}))} \\ &= \sqrt{(6 \cdot \max(105.46, 63.24) / (540.0 \cdot 207.0))} \\ &= 7.4507 \text{ mm}\end{aligned}$$

MDMT Calculations for: LH floating sad, Saddle to Wearplate Junction, Curve: B

$$\begin{aligned}\text{Govrn. thk, tg} &= 16.0, \text{ tr} = 12.758, \text{ c} = 1.0 \text{ mm}, \text{ E*} = 0.85 \\ \text{Thickness Ratio} &= \text{tr} \cdot \text{E*} / (\text{tg} - \text{c}) = 0.723, \text{ Temp. Reduction} = 15 \text{ }^\circ\text{C}\end{aligned}$$

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-29 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8,2026

Input and Calculated Values:

Vessel Mean Radius	Rm	971.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	12688.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2535.00	mm
Saddle Width	b1 or b	515.00	mm
Saddle Bearing Angle	delta or theta	166.00	degrees
Wear Plate Width	b2 or b1	800.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		88.10	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1580.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		171986.16	Kgf
Pressure used in Analysis	P	1319.185	KPa.

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		19.79	138.00
Long. Stress at Bottom of Midspan		78.80	138.00
Long. Stress at Top of Saddles		60.52	138.00
Long. Stress at Bottom of Saddles		24.93	138.00
Tangential Shear in Shell		22.12	110.40
Circ. Compressive Stress in Shell		9.52	138.00
Stiffener Circ. Stress at Shell		2.11	172.50
Stiffener Circ. Stress at Tip		109.91	172.50

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (6053.9/2 + 8630) \cdot 1580.0/1850.0$
 $= 29868.1 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,wind}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(452.6, 11400) \cdot 1580.0/7618.0015$
 $= 2177.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,eq}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(14371.93, 11400) \cdot 1580.0/7618.0015$
 $= 2980.8 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3.0 (14372/2 + 8630) \cdot 1580.0/1850.0$
 $= 40524.1 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8, 2026

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 131462 + \max(2178, 29868, 2981, 40524)$$

$$= 171986.2 \text{ Kgf}$$

Longitudinal Wind Force [F_L]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08(0.0 + \pi/4(2.158 \cdot 1.2)^2)$$

$$= 4438.388 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	173193.67	Kgf
Transverse Shear Load	15816.38	Kgf
Longitudinal Shear Load	14371.93	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1850.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1951	K2 = 0.6681	K3 = 0.3500	K4 = 0.2504
K5 = 0.6435	K6 = 0.0233	K7 = 0.0233	K8 = 0.2756
K9 = 0.1812	K10 = 0.0235	K1* = 0.3314	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -171986 \cdot 2535.0 [1 - (1 - 2535.0/12688.0 + (971.5^2 - 0.0^2) / (2 \cdot 2535.0 \cdot 12688.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 12688.0))]$$

$$= -80712.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 171986 \cdot 12688 / 4 ((1 + 2(971^2 - 0^2) / (12688^2)) / (1 + (4 \cdot 0) / (3 \cdot 12688)) - 4 \cdot 2535 / 12688)$$

$$= 115954.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ₁]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1319.185 \cdot 971.5 / (2 \cdot 13.0) - 115954.1 / (\pi \cdot 971.5^2 \cdot 13.0)$$

$$= 19.79 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ₂]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1319.185 \cdot 971.5 / (2 \cdot 13.0) + 115954.1 / (\pi \cdot 971.5^2 \cdot 13.0)$$

$$= 78.80 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ₃]:

$$= P \cdot Rm / (2t) - M1 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1319.185 \cdot 971.5 / (2 \cdot 15.0) - 80712.6 / (\pi \cdot 971.5^2 \cdot 15.0)$$

$$= 60.52 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ₄]:

$$= P \cdot Rm / (2t) + M1 / (\pi \cdot Rm^2 \cdot t)$$

$$= 1319.185 \cdot 971.5 / (2 \cdot 15.0) + 80712.6 / (\pi \cdot 971.5^2 \cdot 15.0)$$

$$= 24.93 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm / 3))$$

$$= 171986(12688.0 - 2 \cdot 2535.0) / (12688.0 + (4 \cdot 0.0 / 3))$$



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8,2026

= 103262.2 Kgf

Shear Stress in the shell with a single ring (4.15.13) [T1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 103262.18 / (\pi \cdot 971.5 \cdot 15.0)$$

$$= 22.12 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{ (R_m \cdot t) }$$

$$= 0.78 \cdot \sqrt{ (971.5 \cdot 15.0) }$$

$$= 94.159 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.643 \cdot 171986 \cdot 0.1/114.05$$

$$= -9.52 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c1/l$$

$$= -0.2756 \cdot 171986/114.05 + 0.0233 \cdot 171986 \cdot 971.5 \cdot 109.44/9755.48$$

$$= 2.11 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c2/l$$

$$= -0.2756 \cdot 171986/114.05 - 0.0233 \cdot 171986 \cdot 971.5 \cdot 176.56/9755.48$$

$$= -109.91 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	58.0000	mm
Baseplate Width	Bpwid	540.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	620.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{ (R \cdot t) } + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{ (964.0 \cdot 15.0) } + 800.0) 15.0^3) / 12$$

$$= 27.776 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 27.776 + 148.138 (315.792 - 7.5)^2$$

$$= 140823.938 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	987.6	15.0	7.5	148.1	111103.8	0.141E+06
Wearplate	800.0	16.0	23.0	128.0	294400.0	0.110E+06
Web	16.0	527.0	294.5	84.3	2483224.2	0.199E+05
BasePlate	540.0	58.0	587.0	313.2	18384840.0	0.231E+06
Totals	...	...	...	673.7	21273568.0	0.502E+06

Distance to Centroid [C1]:



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8, 2026

$$\begin{aligned}
 &= AY / A \\
 &= 21273568 / 673.658 \\
 &= 315.792 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Angle } [\beta]: \\
 &= 180 - \text{Saddle Angle} / 2 \\
 &= 180 - 166.0 / 2 \\
 &= 97.0
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Coefficient [K1]:} \\
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(97.0) - 0.5 \cdot \sin(97.0)^2) / (\pi - 1.693 + \sin(97.0) \cos(97.0)) \\
 &= 0.2904
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Force [Fh]:} \\
 &= K1 \cdot Q \\
 &= 0.29 \cdot 171986.156 \\
 &= 49945.2500 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 9.3204 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Saddle Splitting Dimension [d]:} \\
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1580.0 - 964.0 \cdot \sin(166.0/2) / 1.4486 \\
 &= 919.501 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 45925.6055 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 28.3470 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{3(171986 + 1208)540.0 / (4 \cdot 1850.0 \cdot 153.599)} \\
 &= 49.201 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

$$\begin{aligned}
 \text{Web Length Dimension [Web Length]:} \\
 &= 2 \cdot \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 166.0/2) (963.0 + 16.0 + 16.0) \\
 &= 1975.167 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Distance between Ribs [e]:} \\
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1975.1669 / (4 - 1) \\
 &= 658.389 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 \text{Baseplate Pressure Area [Ap]:} \\
 &= e \cdot B_{\text{pwid}} / 2 \\
 &= 658.3889 \cdot 540.0 / 2 \\
 &= 1777.650 \text{ cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Bearing Pressure [Bp]:} \\
 &= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth}) \\
 &= 173193.672 / (1850.0 \cdot 540.0) \\
 &= 17.337 \text{ Kgf/cm}^2
 \end{aligned}$$

$$\begin{aligned}
 \text{Axial Load [P]:} \\
 &= Ap \cdot Bp \\
 &= 1777.7 \cdot 17.34 \\
 &= 30818.594 \text{ Kgf}
 \end{aligned}$$

$$\text{Area of the Rib and Web [Ar]:}$$



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Saddle Calcs: Operating Case:

Step: 22

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= Rib Area + Web Area

= 79.84 + 52.671

= 132.511 cm²

Compressive Stress [Sc]:

= P/Ar

= 30818.6/132.5111

= 22.808 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	515.0	...	82.4	...	0.182E+05

Rib dimension [D]:

= Saddle Width - Web Thickness

= 515.0 - 16.0

= 499.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/132.511

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 515.0/2

= 257.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{18212.1/132.511}$

= 117.234 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - cos(saddle bearing angle / 2)(radius + shlthk + wpdthk) - bpthk

= 1580.0 - cos(166.0/2)(963.0 + 16.0 + 16.0) - 58.0

= 1400.740 mm

Length of Outer Rib, step 2, [Lw]:

= 2 • sin(saddle bearing angle / 2)(radius + shlthk + wpdthk)

= 2 • sin(166.0/2)(963.0 + 16.0 + 16.0)

= 1975.167 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 1850.0 - 50.8

= 1799.200 mm

Length of Outer Rib, step 4, [Dd]:

= (Lw - Bd)/2

= (1975.167 - 1799.2)/2

= 87.983 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{1400.74^2 + 87.983^2}$

= 1403.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09/230.4}$



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8,2026

= 130.885

Slenderness ratio [KL/r]:

= KL/r

= 1 • 1403.5/117.234

= 11.972

Bending Moment [Rm]:

= FI / (2 • Bplen) • e • L / 2

= 14371.9 / (2 • 1850.0) • 658.389 • 1403.5 / 2

= 1794.680 Kg-m.

Compressive Allowable, KL/r < Cc (11.9718 < 130.8854) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))

= (1 - (11.97)² / (2 • 130.89²)) 230 /

(5/3 + 3 • (11.97) / (8 • 130.89) - (11.97³) / (8 • 130.89³))

= 134.9 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)

= Sc/Sca + (Rm • C1 / I) / Sba

= 22.81/134.89 + (1794.68 • 257.5 / 1.8212E+09) / 153.6

= 0.331

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	499.0	0.0	79.8	0.0	0.182E+05
Web	658.4	16.0	0.0	105.3	0.0	22.5
Totals	...	...	...	185.2	...	0.182E+05

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/185.182

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 515.0/2

= 257.500 mm

Length of Inner Rib [L]:

= Saddle Height - √((Ro + Wpdthk)² - (Pitch/2)²) - Bpthk

= 1580.0 - √((995.0 + 16.0)² - (658.389/2)²) - 58.0

= 583.035 mm

Radius of Gyration [r]:

= √(Total Inertia / Total Area)

= √(18234.0/185.182)

= 99.230 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 583.035/99.23

= 5.876

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 14371.931 / (2 • 1850.0)

= 3.884 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 3.884 • 658.389 • 583.035



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Saddle Calcs: Operating Case: Step: 22 6:34pm Sep 8,2026

= 1491.073 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 1491.073/708118

= 20.650 N./mm²

Compressive Allowable, $KL/r < C_c$ (5.8756 < 130.8854) per AISC E2-1 [Sca]:

= $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$

= $(1 - (5.88)^2 / (2 \cdot 130.89^2)) 230 /$

$(5/3 + 3 \cdot (5.88) / (8 \cdot 130.89) - (5.88)^3 / (8 \cdot 130.89^3))$

= 136.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= $S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$

= $32.65 / 136.72 + (1491.07 \cdot 257.5 / 18234.025) / 153.6$

= 0.373

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	138.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BlArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	132669.5 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	2980.8 Kg
Maximum Transverse Force	Ft	15816.4 Kg
Maximum Longitudinal Force	Fl	14371.9 Kg
Saddle Bolted to Steel Foundation		No

Friction force due to thermal growth, [Ff]:

= 36549.3 Kg

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

= $\max(F_f, F_l) / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $36549.3 / (9.797 \cdot 6)$

= 61.0 N./mm². Must be less than 82.8 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$

= $15816.4 / (9.797 \cdot 6)$

= 26.4 N./mm². Must be less than 82.8 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

= 0.0 (Q0 > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$

= $14371.93 / (82.8 \cdot 6.0)$

= 2.8370 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $1580.0 \cdot 15816.38 + 0.0$

= 24990.38 Kg-m.



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Saddle Calcs: Operating Case:

Step: 22

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Eccentricity (e):

= Rmom / QO

= 24990.38/132669.55

= 188.36 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0.0, 2.837, 0.0]

= 2.8370 cm²

MDMT Calculations for: New Saddle, Saddle to Wearplate Junction, Curve: B

Govrn. thk, tg = 16.0, tr = 12.758, c = 1.0 mm, E* = 0.85

Thickness Ratio = tr * E*/(tg - c) = 0.723, Temp. Reduction = 15 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -14 °C

Min Metal Temp. at Required thickness (UCS 66.1) -29 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Test Case:

Step: 23

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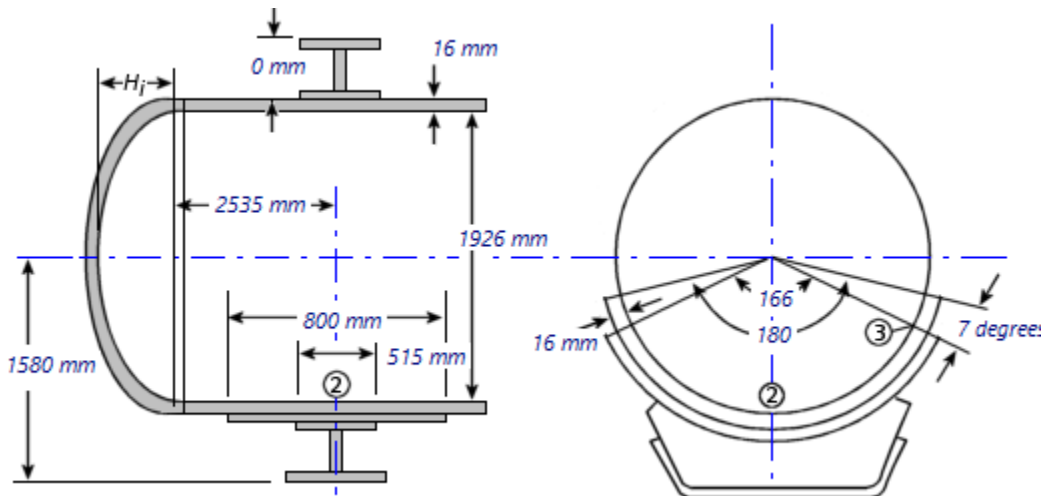
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	971.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	12688.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2535.00	mm
Saddle Width	b1 or b	515.00	mm
Saddle Bearing Angle	delta or theta	166.00	degrees
Wear Plate Width	b2 or b1	800.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1580.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces	118404.58	Kgf
Pressure used in Analysis	P	1712.522 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	43.68	234.00
Long. Stress at Bottom of Midspan	84.30	234.00
Long. Stress at Top of Saddles	67.71	234.00
Long. Stress at Bottom of Saddles	43.21	234.00
Tangential Shear in Shell	15.23	138.58
Circ. Compressive Stress in Shell	6.55	234.00
Stiffener Circ. Stress at Shell	1.45	234.00



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Saddle Calcs: Test Case:

Step: 23

6:34pm

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Stiffener Circ. Stress at Tip	75.67	234.00
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Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (3027.0/2 + 0) \cdot 1580.0/1850.0$$

$$= 3877.8 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(226.3, 0) \cdot 1580.0/7618.0015$$

$$= 46.9 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 114527 + \max(47, 3878, 0, 0)$$

$$= 118404.6 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0.0 + \pi/4 (2.158 \cdot 1.2)^2)$$

$$= 4438.388 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	119612.09	Kgf
Transverse Shear Load	1513.48	Kgf
Longitudinal Shear Load	226.30	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1850.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1951	K2 = 0.6681	K3 = 0.3500	K4 = 0.2504
K5 = 0.6435	K6 = 0.0233	K7 = 0.0233	K8 = 0.2756
K9 = 0.1812	K10 = 0.0235	K1* = 0.3314	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -118405 \cdot 2535.0 [1 - (1 - 2535.0/12688.0 + (971.5^2 - 0.0^2)/(2 \cdot 2535.0 \cdot 12688.0)) / (1 + (4 \cdot 0.0)/(3 \cdot 12688.0))]$$

$$= -55566.9 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 118405 \cdot 12688/4 ((1 + 2(971^2 - 0^2)/(12688^2)) / (1 + (4 \cdot 0)/(3 \cdot 12688)) - 4 \cdot 2535/12688)$$

$$= 79829.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M_2/(\pi \cdot R_m^2 t)$$

$$= 1712.522 \cdot 971.5/(2 \cdot 13.0) - 79829.1/(\pi \cdot 971.5^2 \cdot 13.0)$$

$$= 43.68 \text{ N/mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M_2/(\pi \cdot R_m^2 \cdot t)$$



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$$= 1712.522 \cdot 971.5 / (2 \cdot 13.0) + 79829.1 / (\pi \cdot 971.5^2 \cdot 13.0)$$

$$= 84.30 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$= P \cdot R_m / (2t) - M1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1712.522 \cdot 971.5 / (2 \cdot 15.0) - 55566.9 / (\pi \cdot 971.5^2 \cdot 15.0)$$

$$= 67.71 \text{ N./mm}^2$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 1712.522 \cdot 971.5 / (2 \cdot 15.0) + 55566.9 / (\pi \cdot 971.5^2 \cdot 15.0)$$

$$= 43.21 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 118405(12688.0 - 2 \cdot 2535.0) / (12688.0 + (4 \cdot 0.0 / 3))$$

$$= 71091.3 \text{ Kgf}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$= T / (\pi \cdot R_m \cdot t)$$

$$= 71091.27 / (\pi \cdot 971.5 \cdot 15.0)$$

$$= 15.23 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(971.5 \cdot 15.0)}$$

$$= 94.159 \text{ mm}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$= -K_5 \cdot Q \cdot k/A$$

$$= -0.643 \cdot 118405 \cdot 0.1 / 114.05$$

$$= -6.55 \text{ N./mm}^2$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l$$

$$= -0.2756 \cdot 118405 / 114.05 + 0.0233 \cdot 118405 \cdot 971.5 \cdot 109.44 / 9755.48$$

$$= 1.45 \text{ N./mm}^2$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l$$

$$= -0.2756 \cdot 118405 / 114.05 - 0.0233 \cdot 118405 \cdot 971.5 \cdot 176.56 / 9755.48$$

$$= -75.67 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	58.0000	mm
Baseplate Width	Bpwid	540.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw, c	620.0	mm
Friction Coefficient	mu	0.300	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(964.0 \cdot 15.0)} + 800.0)^3) / 12$$

$$= 27.776 \text{ cm}^4$$

Resolved Inertia for the Shell [I, shell]:



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$$\begin{aligned}
 &= \text{ShellInertia} + A\text{Shell}(C1 - Y)^2 \\
 &= 27.776 + 148.138(315.792 - 7.5)^2 \\
 &= 140823.938 \text{ cm}^4
 \end{aligned}$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² cm ⁴
Shell	987.6	15.0	7.5	148.1	111103.8	0.141E+06
Wearplate	800.0	16.0	23.0	128.0	294400.0	0.110E+06
Web	16.0	527.0	294.5	84.3	2483224.2	0.199E+05
BasePlate	540.0	58.0	587.0	313.2	18384840.0	0.231E+06
Totals	...	...	...	673.7	21273568.0	0.502E+06

Distance to Centroid [C1]:

$$\begin{aligned}
 &= AY / A \\
 &= 21273568 / 673.658 \\
 &= 315.792 \text{ mm}
 \end{aligned}$$

Angle [β]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 166.0/2 \\
 &= 97.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta)\cos(\beta)) \\
 &= (1 + \cos(97.0) - 0.5 \cdot \sin(97.0)^2) / (\pi - 1.693 + \sin(97.0)\cos(97.0)) \\
 &= 0.2904
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 \cdot Q \\
 &= 0.29 \cdot 118404.578 \\
 &= 34385.0117 \text{ Kgf}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 6.4167 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 0.6 \cdot \text{Yield Str} = 155.9972 \text{ N./mm}^2
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 1580.0 - 964.0 \cdot \sin(166.0/2) / 1.4486 \\
 &= 919.501 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh \cdot d = 31617.6719 \text{ Kg-m.}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M \cdot C1 / I) = 19.5156 \text{ N./mm}^2 \\
 \text{Allowed Stress, } Sa &= 2/3 \cdot \text{Yield Str} = 173.3302 \text{ N./mm}^2
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= \sqrt{(3(Q + \text{SaddleWt})\text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})} \\
 &= \sqrt{(3(118405 + 1208)540.0) / (4 \cdot 1850.0 \cdot 173.33)} \\
 &= 38.491 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 \cdot \cos(90 - 166.0/2) (963.0 + 16.0 + 16.0) \\
 &= 1975.167 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (N_{\text{ribs}} - 1) \\
 &= 1975.1669 / (4 - 1) \\
 &= 658.389 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$



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$$= 658.3889 \cdot 540.0/2$$

$$= 1777.650 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 119612.094 / (1850.0 \cdot 540.0)$$

$$= 11.973 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 1777.7 \cdot 11.97$$

$$= 21284.129 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 79.84 + 52.671$$

$$= 132.511 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 21284.1 / 132.511$$

$$= 15.752 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	515.0	...	82.4	...	0.182E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 515.0 - 16.0$$

$$= 499.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 132.511$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 515.0 / 2$$

$$= 257.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(18212.1 / 132.511)}$$

$$= 117.234 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 1580.0 - \cos(166.0/2) (963.0 + 16.0 + 16.0) - 58.0$$

$$= 1400.740 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$$

$$= 2 \cdot \sin(166.0/2) (963.0 + 16.0 + 16.0)$$

$$= 1975.167 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= \text{Baseplate length} - \text{clearance}$$

$$= 1850.0 - 50.8$$

$$= 1799.200 \text{ mm}$$



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Length of Outer Rib, step 4, [Dd]:

$$= (Lw - Bd) / 2$$

$$= (1975.167 - 1799.2) / 2$$

$$= 87.983 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{Lv^2 + Dd^2}$$

$$= \sqrt{1400.74^2 + 87.983^2}$$

$$= 1403.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$$

$$= \sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1403.5 / 117.234$$

$$= 11.972$$

Bending Moment [Rm]:

$$= FI / (2 \cdot Bplen) \cdot e \cdot L / 2$$

$$= 226.3 / (2 \cdot 1850.0) \cdot 658.389 \cdot 1403.5 / 2$$

$$= 28.259 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($11.9718 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (11.97)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (11.97) / (8 \cdot 123.21) - (11.97^3) / (8 \cdot 123.21^3))$$

$$= 151.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc / Sca + (Rm \cdot C1 / I) / Sba$$

$$= 15.75 / 151.95 + (28.26 \cdot 257.5 / 1.8212\text{E}+09) / 173.33$$

$$= 0.106$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	499.0	0.0	79.8	0.0	0.182E+05
Web	658.4	16.0	0.0	105.3	0.0	22.5
Totals	...	...	...	185.2	...	0.182E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 185.182$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 515.0 / 2$$

$$= 257.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1580.0 - \sqrt{(995.0 + 16.0)^2 - (658.389/2)^2} - 58.0$$

$$= 583.035 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{18234.0 / 185.182}$$

$$= 99.230 \text{ mm}$$



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Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 583.035/99.23$$

$$= 5.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 226.298 / (2 \cdot 1850.0)$$

$$= 0.061 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.061 \cdot 658.389 \cdot 583.035$$

$$= 23.478 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 23.478/708118$$

$$= 0.325 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (5.8756 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.88)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (5.88) / (8 \cdot 123.21) - (5.88^3) / (8 \cdot 123.21^3))$$

$$= 154.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 22.55/154.17 + (23.48 \cdot 257.5/18234.025) / 173.33$$

$$= 0.148$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load Q0 (Weight)	Q0	115734.3 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	46.9 Kg
Maximum Transverse Force	Ft	1513.5 Kg
Maximum Longitudinal Force	F1	226.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 226.3 / (9.797 \cdot 6)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1513.5 / (9.797 \cdot 6)$$

$$= 2.5 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:



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= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbolts})$
 = $226.3 / (28.98 \cdot 6.0)$
 = 0.1276 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
 = $B \cdot F_t + \text{Sum of X Moments}$
 = $1580.0 \cdot 1513.48 + 0.0$
 = 2391.34 Kg-m.

Eccentricity (e):
 = R_{mom} / QO
 = $2391.34 / 115734.32$
 = $20.66 \text{ mm} < B_{plen}/6 \rightarrow \text{No Uplift in Transverse direction}$

Bolt Area due to Transverse Load [Bltareart]:
 = 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
 = $\max[Bltarearl, Bltarears, Bltareart]$
 = $\max[0.0, 0.1276, 0.0]$
 = 0.1276 cm^2

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	971.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	12688.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2535.00	mm
Saddle Width	b1 or b	515.00	mm
Saddle Bearing Angle	delta or theta	166.00	degrees
Wear Plate Width	b2 or b1	800.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		234.00	N./mm ²
Head Allowable Stress used in Calculation		234.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1580.00	mm
Coefficient of Friction	mu	0.00	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces 128940.48 KgF
 Pressure used in Analysis P 1712.522 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	41.88	234.00
Long. Stress at Bottom of Midspan	86.11	234.00
Long. Stress at Top of Saddles	68.80	234.00
Long. Stress at Bottom of Saddles	42.12	234.00
Tangential Shear in Shell	16.58	138.58
Circ. Compressive Stress in Shell	7.13	234.00



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Stiffener Circ. Stress at Shell	1.58	234.00
Stiffener Circ. Stress at Tip	82.40	234.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$$

$$= 3.0 (3027.0/2 + 0) \cdot 1580.0/1850.0$$

$$= 3877.8 \text{ Kg}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$$

$$= \max(226.3, 0) \cdot 1580.0/7618.0015$$

$$= 46.9 \text{ Kg}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$$

$$= 125063 + \max(47, 3878, 0, 0)$$

$$= 128940.5 \text{ Kg}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0.0 + \pi/4 (2.158 \cdot 1.2)^2)$$

$$= 4438.388 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	130147.98	Kgf
Transverse Shear Load	1513.48	Kgf
Longitudinal Shear Load	226.30	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1850.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.1951	K2 = 0.6681	K3 = 0.3500	K4 = 0.2504
K5 = 0.6435	K6 = 0.0233	K7 = 0.0233	K8 = 0.2756
K9 = 0.1812	K10 = 0.0235	K1* = 0.3314	K6p = 0.0174
K7p = 0.0174			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L)) / (1 + (4h_m)/3L)]$$

$$= -128940 \cdot 2535.0 [1 - (1 - 2535.0/12688.0 + (971.5^2 - 0.0^2) / (2 \cdot 2535.0 \cdot 12688.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 12688.0))]$$

$$= -60511.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L)$$

$$= 128940 \cdot 12688 / 4 ((1 + 2(971.5^2 - 0^2)/(12688^2)) / (1 + (4 \cdot 0)/(3 \cdot 12688)) - 4 \cdot 2535/12688)$$

$$= 86932.5 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= 1712.522 \cdot 971.5 / (2 \cdot 13.0) - 86932.5 / (\pi \cdot 971.5^2 \cdot 13.0)$$

$$= 41.88 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:



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$$\begin{aligned}
 &= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t) \\
 &= 1712.522 \cdot 971.5 / (2 \cdot 13.0) + 86932.5 / (\pi \cdot 971.5^2 \cdot 13.0) \\
 &= 86.11 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) - M_1 / (\pi \cdot R_m^2 \cdot t) \\
 &= 1712.522 \cdot 971.5 / (2 \cdot 15.0) - 60511.4 / (\pi \cdot 971.5^2 \cdot 15.0) \\
 &= 68.80 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$\begin{aligned}
 &= P \cdot R_m / (2t) + M_1 / (\pi \cdot R_m^2 \cdot t) \\
 &= 1712.522 \cdot 971.5 / (2 \cdot 15.0) + 60511.4 / (\pi \cdot 971.5^2 \cdot 15.0) \\
 &= 42.12 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L-2a) / (L + (4 \cdot h_m / 3)) \\
 &= 128940(12688.0 - 2 \cdot 2535.0) / (12688.0 + (4 \cdot 0.0 / 3)) \\
 &= 77417.1 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$\begin{aligned}
 &= T / (\pi \cdot R_m \cdot t) \\
 &= 77417.12 / (\pi \cdot 971.5 \cdot 15.0) \\
 &= 16.58 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(R_m \cdot t)} \\
 &= 0.78 \cdot \sqrt{(971.5 \cdot 15.0)} \\
 &= 94.159 \text{ mm}
 \end{aligned}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$\begin{aligned}
 &= -K_5 \cdot Q \cdot k/A \\
 &= -0.643 \cdot 128940 \cdot 0.1 / 114.05 \\
 &= -7.13 \text{ N./mm}^2
 \end{aligned}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$\begin{aligned}
 &= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/l \\
 &= -0.2756 \cdot 128940 / 114.05 + 0.0233 \cdot 128940 \cdot 971.5 \cdot 109.44 / 9755.48 \\
 &= 1.58 \text{ N./mm}^2
 \end{aligned}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$\begin{aligned}
 &= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/l \\
 &= -0.2756 \cdot 128940 / 114.05 - 0.0233 \cdot 128940 \cdot 971.5 \cdot 176.56 / 9755.48 \\
 &= -82.40 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	58.0000	mm
Baseplate Width	Bpwid	540.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	260.0	N./mm ²
Height of Web at Center	Hw,c	620.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$\begin{aligned}
 &= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12 \\
 &= ((1.56 \cdot \sqrt{(964.0 \cdot 15.0)} + 800.0)^3) / 12 \\
 &= 27.776 \text{ cm}^4
 \end{aligned}$$



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Resolved Inertia for the Shell [I,shell]:

= ShellInertia + AShell(C1 - Y)²

= 27.776 + 148.138(315.792 - 7.5)²

= 140823.938 cm⁴

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	987.6	15.0	7.5	148.1	111103.8	0.141E+06
Wearplate	800.0	16.0	23.0	128.0	294400.0	0.110E+06
Web	16.0	527.0	294.5	84.3	2483224.2	0.199E+05
BasePlate	540.0	58.0	587.0	313.2	18384840.0	0.231E+06
Totals	...	...	...	673.7	21273568.0	0.502E+06

Distance to Centroid [C1]:

= AY / A

= 21273568/673.658

= 315.792 mm

Angle [β]:

= 180 - Saddle Angle/2

= 180 - 166.0/2

= 97.0

Saddle Splitting Coefficient [K1]:

= (1 + cos(β) - 0.5•sin(β)²)/(π - β + sin(β)cos(β))

= (1 + cos(97.0) - 0.5•sin(97.0)²)/(π - 1.693 + sin(97.0)cos(97.0))

= 0.2904

Saddle Splitting Force [Fh]:

= K1 • Q

= 0.29 • 128940.477

= 37444.6641 Kgf

Tension Stress, St = (Fh/As) = 6.9876 N./mm²

Allowed Stress, Sa = 0.6 * Yield Str = 155.9972 N./mm²

Saddle Splitting Dimension [d]:

= B - R • sin(theta/2)/(theta/2 in radians)

= 1580.0 - 964.0 • sin(166.0/2)/1.4486

= 919.501 mm

Bending Moment, M = Fh * d = 34431.0820 Kg-m.

Bending Stress, Sb = (M * C1 / I) = 21.2521 N./mm²

Allowed Stress, Sa = 2/3 * Yield Str = 173.3302 N./mm²

Minimum Thickness of Baseplate per Moss:

= √(3(Q + SaddleWt)BasePlateWidth / (4 • BasePlateLength • AllStress))

= √(3(128940 + 1208)540.0/(4 • 1850.0 • 173.33))

= 40.150 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)

= 2 • cos(90 - 166.0/2)(963.0 + 16.0 + 16.0)

= 1975.167 mm

Distance between Ribs [e]:

= Web Length / (Nribs - 1)

= 1975.1669/(4 - 1)

= 658.389 mm



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Baseplate Pressure Area [Ap]:

$$= e \cdot Bpwid / 2$$

$$= 658.3889 \cdot 540.0 / 2$$

$$= 1777.650 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + SaddleWt) / (BasePlateLength \cdot BasePlateWidth)$$

$$= 130147.984 / (1850.0 \cdot 540.0)$$

$$= 13.028 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1777.7 \cdot 13.03$$

$$= 23158.916 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= Rib \text{ Area} + Web \text{ Area}$$

$$= 79.84 + 52.671$$

$$= 132.511 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / Ar$$

$$= 23158.9 / 132.5111$$

$$= 17.139 \text{ N/mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	515.0	...	82.4	...	0.182E+05

Rib dimension [D]:

$$= Saddle \text{ Width} - Web \text{ Thickness}$$

$$= 515.0 - 16.0$$

$$= 499.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 132.511$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= Saddle \text{ Width} / 2$$

$$= 515.0 / 2$$

$$= 257.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ (Total \text{ Inertia} / Total \text{ Area}) }$$

$$= \sqrt{ (18212.1 / 132.511) }$$

$$= 117.234 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:

$$= Saddle \text{ Height} - \cos(saddle \text{ bearing angle} / 2) (radius + shlthk + wpdthk) - bpthk$$

$$= 1580.0 - \cos(166.0 / 2) (963.0 + 16.0 + 16.0) - 58.0$$

$$= 1400.740 \text{ mm}$$

Length of Outer Rib, step 2, [Lw]:

$$= 2 \cdot \sin(saddle \text{ bearing angle} / 2) (radius + shlthk + wpdthk)$$

$$= 2 \cdot \sin(166.0 / 2) (963.0 + 16.0 + 16.0)$$

$$= 1975.167 \text{ mm}$$

Length of Outer Rib, step 3, [Bd]:

$$= Baseplate \text{ length} - clearance$$



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Saddle Calcs: Test Case:

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$$= 1850.0 - 50.8$$

$$= 1799.200 \text{ mm}$$

Length of Outer Rib, step 4, [Dd]:

$$= (L_w - B_d) / 2$$

$$= (1975.167 - 1799.2) / 2$$

$$= 87.983 \text{ mm}$$

Length of Outer Rib [L]:

$$= \sqrt{(L_v^2 + D_d^2)}$$

$$= \sqrt{(1400.74^2 + 87.983^2)}$$

$$= 1403.500 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 260.0)}$$

$$= 123.211$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 \cdot 1403.5 / 117.234$$

$$= 11.972$$

Bending Moment [Rm]:

$$= F_l / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 226.3 / (2 \cdot 1850.0) \cdot 658.389 \cdot 1403.5 / 2$$

$$= 28.259 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($11.9718 < 123.2105$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (11.97)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (11.97) / (8 \cdot 123.21) - (11.97^3) / (8 \cdot 123.21^3))$$

$$= 151.9 \text{ N/mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 17.14 / 151.95 + (28.26 \cdot 257.5 / 1.8212\text{E}+09) / 173.33$$

$$= 0.115$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	499.0	0.0	79.8	0.0	0.182E+05
Web	658.4	16.0	0.0	105.3	0.0	22.5
Totals	...	...	...	185.2	...	0.182E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0 / 185.182$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 515.0 / 2$$

$$= 257.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(R_o + W_{pdthk})^2 - (\text{Pitch}/2)^2} - B_{pthk}$$

$$= 1580.0 - \sqrt{(995.0 + 16.0)^2 - (658.389/2)^2} - 58.0$$

$$= 583.035 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$



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Saddle Calcs: Test Case:

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$$= \sqrt{18234.0/185.182}$$

$$= 99.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 583.035/99.23$$

$$= 5.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 226.298 / (2 \cdot 1850.0)$$

$$= 0.061 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.061 \cdot 658.389 \cdot 583.035$$

$$= 23.478 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 23.478/708118$$

$$= 0.325 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ (5.8756 < 123.2105) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (5.88)^2 / (2 \cdot 123.21^2)) 260 /$$

$$(5/3 + 3 \cdot (5.88) / (8 \cdot 123.21) - (5.88)^3 / (8 \cdot 123.21^3))$$

$$= 154.2 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 24.54/154.17 + (23.48 \cdot 257.5/18234.025) / 173.33$$

$$= 0.161$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	6
Total Number of Bolts in Tension/Baseplate	Nbt	3
Bolt Material Specification		SA-307 A
Bolt Allowable Stress	Stba	48.30 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm
Distance from Bolts to Edge	Edgedis	50.8 mm
User Defined Area in a Single Bolt	UserBoltArea	11.5300 cm ²
BasePlate Allowable Stress	S	138.00 N./mm ²
Area Available in a Single Bolt	BltArea	9.7967 cm ²
Saddle Load QO (Weight)	QO	126270.2 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	46.9 Kg
Maximum Transverse Force	Ft	1513.5 Kg
Maximum Longitudinal Force	F1	226.3 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 226.3 / (9.797 \cdot 6)$$

$$= 0.4 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1513.5 / (9.797 \cdot 6)$$

$$= 2.5 \text{ N./mm}^2. \text{ Must be less than } 29.0 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (QO > QL --> No Uplift in Longitudinal direction)}$$



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Bolt Area due to Shear Load [Bltarears]:

= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$

= $226.3 / (28.98 \cdot 6.0)$

= 0.1276 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= $B \cdot F_t + \text{Sum of X Moments}$

= $1580.0 \cdot 1513.48 + 0.0$

= 2391.34 Kg-m.

Eccentricity (e):

= R_{mom} / QO

= $2391.34 / 126270.2$

= 18.94 mm < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$

= $\max[0.0, 0.1276, 0.0]$

= 0.1276 cm²

Review notes about nozzle loadings and supports in the Warnings report.

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Saddle Calcs: Ext. Pressure:

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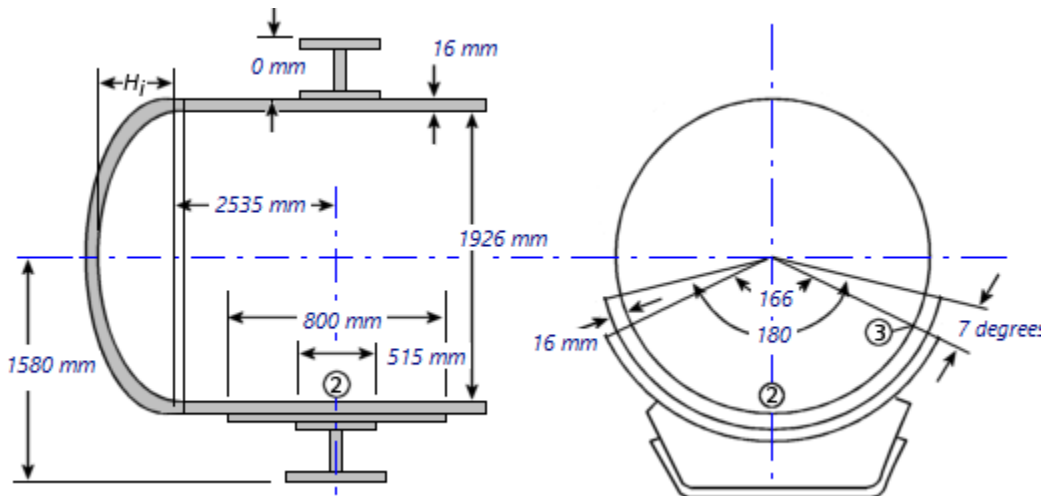
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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	971.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	12688.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2535.00	mm
Saddle Width	b1 or b	515.00	mm
Saddle Bearing Angle	delta or theta	166.00	degrees
Wear Plate Width	b2 or b1	800.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		88.10	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	1580.00	mm
Coefficient of Friction	mu	0.30	



Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case	228532.52	Kgf
Pressure used in Analysis	P	-51.704 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-41.13	-195.48
Long. Stress at Top of Midspan	41.13	138.00
Long. Stress at Bottom of Midspan	37.27	138.00
Long. Stress at Top of Saddles	21.97	138.00
Long. Stress at Bottom of Saddles	-25.32	-207.36
Long. Stress at Bottom of Saddles	25.32	138.00
Tangential Shear in Shell	29.39	110.40



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Saddle Calcs: Ext. Pressure: Step: 24 6:34pm Sep 8, 2026

Circ. Compressive Stress in Shell	12.64	138.00
Stiffener Circ. Stress at Shell	2.80	172.50
Stiffener Circ. Stress at Tip	146.05	292.49

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(6053.9/2 + 34930) \cdot 1580.0/1850.0$$

$$= 97253.0 \text{ Kgf}$$

Longitudinal Saddle Reaction Force [Fwl]:

$$= \max(Fl, wind; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(452.6, 11400) \cdot 1580.0/7618.0015$$

$$= 2177.7 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Fl [Fsl]:

$$= \max(Fl, eq; \text{Sum of X Forces}) \cdot B / Ls$$

$$= \max(14371.93, 11400) \cdot 1580.0/7618.0015$$

$$= 2980.8 \text{ Kgf}$$

Saddle Reaction Force due to Earthquake Ft [Fst]:

$$= Ftr(Ft/Num of Saddles + Z Force Load) \cdot B / E$$

$$= 3.0(14372/2 + 34930) \cdot 1580.0/1850.0$$

$$= 107909.0 \text{ Kgf}$$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(Fwl, Fwt, Fsl, Fst)$$

$$= 120624 + \max(2178, 97253, 2981, 107909)$$

$$= 228532.5 \text{ Kgf}$$

Longitudinal Wind Force [Fl]:

$$= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 766.08 (0.0 + \pi/4 (2.158 \cdot 1.2)^2)$$

$$= 4438.388 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 229740.03 Kgf

Transverse Shear Load 42116.38 Kgf

Longitudinal Shear Load 14371.93 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \text{Baseplate Length}$$

$$= 1850.000 \text{ mm}$$

The Computed K values from Table 4.15.1:

$$K1 = 0.1951 \quad K2 = 0.6681 \quad K3 = 0.3500 \quad K4 = 0.2504$$

$$K5 = 0.6435 \quad K6 = 0.0233 \quad K7 = 0.0233 \quad K8 = 0.2756$$

$$K9 = 0.1812 \quad K10 = 0.0235 \quad K1* = 0.3314 \quad K6p = 0.0174$$

$$K7p = 0.0174$$

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm)/3L)]$$

$$= -228533 \cdot 2535.0 [1 - (1 - 2535.0/12688.0 + (971.5^2 - 0.0^2) / (2 \cdot 2535.0 \cdot 12688.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 12688.0))]$$

$$= -107249.6 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:



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Saddle Calcs: Ext. Pressure:

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$$\begin{aligned}
 &= Q \cdot L / 4 \cdot ((1 + 2(Rm^2 - hm^2) / (L^2)) / (1 + (4hm^2) / (3L)) - 4a / L) \\
 &= 228533 \cdot 12688 / 4 \cdot ((1 + 2(971^2 - 0^2) / (12688^2)) / (1 + (4 \cdot 0) / (3 \cdot 12688)) - 4 \cdot 2535 / 12688) \\
 &= 154078.0 \text{ Kg-m.}
 \end{aligned}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -51.704 \cdot 971.5 / (2 \cdot 13.0) - 154078.0 / (\pi \cdot 971.5^2 \cdot 13.0) \\
 &= -41.13 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t) \\
 &= -51.704 \cdot 971.5 / (2 \cdot 13.0) + 154078.0 / (\pi \cdot 971.5^2 \cdot 13.0) \\
 &= 37.27 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) - M1 / (\pi \cdot Rm^2 \cdot t) \\
 &= -51.704 \cdot 971.5 / (2 \cdot 15.0) - 107249.6 / (\pi \cdot 971.5^2 \cdot 15.0) \\
 &= 21.97 \text{ N./mm}^2
 \end{aligned}$$

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

$$\begin{aligned}
 &= P \cdot Rm / (2t) + M1 / (\pi \cdot Rm^2 \cdot t) \\
 &= -51.704 \cdot 971.5 / (2 \cdot 15.0) + 107249.6 / (\pi \cdot 971.5^2 \cdot 15.0) \\
 &= -25.32 \text{ N./mm}^2
 \end{aligned}$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$\begin{aligned}
 &= Q(L - 2a) / (L + (4 \cdot hm / 3)) \\
 &= 228533(12688.0 - 2 \cdot 2535.0) / (12688.0 + (4 \cdot 0.0 / 3)) \\
 &= 137213.2 \text{ Kg}
 \end{aligned}$$

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

$$\begin{aligned}
 &= T / (\pi \cdot Rm \cdot t) \\
 &= 137213.16 / (\pi \cdot 971.5 \cdot 15.0) \\
 &= 29.39 \text{ N./mm}^2
 \end{aligned}$$

Decay Length (4.15.20) [x_1, x_2]:

$$\begin{aligned}
 &= 0.78 \cdot \sqrt{(Rm \cdot t)} \\
 &= 0.78 \cdot \sqrt{(971.5 \cdot 15.0)} \\
 &= 94.159 \text{ mm}
 \end{aligned}$$

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

$$\begin{aligned}
 &= -K5 \cdot Q \cdot k / A \\
 &= -0.643 \cdot 228533 \cdot 0.1 / 114.05 \\
 &= -12.64 \text{ N./mm}^2
 \end{aligned}$$

Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:

$$\begin{aligned}
 &= -K8 \cdot Q / A + K6 \cdot Q \cdot Rm \cdot c1 / I \\
 &= -0.2756 \cdot 228533 / 114.05 + 0.0233 \cdot 228533 \cdot 971.5 \cdot 109.44 / 9755.48 \\
 &= 2.80 \text{ N./mm}^2
 \end{aligned}$$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:

$$\begin{aligned}
 &= -K8 \cdot Q / A - K6 \cdot Q \cdot Rm \cdot c2 / I \\
 &= -0.2756 \cdot 228533 / 114.05 - 0.0233 \cdot 228533 \cdot 971.5 \cdot 176.56 / 9755.48 \\
 &= -146.05 \text{ N./mm}^2
 \end{aligned}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	58.0000	mm
Baseplate Width	Bpwid	540.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	



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Saddle Design Temperature		21.1 °C
Saddle Yield Stress	Sy	230.4 N./mm ²
Height of Web at Center	Hw, c	620.0 mm
Friction Coefficient	mu	0.300

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3) / 12$$

$$= ((1.56 \cdot \sqrt{(964.0 \cdot 15.0)} + 800.0)^3) / 12$$

$$= 27.776 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 27.776 + 148.138(315.792 - 7.5)^2$$

$$= 140823.938 \text{ cm}^4$$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ² (2) cm ⁴
Shell	987.6	15.0	7.5	148.1	111103.8	0.141E+06
Wearplate	800.0	16.0	23.0	128.0	294400.0	0.110E+06
Web	16.0	527.0	294.5	84.3	2483224.2	0.199E+05
BasePlate	540.0	58.0	587.0	313.2	18384840.0	0.231E+06
Totals	...	...	...	673.7	21273568.0	0.502E+06

Distance to Centroid [C1]:

$$= AY / A$$

$$= 21273568 / 673.658$$

$$= 315.792 \text{ mm}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 166.0 / 2$$

$$= 97.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(97.0) - 0.5 \cdot \sin(97.0)^2) / (\pi - 1.693 + \sin(97.0) \cos(97.0))$$

$$= 0.2904$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.29 \cdot 228532.516$$

$$= 66366.4609 \text{ Kg}$$

Tension Stress, St = (Fh/As)	=	12.3848	N./mm ²
Allowed Stress, Sa = 0.6 * Yield Str	=	138.2389	N./mm ²

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1580.0 - 964.0 \cdot \sin(166.0/2) / 1.4486$$

$$= 919.501 \text{ mm}$$

Bending Moment, M = Fh * d	=	61025.2227	Kg-m.
----------------------------	---	------------	-------

Bending Stress, Sb = (M * C1 / I)	=	37.6670	N./mm ²
Allowed Stress, Sa = 2/3 * Yield Str	=	153.5988	N./mm ²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(228533 + 1208) \cdot 540.0) / (4 \cdot 1850.0 \cdot 153.599)}$$

$$= 56.667 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:



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Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 166.0/2) (963.0 + 16.0 + 16.0)$$

$$= 1975.167 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1975.1669 / (4 - 1)$$

$$= 658.389 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 658.3889 \cdot 540.0 / 2$$

$$= 1777.650 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 229740.016 / (1850.0 \cdot 540.0)$$

$$= 22.997 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= \text{Ap} \cdot \text{Bp}$$

$$= 1777.7 \cdot 23.0$$

$$= 40880.617 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 79.84 + 52.671$$

$$= 132.511 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / \text{Ar}$$

$$= 40880.6 / 132.5111$$

$$= 30.255 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	515.0	...	82.4	...	0.182E+05

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 515.0 - 16.0$$

$$= 499.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= \text{AY} / \text{A}$$

$$= 0.0 / 132.511$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 515.0 / 2$$

$$= 257.500 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(18212.1 / 132.511)}$$

$$= 117.234 \text{ mm}$$

Length of Outer Rib, step 1, [Lv]:



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$$\begin{aligned}
 &= \text{Saddle Height} - \cos(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk} \\
 &= 1580.0 - \cos(166.0/2)(963.0 + 16.0 + 16.0) - 58.0 \\
 &= 1400.740 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Length of Outer Rib, step 2, [Lw]:} \\
 &= 2 \cdot \sin(\text{saddle bearing angle} / 2)(\text{radius} + \text{shlthk} + \text{wpdthk}) \\
 &= 2 \cdot \sin(166.0/2)(963.0 + 16.0 + 16.0) \\
 &= 1975.167 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Length of Outer Rib, step 3, [Bd]:} \\
 &= \text{Baseplate length} - \text{clearance} \\
 &= 1850.0 - 50.8 \\
 &= 1799.200 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Length of Outer Rib, step 4, [Dd]:} \\
 &= (\text{Lw} - \text{Bd})/2 \\
 &= (1975.167 - 1799.2)/2 \\
 &= 87.983 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Length of Outer Rib [L]:} \\
 &= \sqrt{(\text{Lv}^2 + \text{Dd}^2)} \\
 &= \sqrt{(1400.74^2 + 87.983^2)} \\
 &= 1403.500 \text{ mm}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Intermediate Term [Cc]:} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})} \\
 &= \sqrt{(2 \cdot \pi^2 \cdot 0.19994\text{E}+09/230.4)} \\
 &= 130.885
 \end{aligned}$$

$$\begin{aligned}
 &\text{Slenderness ratio [KL/r]:} \\
 &= \text{KL/r} \\
 &= 1 \cdot 1403.5/117.234 \\
 &= 11.972
 \end{aligned}$$

$$\begin{aligned}
 &\text{Bending Moment [Rm]:} \\
 &= \text{Fl} / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2 \\
 &= 14371.9 / (2 \cdot 1850.0) \cdot 658.389 \cdot 1403.5/2 \\
 &= 1794.680 \text{ Kg-m.}
 \end{aligned}$$

$$\begin{aligned}
 &\text{Compressive Allowable, KL/r < Cc (11.9718 < 130.8854) per AISC E2-1 [Sca]:} \\
 &= (1 - (\text{KL/r})^2 / (2 \cdot \text{Cc}^2)) \text{Fy} / (5/3 + 3 \cdot (\text{KL/r}) / (8 \cdot \text{Cc}) - (\text{KL/r})^3 / (8 \cdot \text{Cc}^3)) \\
 &= (1 - (11.97)^2 / (2 \cdot 130.89^2)) 230 / \\
 &\quad (5/3 + 3 \cdot (11.97) / (8 \cdot 130.89) - (11.97^3) / (8 \cdot 130.89^3)) \\
 &= 134.9 \text{ N./mm}^2
 \end{aligned}$$

$$\begin{aligned}
 &\text{AISC Unity Check of Outside Ribs (must be } \leq 1) \\
 &= \text{Sc/Sca} + (\text{Rm} \cdot \text{C1} / \text{I}) / \text{Sba} \\
 &= 30.25/134.89 + (1794.68 \cdot 257.5 / 1.8212\text{E}+09) / 153.6 \\
 &= 0.386
 \end{aligned}$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	499.0	0.0	79.8	0.0	0.182E+05
Web	658.4	16.0	0.0	105.3	0.0	22.5
Totals	...	...	...	185.2	...	0.182E+05

$$\begin{aligned}
 &\text{Distance to Centroid from Datum [ytot]:} \\
 &= \text{AY} / \text{A} \\
 &= 0.0/185.182 \\
 &= 0.000 \text{ mm}
 \end{aligned}$$

Distance to Centroid [C1]:



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= Saddle Width / 2

= 515.0/2

= 257.500 mm

Length of Inner Rib [L]:

= Saddle Height - $\sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2}$ - Bpthk

= 1580.0 - $\sqrt{(995.0 + 16.0)^2 - (658.389/2)^2}$ - 58.0

= 583.035 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{18234.0/185.182}$

= 99.230 mm

Slenderness ratio [KL/r]:

= KL/r

= 1 • 583.035/99.23

= 5.876

Unit Force [Force,u]:

= FI / (2 • Baseplate Length)

= 14371.931/(2 • 1850.0)

= 3.884 Kg/mm

Moment at base of inner Rib [Mbase,c]:

= Unit Force • e • L

= 3.884 • 658.389 • 583.035

= 1491.073 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

= Bending Moment / Section Modulus

= 1491.073/708118

= 20.650 N./mm²

Compressive Allowable, KL/r < Cc (5.8756 < 130.8854) per AISC E2-1 [Sca]:

= (1 - (KL/r)² / (2 • Cc²)) Fy / (5/3 + 3 • (KL/r) / (8 • Cc) - (KL/r)³ / (8 • Cc³))

= (1 - (5.88)² / (2 • 130.89²)) 230 /

(5/3 + 3 • (5.88) / (8 • 130.89) - (5.88³) / (8 • 130.89³))

= 136.7 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)

= Sc/Sca + (Mbase,c • C1/I) / Sba

= 43.31/136.72 + (1491.07 • 257.5/18234.025) / 153.6

= 0.451

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	971.50	mm
Shell Thickness used in this Case	t	16.000	mm
Stiffened Vessel Length per 4.15.6	L	12688.00	mm
Distance from Saddle to Vessel tangent	a1 or a	2535.00	mm
Saddle Width	b1 or b	515.00	mm
Saddle Bearing Angle	delta or theta	166.00	degrees
Wear Plate Width	b2 or b1	800.00	mm
Wear Plate Bearing Angle	delta2 or theta1	180.00	degrees
Wear Plate Thickness	e2 or tr	16.0	mm
Wear Plate Allowable Stress	fw or Sr	138.00	N./mm ²
Shell Allowable Stress used in Calculation		138.00	N./mm ²
Head Allowable Stress used in Calculation		88.10	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	



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Circumferential Efficiency at Mid-Span 1.00
 Distance from Saddle Base to Centerline B 1580.00 mm
 Coefficient of Friction mu 0.00

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 171986.16 KgF
 Pressure used in Analysis P -51.704 KPa.

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-31.43	-195.48
Long. Stress at Top of Midspan	31.43	138.00
Long. Stress at Bottom of Midspan	27.57	138.00
Long. Stress at Top of Saddles	16.12	138.00
Long. Stress at Bottom of Saddles	-19.47	-207.36
Long. Stress at Bottom of Saddles	19.47	138.00
Tangential Shear in Shell	22.12	110.40
Circ. Compressive Stress in Shell	9.52	138.00
Stiffener Circ. Stress at Shell	2.11	172.50
Stiffener Circ. Stress at Tip	109.91	292.49

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 = $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 = $3.0 (6053.9/2 + 8630) \cdot 1580.0/1850.0$
 = 29868.1 KgF

Longitudinal Saddle Reaction Force [Fwl]:
 = $\max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 = $\max(452.6, 11400) \cdot 1580.0/7618.0015$
 = 2177.7 KgF

Saddle Reaction Force due to Earthquake F_l [Fsl]:
 = $\max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 = $\max(14371.93, 11400) \cdot 1580.0/7618.0015$
 = 2980.8 KgF

Saddle Reaction Force due to Earthquake F_t [Fst]:
 = $F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 = $3.0 (14372/2 + 8630) \cdot 1580.0/1850.0$
 = 40524.1 KgF

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + $\max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 = $131462 + \max(2178, 29868, 2981, 40524)$
 = 171986.2 KgF

Longitudinal Wind Force [Fl]:
 = $\text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 = $1.1 \cdot 766.08 (0.0 + \pi/4 (2.158 \cdot 1.2)^2)$
 = 4438.388 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 173193.67 KgF
 Transverse Shear Load 15816.38 KgF
 Longitudinal Shear Load 14371.93 KgF

Note: Wear Plate is Welded to the Shell, k = 0.1



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Saddle Dimension [E]:

= Baseplate Length

= 1850.000 mm

The Computed K values from Table 4.15.1:

K1 = 0.1951 K2 = 0.6681 K3 = 0.3500 K4 = 0.2504

K5 = 0.6435 K6 = 0.0233 K7 = 0.0233 K8 = 0.2756

K9 = 0.1812 K10 = 0.0235 K1* = 0.3314 K6p = 0.0174

K7p = 0.0174

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= $-Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$

= $-171986 \cdot 2535.0 [1 - (1 - 2535.0/12688.0 + (971.5^2 - 0.0^2)/$

$(2 \cdot 2535.0 \cdot 12688.0)/(1 + (4 \cdot 0.0)/(3 \cdot 12688.0))]$

= -80712.6 Kg-m.

Moment per Equation 4.15.2 [M2]:

= $Q \cdot L/4 ((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$

= $171986 \cdot 12688/4 ((1 + 2(971.5^2 - 0^2)/(12688^2))/(1 + (4 \cdot 0)/$

$(3 \cdot 12688)) - 4 \cdot 2535/12688)$

= 115954.1 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= $P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$

= $-51.704 \cdot 971.5/(2 \cdot 13.0) - 115954.1/(\pi \cdot 971.5^2 \cdot 13.0)$

= -31.43 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= $P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$

= $-51.704 \cdot 971.5/(2 \cdot 13.0) + 115954.1/(\pi \cdot 971.5^2 \cdot 13.0)$

= 27.57 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.6) [σ_3]:

= $P \cdot Rm/(2t) - M1/(\pi \cdot Rm^2 \cdot t)$

= $-51.704 \cdot 971.5/(2 \cdot 15.0) - 80712.6/(\pi \cdot 971.5^2 \cdot 15.0)$

= 16.12 N./mm²

Longitudinal Stress at bottom of Shell at Support (4.15.7) [σ_4]:

= $P \cdot Rm/(2t) + M1/(\pi \cdot Rm^2 \cdot t)$

= $-51.704 \cdot 971.5/(2 \cdot 15.0) + 80712.6/(\pi \cdot 971.5^2 \cdot 15.0)$

= -19.47 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= $Q(L - 2a)/(L + (4 \cdot hm/3))$

= $171986(12688.0 - 2 \cdot 2535.0)/(12688.0 + (4 \cdot 0.0/3))$

= 103262.2 Kgf

Shear Stress in the shell with a single ring (4.15.13) [τ_1]:

= $T/(\pi \cdot Rm \cdot t)$

= $103262.18/(\pi \cdot 971.5 \cdot 15.0)$

= 22.12 N./mm²

Decay Length (4.15.20) [x1,x2]:

= $0.78 \cdot \sqrt{Rm \cdot t}$

= $0.78 \cdot \sqrt{971.5 \cdot 15.0}$

= 94.159 mm

Circ. Stress in shell w/ring in Plane of Saddle (4.15.32) [σ_6^*]:

= $-K5 \cdot Q \cdot k/A$

= $-0.643 \cdot 171986 \cdot 0.1/114.05$

= -9.52 N./mm²



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Note: Single Ring in Plane of Saddle Outside the Shell.

Circ. + Bending Stress in shell w/ring in Plane of Saddle (4.15.35) [σ_8^*]:
 $= -K_8 \cdot Q/A + K_6 \cdot Q \cdot R_m \cdot c_1/I$
 $= -0.2756 \cdot 171986/114.05 + 0.0233 \cdot 171986 \cdot 971.5 \cdot 109.44/9755.48$
 $= 2.11 \text{ N./mm}^2$

Circ. + Bending Stress in ring w/ring in Plane of Saddle (4.15.36) [σ_9^*]:
 $= -K_8 \cdot Q/A - K_6 \cdot Q \cdot R_m \cdot c_2/I$
 $= -0.2756 \cdot 171986/114.05 - 0.0233 \cdot 171986 \cdot 971.5 \cdot 176.56/9755.48$
 $= -109.91 \text{ N./mm}^2$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	1850.0000	mm
Baseplate Thickness	Bpthk	58.0000	mm
Baseplate Width	Bpwid	540.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	4	
Rib Thickness	Ribtk	16.0000	mm
Web Thickness	Webtk	16.0000	mm
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	230.4	N./mm ²
Height of Web at Center	Hw,c	620.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:
 $= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})^3)/12$
 $= ((1.56 \cdot \sqrt{(964.0 \cdot 15.0)} + 800.0)^3)/12$
 $= 27.776 \text{ cm}^4$

Resolved Inertia for the Shell [I,shell]:
 $= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$
 $= 27.776 + 148.138(315.792 - 7.5)^2$
 $= 140823.938 \text{ cm}^4$

	B mm	D mm	Y mm	A cm ²	AY mm ³	I + AD ⁽²⁾ cm ⁴
Shell	987.6	15.0	7.5	148.1	111103.8	0.141E+06
Wearplate	800.0	16.0	23.0	128.0	294400.0	0.110E+06
Web	16.0	527.0	294.5	84.3	2483224.2	0.199E+05
BasePlate	540.0	58.0	587.0	313.2	18384840.0	0.231E+06
Totals	...	...	...	673.7	21273568.0	0.502E+06

Distance to Centroid [C1]:
 $= AY / A$
 $= 21273568/673.658$
 $= 315.792 \text{ mm}$

Angle [β]:
 $= 180 - \text{Saddle Angle}/2$
 $= 180 - 166.0/2$
 $= 97.0$

Saddle Splitting Coefficient [K1]:
 $= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 $= (1 + \cos(97.0) - 0.5 \cdot \sin(97.0)^2) / (\pi - 1.693 + \sin(97.0) \cos(97.0))$
 $= 0.2904$

Saddle Splitting Force [Fh]:
 $= K_1 \cdot Q$



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Saddle Calcs: Ext. Pressure:

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$$= 0.29 \cdot 171986.156$$

$$= 49945.2500 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh/As) = 9.3204 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 138.2389 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 1580.0 - 964.0 \cdot \sin(166.0/2) / 1.4486$$

$$= 919.501 \text{ mm}$$

$$\text{Bending Moment, } M = Fh \cdot d = 45925.6055 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 28.3470 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 153.5988 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(171986 + 1208) \cdot 540.0) / (4 \cdot 1850.0 \cdot 153.599)}$$

$$= 49.201 \text{ mm}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 166.0/2) (963.0 + 16.0 + 16.0)$$

$$= 1975.167 \text{ mm}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 1975.1669 / (4 - 1)$$

$$= 658.389 \text{ mm}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 658.3889 \cdot 540.0 / 2$$

$$= 1777.650 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 173193.672 / (1850.0 \cdot 540.0)$$

$$= 17.337 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 1777.7 \cdot 17.34$$

$$= 30818.594 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 79.84 + 52.671$$

$$= 132.511 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 30818.6 / 132.5111$$

$$= 22.808 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	16.0	515.0	...	82.4	...	0.182E+05



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Saddle Calcs: Ext. Pressure:

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Rib dimension [D]:

= Saddle Width - Web Thickness

= 515.0 - 16.0

= 499.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A

= 0.0/132.511

= 0.000 mm

Distance to Centroid [C1]:

= Saddle Width / 2

= 515.0/2

= 257.500 mm

Radius of Gyration [r]:

= $\sqrt{\text{Total Inertia} / \text{Total Area}}$

= $\sqrt{18212.1/132.511}$

= 117.234 mm

Length of Outer Rib, step 1, [Lv]:

= Saddle Height - $\cos(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$

= 1580.0 - $\cos(166.0/2) (963.0 + 16.0 + 16.0) - 58.0$

= 1400.740 mm

Length of Outer Rib, step 2, [Lw]:

= $2 \cdot \sin(\text{saddle bearing angle} / 2) (\text{radius} + \text{shlthk} + \text{wpdthk})$

= $2 \cdot \sin(166.0/2) (963.0 + 16.0 + 16.0)$

= 1975.167 mm

Length of Outer Rib, step 3, [Bd]:

= Baseplate length - clearance

= 1850.0 - 50.8

= 1799.200 mm

Length of Outer Rib, step 4, [Dd]:

= $(Lw - Bd) / 2$

= $(1975.167 - 1799.2) / 2$

= 87.983 mm

Length of Outer Rib [L]:

= $\sqrt{Lv^2 + Dd^2}$

= $\sqrt{1400.74^2 + 87.983^2}$

= 1403.500 mm

Intermediate Term [Cc]:

= $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$

= $\sqrt{2 \cdot \pi^2 \cdot 0.19994\text{E}+09 / 230.4}$

= 130.885

Slenderness ratio [KL/r]:

= KL/r

= $1 \cdot 1403.5 / 117.234$

= 11.972

Bending Moment [Rm]:

= $Fl / (2 \cdot Bplen) \cdot e \cdot L / 2$

= $14371.9 / (2 \cdot 1850.0) \cdot 658.389 \cdot 1403.5 / 2$

= 1794.680 Kg-m.

Compressive Allowable, $KL/r < Cc$ ($11.9718 < 130.8854$) per AISC E2-1 [Sca]:

= $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$

= $(1 - (11.97)^2 / (2 \cdot 130.89^2)) 230 /$

$(5/3 + 3 \cdot (11.97) / (8 \cdot 130.89) - (11.97^3) / (8 \cdot 130.89^3))$



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Saddle Calcs: Ext. Pressure:

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$$= 134.9 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 22.81/134.89 + (1794.68 \cdot 257.5 / 18212E+09) / 153.6$$

$$= 0.331$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	16.0	499.0	0.0	79.8	0.0	0.182E+05
Web	658.4	16.0	0.0	105.3	0.0	22.5
Totals	...	...	...	185.2	...	0.182E+05

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0.0/185.182$$

$$= 0.000 \text{ mm}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 515.0/2$$

$$= 257.500 \text{ mm}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \sqrt{(Ro + Wpdthk)^2 - (Pitch/2)^2} - Bpthk$$

$$= 1580.0 - \sqrt{(995.0 + 16.0)^2 - (658.389/2)^2} - 58.0$$

$$= 583.035 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(18234.0/185.182)}$$

$$= 99.230 \text{ mm}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 583.035/99.23$$

$$= 5.876$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 14371.931 / (2 \cdot 1850.0)$$

$$= 3.884 \text{ Kg/mm}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 3.884 \cdot 658.389 \cdot 583.035$$

$$= 1491.073 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [$\sigma B, \text{base}, c$]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 1491.073/708118$$

$$= 20.650 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < Cc$ (5.8756 < 130.8854) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (5.88)^2 / (2 \cdot 130.89^2)) 230 /$$

$$(5/3 + 3 \cdot (5.88) / (8 \cdot 130.89) - (5.88^3) / (8 \cdot 130.89^3))$$

$$= 136.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (Mbase, c \cdot C1 / I) / Sba$$

$$= 32.65/136.72 + (1491.07 \cdot 257.5 / 18234.025) / 153.6$$



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Saddle Calcs: Ext. Pressure:

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= 0.373

Review notes about nozzle loadings and supports in the Warnings report.

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Nozzle Calcs.: N1-Tube Inlet

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Input, Nozzle Desc: N1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	520.000	KPa.
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	51.70	KPa.
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-240 304L	
Parent Allowable Stress at Temperature	Sv	115.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	115.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1920.00	mm
Design Length of Section	L	1499.9696	mm
Shell Finished (Minimum) Thickness	t	22.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		1085.18	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-182 F304L	
Material UNS Number		S30403	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	88.10	N./mm ²
Allowable Stress At Ambient	Sna	115.00	N./mm ²
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	12.7000	mm
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	708.0000	mm
Thickness of Pad	te	26.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	20.0000	mm
Reinforcing Pad Width		100.0000	mm
Flange Class		150	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.



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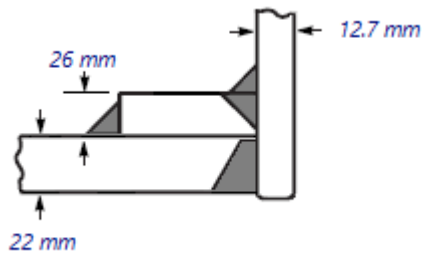
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Nozzle Calcs.: N1-Tube Inlet

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Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	20.000 in.
Actual Thickness Used in Calculation	0.500 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 520.0 \cdot 961.0 / (115 \cdot 1.0 - 0.6 \cdot 520.0)$
 $= 4.3575 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 520.0 \cdot 254.0 / (88 \cdot 1.0 + 0.4 \cdot 520.0)$
 $= 1.4957 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0325 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	977.2000 mm
Parallel to Vessel Wall, opening length	d	488.6000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		50.2500 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.1 / 115.0)$
 $= 0.766$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 88.1 / 115.0)$
 $= 0.766$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 115.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.766, 1.0)$
 $= 0.766$



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Nozzle Calcs.: N1-Tube Inlet

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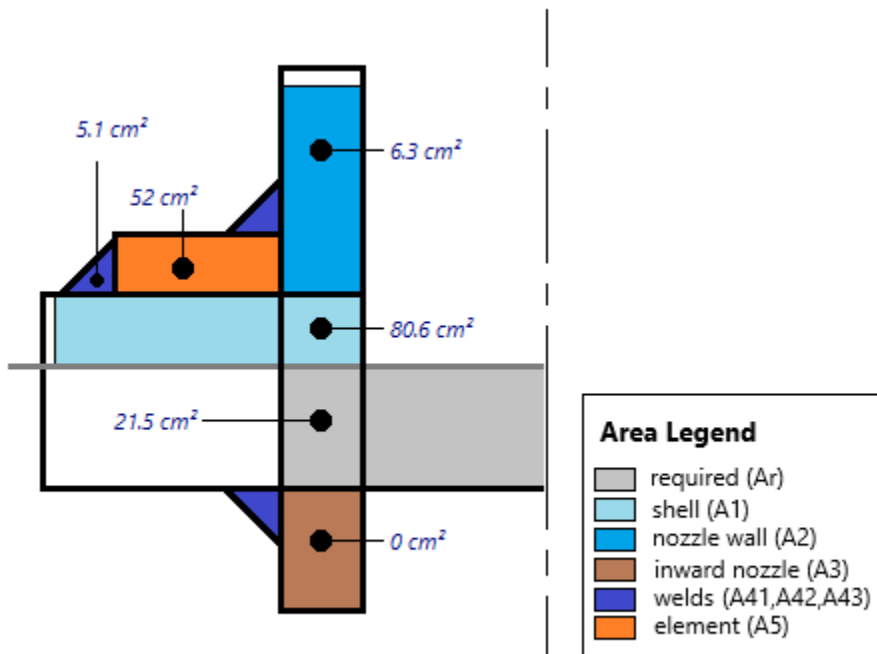


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	21.489	10.856	NA
Area in Shell	A1	80.560	80.341	NA
Area in Nozzle Wall	A2	6.317	6.673	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	5.103	5.103	NA
Area in Element	A5	52.000	52.000	NA
TOTAL AREA AVAILABLE	Atot	143.980	144.118	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (488.6 \cdot 4.3575 \cdot 1.0 + 2 \cdot 9.7 \cdot 4.3575 \cdot 1.0 \cdot (1 - 0.77)) \\
 &= 21.489 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 488.6(1.0 \cdot 21.0 - 1.0 \cdot 4.358) - 2 \cdot 9.7 \\
 &\quad (1.0 \cdot 21.0 - 1.0 \cdot 4.3575) \cdot (1 - 0.766) \\
 &= 80.560 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 50.25) \cdot (9.7 - 1.5) \cdot 0.7661
 \end{aligned}$$



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Nozzle Calcs.: N1-Tube Inlet

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= 6.317 cm²

Area Available in Welds [A41 + A42 + A43]:

= $W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$

= $12.0^2 \cdot 0.77 + (0.0)^2 \cdot 0.77 + 20.0^2 \cdot 1.0$

= 5.103 cm²

Area Available in Element [A5]:

= $(\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$

= $(708.0 - 508.0) \cdot 26.0 \cdot 1.0$

= 52.000 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.4957$ mm

Wall Thickness per UG16(b), $tr_{16b} = 4.5000$ mm

Wall Thickness, shell/head, internal pressure $trb_1 = 5.3575$ mm

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 5.3575$ mm

Wall Thickness, shell/head, external pressure $trb_2 = 1.4322$ mm

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000$ mm

Wall Thickness per table UG-45 $tb_3 = 11.3312$ mm

Determine Nozzle Thickness candidate [tb]:

= $\min[tb_3, \max(tb_1, tb_2)]$

= $\min[11.331, \max(5.3575, 4.5)]$

= 5.3575 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= $\max(t_a, t_b)$

= $\max(4.4957, 5.3575)$

= 5.3575 mm

Available Nozzle Neck Thickness = 12.7000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	30.6,	Allowable	:	88.1 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	223.3 N./mm ²	Passed
Occasional	:	6.4,	Allowable	:	117.2 N./mm ²	Passed
Shear	:	18.5,	Allowable	:	61.7 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle to Flange Weld skipped as Nozzle is not a Carbon Steel material.

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 12.7$, $t_r = 1.147$, $c = 3.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.118$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -21 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 22.0$, $t_r = 4.358$, $c = 1.0$ mm, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.208$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -5 °C



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Nozzle Calcs.: N1-Tube Inlet

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Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 9.7000 mm
Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.7900 = 0.7 * tmin.	8.4840 = 0.7 * Wo mm
Pad Weld	9.5000 = 0.5 * TminPad	14.1400 = 0.7 * Wp mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2*tn*fr1*(E1*t-tr))Sv)
= max(0, (21.4887 - 80.5599 + 2 * 9.7 * 0.7661 *
(1.0 * 21.0 - 4.3575))115)
= max(0, -66368.23) Kgf

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A5+A4-(Wi-Can/.707)^2*fr2)*Sv
= (6.3168 + 52.0 + 5.1032 - 0.0 * 0.77) * 115
= 74368.19 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 * tn * t * fr1)) * Sv
= (6.3168 + 0.0 + 1.1032 + (3.1211)) * 115
= 12360.78 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2*tn*t*fr1))*S
= (6.3168 + 0.0 + 5.1032 + 52.0 + (3.1211)) * 115
= 78028.11 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) * Dlo * Wo * 0.49 * Snw
= (3.1416/2.0) * 508.0 * 12.0 * 0.49 * 88
= 42151. Kgf

Shear, Pad Element Weld [Spew]:
= (π/2) * DP * WP * 0.49 * SEW
= (3.1416/2.0) * 708.0 * 20.0 * 0.49 * 115
= 127803. Kgf

Shear, Nozzle Wall [Snw]:
= (π * (Dlr + Dlo)/4) * (Thk - Can) * 0.7 * Sn
= (3.1416 * 249.15) * (12.7 - 3.0) * 0.7 * 88
= 47745. Kgf

Tension, Pad Groove Weld [Tpgw]:
= (π/2) * Dlo * Wgpn * 0.74 * Seg
= (3.1416/2) * 508.0 * 20.0 * 0.74 * 138
= 166183. Kgf



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 Nozzle Calcs.: N1-Tube Inlet Nozl: 9 6:34pm Sep 8,2026

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$
 $= (3.1416/2.0) \cdot 508.0 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 115$
 $= 96940. \text{ Kg}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SPEW} + \text{SNW}) = (127803 + 47745) = 175548 \text{ Kg}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (42151 + 166183 + 96940 + 0) = 305275 \text{ Kg}$

Failure Path [Path 3-3]:
 $= (\text{Spew} + \text{Tngw} + \text{Sinw})$
 $= (127803 + 96940 + 0) = 224743 \text{ Kg}$

Summary of Failure Path Calculations:

Path 1-1 = 175548 Kg, must exceed W = 0 Kg	or W1 = 74368 Kg
Path 2-2 = 305275 Kg, must exceed W = 0 Kg	or W2 = 12360 Kg
Path 3-3 = 224743 Kg, must exceed W = 0 Kg	or W3 = 78028 Kg

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1293 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 KPa.

The Drop for this Nozzle is : 34.2117 mm
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 356.2117 mm

This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 488.60 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1920.000	mm
Shell Thickness	22.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-240 304L	
Vessel Cold S.I. Allowable	Smc 115.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 115.00	N./mm ²
Diameter Basis for Nozzle	OD	
Nozzle Corrosion Allowance	3.0000	mm
Nozzle Diameter	508.000	mm
Nozzle Thickness	12.7000	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-182 F304L	
Nozzle Cold S.I. Allowable	SNmc 115.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 88.10	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	708.000	mm
Thickness of Reinforcing Pad	26.0000	mm



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Nozzle Calcs.: N1-Tube Inlet

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Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUSTAINED loads.

Design Internal Pressure	Dp	520.00	KPa.
Radial Load	P	-3824.19	Kgf
Circumferential Shear	Vc	2549.46	Kgf
Longitudinal Shear	Vl	2549.46	Kgf
Circumferential Moment	Mc	3567.87	Kg-m.
Longitudinal Moment	Ml	3567.87	Kg-m.
Torsional Moment	Mt	5096.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{984.5(22.0 - 1.0)}$$

$$= 982.495 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1969.000 mm
Nozzle Outside Diameter	(d) =	508.000 mm
Vessel Thickness used	(T) =	47.000 mm
Nozzle Thickness used	(t) =	9.700 mm
T / t	=	4.845
d / t	=	52.371

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.670$$

Nr/P	=	0.033
Mr/P	=	0.032
M0/P	=	0.067
N0/P	=	0.261
MrD/Mc	=	0.079
NrDL/Mc	=	0.072
M0d/Mc	=	0.209
N0DL/Mc	=	0.397
MrD/Ml	=	0.055
NrDL/Ml	=	0.052
M0D/Ml	=	0.114
N0DL/Ml	=	0.579

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-18	-18	18	18
Outplane Bending (Ml)	-21	21	21	-21
Normal Pressure Stress	10	10	10	10
Outplane Stress Summary	-17	11	60	5

Vessel Stresses (N./mm²):



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Nozzle Calcs.: N1-Tube Inlet

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LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-10	10	10	-10
Inplane Pressure Stress	5	5	5	5
Inplane Stress Summary	-2	11	20	-6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress	18	14	61	12

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-12	-12	12	12
Outplane Bending (Mc)	-39	39	39	-39
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	5	5	5	5
Outplane Stress Summary	-34	29	67	-23

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	3	-3	3	-3
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-14	14	14	-14
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	10	10	10	10
Inplane Stress Summary	-2	20	31	-4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to	Cu Left	Cl Left	Du Right	D1 Right
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Nozzle Calcs.: N1-Tube Inlet

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circumferential plane)		Outside	Inside	Outside	Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2
Shear Stress Summary		1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stress Intensities		Left	Left	Right	Right
		Outside	Inside	Outside	Inside
Two * Max Shear Stress		35	29	68	24

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses in the		Top	Top	Bottom	Bottom
hoop direction)		Outside	Inside	Outside	Inside
Hoop Membrane (P)		4	4	4	4
Hoop Bending (P)		0	0	0	0
Hoop Membrane (Mc)		0	0	0	0
Hoop Bending (Mc)		0	0	0	0
Hoop Membrane (ML)		-18	-18	18	18
Hoop Bending (ML)		0	0	0	0
Hoop Pressure Stress		15	16	15	16
Hoop Stress Summary		1	2	37	38

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stresses Normal to		Top	Top	Bottom	Bottom
pipe cross-section)		Outside	Inside	Outside	Inside
Axial Membrane (P)		2	2	2	2
Axial Bending (P)		37	-37	37	-37
Axial Membrane (Mc)		0	0	0	0
Axial Bending (Mc)		0	0	0	0
Axial Membrane (ML)		-18	-18	18	18
Axial Bending (ML)		-127	127	127	-127
Axial Pressure Stress		7	7	7	7
Axial Stress Summary		-98	80	194	-136

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Shear stress)		Outside	Inside	Outside	Inside
Shear due to (Vc)		3	3	-3	-3
Shear due to (Vl)		0	0	0	0
Shear due to Torsion		12	12	12	12
Shear Stress Summary		15	15	9	9

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE		Au	Al	Bu	Bl
(Stress Intensities		Outside	Inside	Outside	Inside
Two * Max Shear Stress		104	84	195	175

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE		Cu	Cl	Du	Dl
(Stresses in the		Left	Left	Right	Right
hoop direction)		Outside	Inside	Outside	Inside



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Nozzle Calcs.: N1-Tube Inlet

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Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-12	-12	12	12
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	15	16	15	16

Hoop Stress Summary	7	8	32	32
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	2	2	2	2
Axial Bending (P)	37	-37	37	-37
Axial Membrane (Mc)	-18	-18	18	18
Axial Bending (Mc)	-188	188	188	-188
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	7	7	7	7

Axial Stress Summary	-159	141	255	-197
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-3	-3	3	3
Shear due to Torsion	12	12	12	12

Shear Stress Summary	9	9	15	15
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	167	142	256	232

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{971.5(22.0 - 1.0)}$$

$$= 979.352 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1943.000 mm
Nozzle Outside Diameter	(d) =	708.000 mm
Vessel Thickness used	(T) =	21.000 mm
Nozzle Thickness used	(t) =	109.700 mm
T / t	=	0.191
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 3.505$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

$$Nr/P = 0.053$$

$$Mr/P = 0.093$$



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Nozzle Calcs.: N1-Tube Inlet

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MØ/P = 0.028
 NØ/P = 0.055
 MrD/Mc = 0.260
 NrDL/Mc = 0.092
 MØd/Mc = 0.078
 NØDL/Mc = 0.083
 MrD/Ml = 0.073
 NrDL/Ml = 0.056
 MØD/Ml = 0.022
 NØDL/Ml = 0.063

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-7	-7	7	7
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	23	24	23	24
Outplane Stress Summary	20	21	64	6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-6	-6	6	6
Inplane Bending (ML)	-49	49	49	-49
Inplane Pressure Stress	11	11	11	11
Inplane Stress Summary	7	12	119	-74

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	22	23	119	81

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4



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Outplane Bending (P)	14	-14	14	-14
Outplane Membrane (Mc)	-9	-9	9	9
Outplane Bending (Mc)	-52	52	52	-52
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	11	11	11	11
Outplane Stress Summary	-30	45	92	-40

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Inplane Membrane (P)	4	4	4	4
Inplane Bending (P)	47	-47	47	-47
Inplane Membrane (Mc)	-10	-10	10	10
Inplane Bending (Mc)	-174	174	174	-174
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	23	24	23	24
Inplane Stress Summary	-109	145	260	-183

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	1	1	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Two * Max Shear Stress	109	145	260	183

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	10	10	10	10	10	10	10
Circ. Pl (SUS)	-14	-14	22	22	-2	-2	2	2
Circ. Q (SUS)	-15	15	27	-27	-11	11	17	-17
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-1	-1	1	1	-8	-8	16	16
Long. Q (SUS)	-7	7	13	-13	-33	33	45	-45
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
Pm+Pl (SUS)	8.0	8.0	32.0	32.0	11.0	11.0	21.0	21.0



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 Nozzle Calcs.: N1-Tube Inlet Nozl: 9 6:34pm Sep 8,2026

Pm+Pl+Q (Total)	19.2	13.0	59.1	12.6	36.1	30.4	66.1	24.2
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Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	115.00	Passed
Pm+Pl (SUS)	32.00	172.50	Passed
Pm+Pl+Q (TOTAL)	66.11	344.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	23	24	23	24	23	24	23	24
Circ. Pl (SUS)	-3	-3	11	11	-6	-6	14	14
Circ. Q (SUS)	0	0	28	-28	-127	127	221	-221
Long. Pm (SUS)	11	11	11	11	11	11	11	11
Long. Pl (SUS)	-2	-2	10	10	-5	-5	13	13
Long. Q (SUS)	-2	2	96	-96	-38	38	66	-66
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	23.0	24.0	23.0	24.0	23.0	24.0	23.0	24.0
Pm+Pl (SUS)	20.1	21.1	34.1	35.1	17.1	18.1	37.1	38.1
Pm+Pl+Q (Total)	21.1	22.4	117.1	82.1	110.1	145.0	258.1	183.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	24.00	115.00	Passed
Pm+Pl (SUS)	38.07	172.50	Passed
Pm+Pl+Q (TOTAL)	258.10	344.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	15	16	15	16	15	16	15	16
Circ. Pl (SUS)	-14	-14	22	22	-8	-8	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0



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Nozzle Calcs.: N1-Tube Inlet

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Long. Pm (SUS)	7	7	7	7	7	7	7	7
Long. Pl (SUS)	-16	-16	20	20	-16	-16	20	20
Long. Q (SUS)	-90	90	164	-164	-151	151	225	-225
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	3	3	-3	-3	-3	-3	3	3
Shear Q (SUS)	12	12	12	12	12	12	12	12
Pm (SUS)	15.0	16.0	15.0	16.0	15.0	16.0	15.0	16.0
Pm+Pl (SUS)	11.7	12.5	37.8	38.8	17.1	18.0	32.6	33.4
Pm+Pl+Q (Total)	104.4	84.5	191.5	175.9	168.0	142.6	253.0	231.9

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	16.00	88.10	Passed
Pm+Pl (SUS)	38.76	132.15	Passed
Pm+Pl+Q (TOTAL)	253.01	264.30	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N3-Shell Inlet

Nozl: 10

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Input, Nozzle Desc: N3-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	1310.000	KPa.
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.70	KPa.
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1926.00	mm
Design Length of Section	L	2535.0308	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		3142.40	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	948.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		220.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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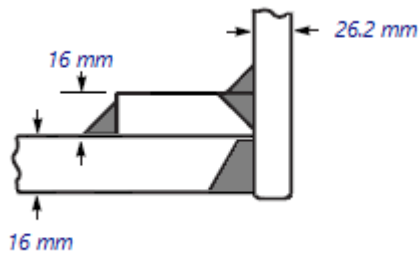
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Nozzle Calcs.: N3-Shell Inlet

Nozl: 10

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Reinforcement CALCULATION, Description: N3-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	17.938 in.
Actual Thickness Used in Calculation	1.031 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1310.0 \cdot 964.0 / (138 \cdot 1.0 - 0.6 \cdot 1310.0)$
 $= 9.2041 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1310.0 \cdot 228.81 / (118 \cdot 1.0 - 0.6 \cdot 1310.0)$
 $= 2.5574 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0009 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	915.2504 mm
Parallel to Vessel Wall, opening length	d	457.6252 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 138.0)$
 $= 0.855$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(0.855, 1.0)$
 $= 0.855$



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Nozzle Calcs.: N3-Shell Inlet

Nozl: 10

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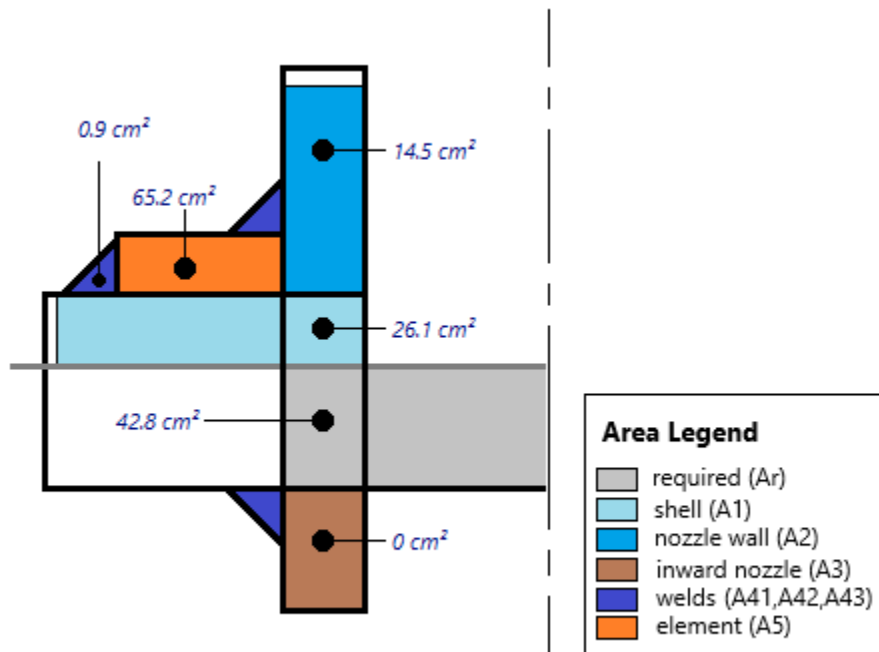


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	42.792	12.303	NA
Area in Shell	A1	26.100	43.715	NA
Area in Nozzle Wall	A2	14.513	15.511	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	0.855	0.855	NA
Area in Element	A5	65.160	65.160	NA
TOTAL AREA AVAILABLE	Atot	106.628	125.241	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Insufficient.

The area available with the given pad is Sufficient.

SELECTION OF POSSIBLE REINFORCING PADS: Diameter Thickness
 Based on given Pad Thickness: 516.2780 16.0000 mm
 Based on given Pad Diameter: 948.0001 0.3252 mm
 Based on Shell or Nozzle Thickness: 516.2780 16.0000 mm

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (457.6252 \cdot 9.2041 \cdot 1.0 + 2 \cdot 25.1874 \cdot 9.2041 \cdot 1.0 \cdot (1 - 0.86))$$

$$= 42.792 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 457.625(1.0 \cdot 15.0 - 1.0 \cdot 9.204) - 2 \cdot 25.187$$

$$(1.0 \cdot 15.0 - 1.0 \cdot 9.2041) \cdot (1 - 0.855)$$



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$$= 26.100 \text{ cm}^2$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$= (2 \cdot T_{lwp}) \cdot (t_n - t_{rn}) \cdot f_{r2}$$

$$= (2 \cdot 37.5) \cdot (25.19 - 2.56) \cdot 0.8551$$

$$= 14.513 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot f_{r3} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} + W_p^2 \cdot f_{r4}$$

$$= 10.0^2 \cdot 0.86 + (0.0)^2 \cdot 0.86 + 0.0^2 \cdot 1.0$$

$$= 0.855 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot f_{r4}$$

$$= (915.2504 - 508.0) \cdot 16.0 \cdot 1.0$$

$$= 65.160 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.5574 \text{ mm}$

Wall Thickness per UG16(b), $t_{r16b} = 3.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $t_{rb1} = 10.2041 \text{ mm}$

Wall Thickness $t_{b1} = \max(t_{rb1}, t_{r16b}) = 10.2041 \text{ mm}$

Wall Thickness, shell/head, external pressure $t_{rb2} = 1.3613 \text{ mm}$

Wall Thickness $t_{b2} = \max(t_{rb2}, t_{r16b}) = 3.5000 \text{ mm}$

Wall Thickness per table UG-45 $t_{b3} = 9.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[9.331, \max(10.2041, 3.5)]$$

$$= 9.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.5574, 9.3312)$$

$$= 9.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 26.187 = 22.914 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	15.9	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0	Allowable	:	279.1 N./mm ²	Passed
Occasional	:	5.6	Allowable	:	156.9 N./mm ²	Passed
Shear	:	8.0	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 22.914$, $t_r = 2.557$, $c = 1.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.117$, Temp. Reduction = 78°C

Min Metal Temp. w/o impact per UCS-66, Curve A 16°C

Min Metal Temp. at Required thickness (UCS 66.1) -104°C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16.0$, $c = 1.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.614$, Temp. Reduction = 21°C

Attachment governing, Attachment stress = Shell stress per Code



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Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 16.0$, $c = 1.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr * E^* / (t_g - c) = 0.614$, Temp. Reduction = 21 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 16.0$, $tr = 9.204$, $c = 1.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr * E^* / (t_g - c) = 0.614$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $t_g = 16.0$, $tr = 9.204$, $c = 1.0$ mm, $E^* = 1.0$

Thickness Ratio = $tr * E^* / (t_g - c) = 0.614$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-36 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-36 °C
Gov. MDMT of the Reinforcement Pad	:	-36 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-36 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-36 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1310.00 / 1960.01 = 0.668$

Weld Size Calculations, Description: N3-Shell Inlet

Intermediate Calc. for nozzle/shell Welds	T_{min}	16.0000 mm
Intermediate Calc. for pad/shell Welds	T_{minPad}	15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 * W_o \text{ mm}$
Pad Weld	$7.5000 = 0.5 * T_{minPad}$	$8.4840 = 0.7 * W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - tr)) S_v)$$

$$= \max(0, (42.7924 - 26.1002 + 2 \cdot 25.1874 \cdot 0.8551 \cdot (1.0 \cdot 15.0 - 9.2041)) 138)$$

$$= 27001.60 \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:



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$$= (A2 + A5 + A4 - (W_i - Can / .707)^2 \cdot fr2) \cdot Sv$$

$$= (14.5127 + 65.1601 + 0.8551 - 0.0 \cdot 0.86) \cdot 138$$

$$= 113315.30 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (14.5127 + 0.0 + 0.8551 + (6.4611)) \cdot 138$$

$$= 30716.66 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot tn \cdot t \cdot fr1)) \cdot S$$

$$= (14.5127 + 0.0 + 0.8551 + 65.1601 + (6.4611)) \cdot 138$$

$$= 122407.10 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 118$$

$$= 47046. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 948.0 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 123211. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 241.4063) \cdot (26.1874 - 1.0) \cdot 0.7 \cdot 118$$

$$= 160889. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 508.0 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 132947. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 116328. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (123211 + 160889) = 284099 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (47046 + 132947 + 116328 + 0) = 296322 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (123211 + 116328 + 0) = 239539 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 284099 Kgf, must exceed W = 27001 Kgf	or W1 = 113315 Kgf
Path 2-2 = 296321 Kgf, must exceed W = 27001 Kgf	or W2 = 30716 Kgf
Path 3-3 = 239539 Kgf, must exceed W = 27001 Kgf	or W3 = 122407 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1789 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.



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Nozzle Calcs.: N3-Shell Inlet

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Nozzle is O.K. for the External Pressure: 52 KPa.

The Drop for this Nozzle is : 34.1012 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 350.1012 mm

Input Echo, WRC297 Item 1, Description: N3-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1926.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	455.625	mm
Nozzle Thickness	26.1874	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-106 B	
Nozzle Cold S.I. Allowable	SNmc 118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	948.000	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	1310.00	KPa.
Radial Load	P	-3824.19	Kgf
Circumferential Shear	Vc	2549.46	Kgf
Longitudinal Shear	Vl	2549.46	Kgf
Circumferential Moment	Mc	3567.87	Kg-m.
Longitudinal Moment	Ml	3567.87	Kg-m.
Torsional Moment	Mt	5096.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{979.5(16.0 - 1.0)}$$

$$= 908.002 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1959.000	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	31.000	mm



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Nozzle Calcs.: N3-Shell Inlet

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Nozzle Thickness used (t) = 25.187 mm
 T / t = 1.231
 d / t = 20.169

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.061$$

Nr/P = 0.060
 Mr/P = 0.109
 M0/P = 0.042
 N0/P = 0.121
 MrD/Mc = 0.237
 NrDL/Mc = 0.107
 M0d/Mc = 0.100
 N0DL/Mc = 0.112
 MrD/Ml = 0.120
 NrDL/Ml = 0.062
 M0D/Ml = 0.051
 N0DL/Ml = 0.186

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	40	41	40	41
Outplane Stress Summary	19	44	89	27

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	-8	44	103	-50

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
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Two * Max Shear Stress	29	49	104	78
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-43	43	43	-43
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20
Outplane Stress Summary	-16	50	85	-20

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-101	101	101	-101
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	41	40	41
Inplane Stress Summary	-41	112	177	-75

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu	Cl	Du	Dl
	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	42	112	177	76

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au	Al	Bu	B1
	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	15	13	15
Hoop Stress Summary	4	6	31	33



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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au	Al	Bu	Bl
	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-57	57	57	-57
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-30	29	101	-70

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	6	6	3	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au	Al	Bu	Bl
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	37	31	101	103

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	15	13	15
Hoop Stress Summary	10	11	26	27

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu	Cl	Du	Dl
	Left Outside	Left Inside	Right Outside	Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-119	119	119	-119
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-91	91	163	-131

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu	Cl	Du	Dl
	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N3-Shell Inlet Nozl: 10 6:34pm Sep 8,2026

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4
Shear Stress Summary	3	3	6	6

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	102	91	163	160

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{971.5(16.0 - 1.0)}$$

$$= 906.365 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1943.000 mm
Nozzle Outside Diameter	(d) =	948.000 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	245.187 mm
T / t	=	0.061
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.553$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.024
MrD/Mc	=	0.238
NrDL/Mc	=	0.098
M0d/Mc	=	0.070
N0DL/Mc	=	0.047
MrD/Ml	=	0.044
NrDL/Ml	=	0.044
M0D/Ml	=	0.014
N0DL/Ml	=	0.022

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	83	84	83	84
Outplane Stress Summary	89	79	123	60

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------



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Nozzle Calcs.: N3-Shell Inlet

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Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	41	41	41	41

Inplane Stress Summary	69	14	170	-57
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside

Two * Max Shear Stress	90	79	170	117
------------------------	----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-68	68	68	-68
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	41	41	41	41

Outplane Stress Summary	-11	87	140	-34
-------------------------	-----	----	-----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-234	234	234	-234
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	83	84	83	84

Inplane Stress Summary	-88	239	412	-196
------------------------	-----	-----	-----	------

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside

Outplane Shear (Vc)	0	0	0	0
---------------------	---	---	---	---



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Nozzle Calcs.: N3-Shell Inlet

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Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2
Shear Stress Summary	1	1	3	3

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	88	239	412	196

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	41	40	41	40	41	40	41
Circ. Pl (SUS)	-9	-9	17	17	-5	-5	9	9
Circ. Q (SUS)	-12	12	30	-30	-76	76	126	-126
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-2	-2	6	6	-4	-4	12	12
Long. Q (SUS)	-26	26	76	-76	-34	34	52	-52
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	40.0	41.0	40.0	41.0	40.0	41.0	40.0	41.0
Pm+Pl (SUS)	31.1	32.1	57.0	58.0	35.1	36.0	49.1	50.1
Pm+Pl+Q (Total)	28.2	48.0	102.3	78.1	41.2	112.1	175.2	76.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	58.03	207.00	Passed
Pm+Pl+Q (TOTAL)	175.18	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	83	84	83	84	83	84	83	84
Circ. Pl (SUS)	1	1	7	7	-9	-9	23	23
Circ. Q (SUS)	5	-5	31	-31	-164	164	304	-304
Long. Pm (SUS)	41	41	41	41	41	41	41	41
Long. Pl (SUS)	0	0	14	14	-3	-3	11	11
Long. Q (SUS)	27	-27	113	-113	-50	50	86	-86



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Nozzle Calcs.: N3-Shell Inlet

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Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	83.0	84.0	83.0	84.0	83.0	84.0	83.0	84.0
Pm+P1 (SUS)	84.0	85.0	90.0	91.0	74.0	75.0	106.0	107.0
Pm+P1+Q (Total)	89.4	80.1	168.0	118.0	90.0	239.0	410.0	197.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	84.00	138.00	Passed
Pm+P1 (SUS)	107.02	207.00	Passed
Pm+P1+Q (TOTAL)	410.03	413.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	15	13	15	13	15	13	15
Circ. P1 (SUS)	-9	-9	17	17	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. P1 (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-30	30	84	-84	-92	92	146	-146
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	13.0	15.0	13.0	15.0	13.0	15.0	13.0	15.0
Pm+P1 (SUS)	5.4	7.3	30.1	32.1	10.2	12.2	25.1	27.1
Pm+P1+Q (Total)	36.4	30.0	97.1	103.2	102.2	91.1	159.2	160.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	118.00	Passed
Pm+P1 (SUS)	32.05	177.00	Passed
Pm+P1+Q (TOTAL)	160.31	353.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.



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Nozzle Calcs.: N4-Shell Outlet

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Input, Nozzle Desc: N4-Shell Outlet

From: 60

Pressure for Reinforcement Calculations	P	1328.624	KPa.
Temperature for Internal Pressure	Temp	160	°C
Design External Pressure	Pext	51.70	KPa.
Temperature for External Pressure	Tempex	160	°C
Parent Material		SA-516 70	
Parent Allowable Stress at Temperature	Sv	138.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	138.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1926.00	mm
Design Length of Section	L	2534.9719	mm
Shell Finished (Minimum) Thickness	t	16.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		13575.31	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		20.0000	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	948.0000	mm
Thickness of Pad	te	16.0000	mm
Weld leg size between Pad and Shell	Wp	12.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		220.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



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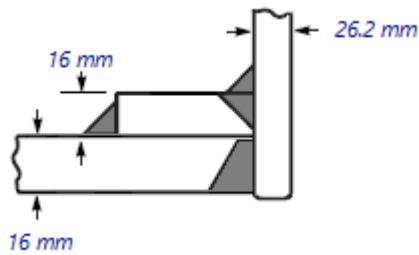
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Nozzle Calcs.: N4-Shell Outlet

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Reinforcement CALCULATION, Description: N4-Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	17.938 in.
Actual Thickness Used in Calculation	1.031 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1328.62 \cdot 964.0 / (138 \cdot 1.0 - 0.6 \cdot 1328.62)$
 $= 9.3358 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1328.62 \cdot 228.81 / (138 \cdot 1.0 - 0.6 \cdot 1328.62)$
 $= 2.2159 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.0009 mm

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	915.2504 mm
Parallel to Vessel Wall, opening length	d	457.6252 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		37.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 138.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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Nozzle Calcs.: N4-Shell Outlet

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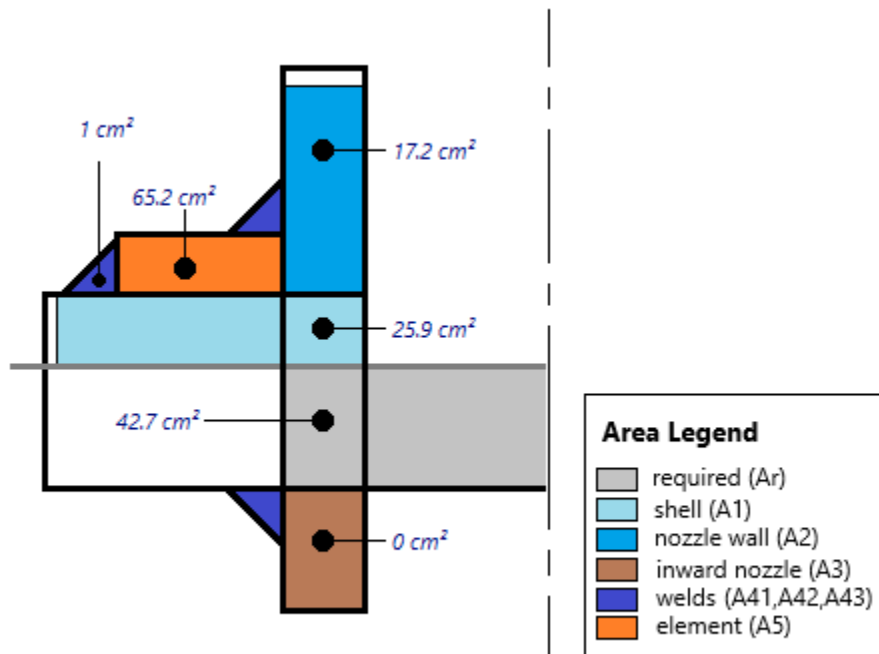


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	42.723	12.110	NA
Area in Shell	A1	25.921	44.424	NA
Area in Nozzle Wall	A2	17.229	18.140	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	1.000	1.000	NA
Area in Element	A5	65.160	65.160	NA
TOTAL AREA AVAILABLE	Atot	109.310	128.724	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (457.6252 \cdot 9.3358 \cdot 1.0 + 2 \cdot 25.1874 \cdot 9.3358 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 42.723 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 457.625(1.0 \cdot 15.0 - 1.0 \cdot 9.336) - 2 \cdot 25.187 \\
 &\quad (1.0 \cdot 15.0 - 1.0 \cdot 9.3358) \cdot (1 - 1.0) \\
 &= 25.921 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 37.5) \cdot (25.19 - 2.22) \cdot 1.0
 \end{aligned}$$



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Nozzle Calcs.: N4-Shell Outlet

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$$= 17.229 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 10.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 0.0^2 \cdot 1.0$$

$$= 1.000 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, D_L) - (\text{Nozzle OD})) \cdot (\min(t_p, t_{lp}, t_e)) \cdot fr_4$$

$$= (915.2504 - 508.0) \cdot 16.0 \cdot 1.0$$

$$= 65.160 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 3.2159 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 3.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 10.3358 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 10.3358 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 1.3613 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 3.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 9.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[9.331, \max(10.3358, 3.5)]$$

$$= 9.3312 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(3.2159, 9.3312)$$

$$= 9.3312 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 \cdot 26.187 = 22.914 \text{ mm} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 16.0,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 329.0 N./mm ²	Passed
Occasional	: 5.7,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 8.0,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 22.914$, $t_r = 2.216$, $c = 1.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.101$, Temp. Reduction = $78 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A $16 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-104 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, $t_g = 16.0$, $c = 1.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.622$, Temp. Reduction = $21 \text{ }^\circ\text{C}$

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B $-14 \text{ }^\circ\text{C}$

Min Metal Temp. at Required thickness (UCS 66.1) $-35 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UG-20(f) $-29 \text{ }^\circ\text{C}$

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B



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Nozzle Calcs.: N4-Shell Outlet

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Govrn. thk, tg = 16.0, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.622$, Temp. Reduction = 21 °C

Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-35 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, tg = 16.0, tr = 9.336, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.622$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-35 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 16.0, tr = 9.336, c = 1.0 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^*/(tg - c) = 0.622$, Temp. Reduction = 21 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-35 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Gov. MDMT of the Nozzle	:	-35 °C
Gov. MDMT of the Reinforcement Pad	:	-35 °C
Gov. MDMT of the nozzle to shell joint welded assembly	:	-35 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-36 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $1328.62/1960.01 = 0.678$

Weld Size Calculations, Description: N4-Shell Outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	16.0000 mm
Intermediate Calc. for pad/shell Welds	TminPad	15.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$6.0000 = \text{Min per Code}$	$7.0700 = 0.7 \cdot W_o \text{ mm}$
Pad Weld	$7.5000 = 0.5 \cdot \text{TminPad}$	$8.4840 = 0.7 \cdot W_p \text{ mm}$

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr))Sv)$$

$$= \max(0, (42.7228 - 25.921 + 2 \cdot 25.1874 \cdot 1.0 \cdot (1.0 \cdot 15.0 - 9.3358))138)$$

$$= 27657.80 \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (17.2286 + 65.1601 + 1. - 0.0 \cdot 1.0) \cdot 138$$

$$= 117340.98 \text{ Kgf}$$

Weld Load [W2]:



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Nozzle Calcs.: N4-Shell Outlet

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$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$$

$$= (17.2286 + 0.0 + 1. + (7.5562)) \cdot 138$$

$$= 36283.31 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot S$$

$$= (17.2286 + 0.0 + 1. + 65.1601 + (7.5562)) \cdot 138$$

$$= 127973.77 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 508.0 \cdot 10.0 \cdot 0.49 \cdot 138$$

$$= 55020. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 948.0 \cdot 12.0 \cdot 0.49 \cdot 138$$

$$= 123211. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo) / 4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 241.4063) \cdot (26.1874 - 1.0) \cdot 0.7 \cdot 138$$

$$= 188158. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 508.0 \cdot 16.0 \cdot 0.74 \cdot 138$$

$$= 132947. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 508.0 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 138$$

$$= 116328. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (123211 + 188158) = 311369 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (55020 + 132947 + 116328 + 0) = 304295 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (123211 + 116328 + 0) = 239539 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 311368 Kgf, must exceed W = 27657 Kgf	or W1 = 117340 Kgf
Path 2-2 = 304295 Kgf, must exceed W = 27657 Kgf	or W2 = 36283 Kgf
Path 3-3 = 239539 Kgf, must exceed W = 27657 Kgf	or W3 = 127973 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1790 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 KPa.

The Drop for this Nozzle is : 34.1012 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 350.1012 mm



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 Nozzle Calcs.: N4-Shell Outlet Nozl: 11 6:34pm Sep 8,2026

Input Echo, WRC297 Item 1, Description: N4-Shell Outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	1.0000	mm
Shell External Corrosion Allowance	0.0000	mm
Shell Diameter	1926.000	mm
Shell Thickness	16.0000	mm
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 70	
Vessel Cold S.I. Allowable	Smc 138.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	1.0000	mm
Nozzle Diameter	455.625	mm
Nozzle Thickness	26.1874	mm
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	948.000	mm
Thickness of Reinforcing Pad	16.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	1328.62	KPa.
Radial Load	P	-3824.19	Kgf
Circumferential Shear	Vc	2549.46	Kgf
Longitudinal Shear	Vl	2549.46	Kgf
Circumferential Moment	Mc	3567.87	Kg-m.
Longitudinal Moment	Ml	3567.87	Kg-m.
Torsional Moment	Mt	5096.96	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 $= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$
 $= 508.0 + 2 \cdot 1.65 \cdot \sqrt{979.5(16.0 - 1.0)}$
 $= 908.002 \text{ mm}$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1959.000	mm
Nozzle Outside Diameter	(d) =	508.000	mm
Vessel Thickness used	(T) =	31.000	mm
Nozzle Thickness used	(t) =	25.187	mm
T / t	=	1.231	
d / t	=	20.169	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.061$$



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Nozzle Calcs.: N4-Shell Outlet

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Nr/P = 0.060
 Mr/P = 0.109
 MØ/P = 0.042
 NØ/P = 0.121
 MrD/Mc = 0.237
 NrDL/Mc = 0.107
 MØd/Mc = 0.100
 NØDL/Mc = 0.112
 MrD/Ml = 0.120
 NrDL/Ml = 0.062
 MØD/Ml = 0.051
 NØDL/Ml = 0.186

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-13	-13	13	13
Outplane Bending (ML)	-21	21	21	-21
Normal Pressure Stress	40	41	40	41
Outplane Stress Summary	20	45	90	28

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-4	-4	4	4
Inplane Bending (ML)	-51	51	51	-51
Inplane Pressure Stress	20	20	20	20
Inplane Stress Summary	-7	44	104	-49

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3
Shear Stress Summary	4	4	2	2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	29	49	104	78

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to	Cu Left	Cl Left	Du Right	Dl Right
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Nozzle Calcs.: N4-Shell Outlet

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circumferential plane)	Outside	Inside	Outside	Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	9	-9	9	-9
Outplane Membrane (Mc)	-8	-8	8	8
Outplane Bending (Mc)	-43	43	43	-43
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	20	20	20	20
Outplane Stress Summary	-16	50	86	-19

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	25	-25	25	-25
Inplane Membrane (Mc)	-7	-7	7	7
Inplane Bending (Mc)	-101	101	101	-101
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	40	41	40	41
Inplane Stress Summary	-41	113	178	-75

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3
Shear Stress Summary	2	2	4	4

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left	Left	Right	Right
	Outside	Inside	Outside	Inside
Two * Max Shear Stress	41	113	178	75

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-13	-13	13	13
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	15	13	15
Hoop Stress Summary	5	6	31	33

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top	Top	Bottom	Bottom
	Outside	Inside	Outside	Inside



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Nozzle Calcs.: N4-Shell Outlet

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Axial Membrane (P)	0	0	0	0
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-7	-7	7	7
Axial Bending (ML)	-57	57	57	-57
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-30	29	101	-69

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	4	4	4	4
Shear Stress Summary	6	6	3	3

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE				
(Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	37	31	101	103

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-8	-8	8	8
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	13	15	13	15
Hoop Stress Summary	10	11	26	28

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	27	-27	27	-27
Axial Membrane (Mc)	-7	-7	7	7
Axial Bending (Mc)	-119	119	119	-119
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	6	6	6	6
Axial Stress Summary	-91	91	163	-131

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE				
(Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	4	4	4	4
Shear Stress Summary	3	3	6	6



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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	102	91	163	160

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 508.0 + 2 \cdot 1.65 \cdot \sqrt{971.5(16.0 - 1.0)}$$

$$= 906.365 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1943.000 mm
Nozzle Outside Diameter	(d) =	948.000 mm
Vessel Thickness used	(T) =	15.000 mm
Nozzle Thickness used	(t) =	245.187 mm
T / t	=	0.061
d / t	=	10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 5.553$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P	=	0.045
Mr/P	=	0.071
M0/P	=	0.019
N0/P	=	0.024
MrD/Mc	=	0.238
NrDL/Mc	=	0.098
M0d/Mc	=	0.070
N0DL/Mc	=	0.047
MrD/Ml	=	0.044
NrDL/Ml	=	0.044
M0D/Ml	=	0.014
N0DL/Ml	=	0.022

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-3	-3	3	3
Outplane Bending (ML)	-13	13	13	-13
Normal Pressure Stress	84	86	84	86
Outplane Stress Summary	90	80	124	61

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0



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Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-43	43	43	-43
Inplane Pressure Stress	42	42	42	42

Inplane Stress Summary	70	15	171	-57
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress normal to longitudinal plane)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2

Shear Stress Summary	3	3	1	1
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Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Two * Max Shear Stress	91	81	171	118
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	18	-18	18	-18
Outplane Membrane (Mc)	-7	-7	7	7
Outplane Bending (Mc)	-68	68	68	-68
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	42	42	42	42

Outplane Stress Summary	-10	88	141	-33
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	7	7	7	7
Inplane Bending (P)	70	-70	70	-70
Inplane Membrane (Mc)	-16	-16	16	16
Inplane Bending (Mc)	-234	234	234	-234
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	84	86	84	86

Inplane Stress Summary	-87	241	413	-195
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	87	241	413	195

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	40	41	40	41	40	41	40	41
Circ. Pl (SUS)	-9	-9	17	17	-5	-5	9	9
Circ. Q (SUS)	-12	12	30	-30	-76	76	126	-126
Long. Pm (SUS)	20	20	20	20	20	20	20	20
Long. Pl (SUS)	-2	-2	6	6	-4	-4	12	12
Long. Q (SUS)	-26	26	76	-76	-34	34	52	-52
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	40.0	41.0	40.0	41.0	40.0	41.0	40.0	41.0
Pm+Pl (SUS)	31.1	32.1	57.0	58.0	35.1	36.0	49.1	50.1
Pm+Pl+Q (Total)	28.2	48.0	102.3	78.1	41.2	112.1	175.2	76.3

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	41.00	138.00	Passed
Pm+Pl (SUS)	58.03	207.00	Passed
Pm+Pl+Q (TOTAL)	175.18	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	84	86	84	86	84	86	84	86
Circ. Pl (SUS)	1	1	7	7	-9	-9	23	23
Circ. Q (SUS)	5	-5	31	-31	-164	164	304	-304
Long. Pm (SUS)	42	42	42	42	42	42	42	42
Long. Pl (SUS)	0	0	14	14	-3	-3	11	11
Long. Q (SUS)	27	-27	113	-113	-50	50	86	-86
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	84.0	86.0	84.0	86.0	84.0	86.0	84.0	86.0



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Pm+Pl (SUS)	85.0	87.0	91.0	93.0	75.0	77.0	107.0	109.0
Pm+Pl+Q (Total)	90.4	82.1	169.0	119.0	89.0	241.0	411.0	195.1

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	86.00	138.00	Passed
Pm+Pl (SUS)	109.02	207.00	Passed
Pm+Pl+Q (TOTAL)	411.03	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	13	15	13	15	13	15	13	15
Circ. Pl (SUS)	-9	-9	17	17	-4	-4	12	12
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	6	6	6	6	6	6	6	6
Long. Pl (SUS)	-7	-7	7	7	-7	-7	7	7
Long. Q (SUS)	-30	30	84	-84	-92	92	146	-146
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	4	4	4	4	4	4	4	4
Pm (SUS)	13.0	15.0	13.0	15.0	13.0	15.0	13.0	15.0
Pm+Pl (SUS)	5.4	7.3	30.1	32.1	10.2	12.2	25.1	27.1
Pm+Pl+Q (Total)	36.4	30.0	97.1	103.2	102.2	91.1	159.2	160.3

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	15.00	138.00	Passed
Pm+Pl (SUS)	32.05	207.00	Passed
Pm+Pl+Q (TOTAL)	160.31	413.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2-Tube Outlet

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Input, Nozzle Desc: N2-Tube Outlet

From: 70

Pressure for Reinforcement Calculations	P	538.624	KPa.
Temperature for Internal Pressure	Temp	150	°C
Design External Pressure	Pext	51.70	KPa.
Temperature for External Pressure	Tempex	150	°C
Parent Material		SA-240 304L	
Parent Allowable Stress at Temperature	Sv	115.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	115.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	1920.00	mm
Design Length of Section	L	2000.0278	mm
Shell Finished (Minimum) Thickness	t	20.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		15635.17	mm
User Entered Minimum Design Metal Temperature		-12.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-312 TP304	
Material UNS Number		S30400	
Material Specification/Type		Wld. pipe	
Allowable Stress at Temperature	Sn	130.59	N./mm ²
Allowable Stress At Ambient	Sna	137.64	N./mm ²

Note: The allowables above have been divided by 0.85, see UG-37 [S].

Diameter Basis (for tr calc only)		Outside	
Layout Angle		180.00	deg
Diameter		30.0000	in.

Size and Thickness Basis		Actual	
Actual Thickness	tn	14.0000	mm

Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	

Corrosion Allowance	can	3.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	

Outside Projection	ho	300.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	12.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	14.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	1362.0000	mm
Thickness of Pad	te	30.0000	mm
Weld leg size between Pad and Shell	Wp	20.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	25.0000	mm
Reinforcing Pad Width		300.0000	mm

Flange Class		150	
Flange Grade		GR 2.1	
Flange Series		B	



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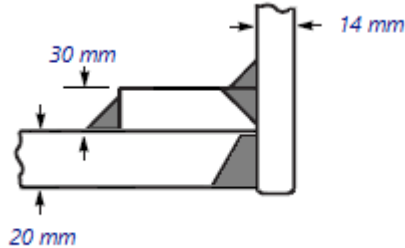
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Nozzle Calcs.: N2-Tube Outlet

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The Pressure Design option was Design Pressure + static head.



Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	30.000 in.
Actual Thickness Used in Calculation	0.551 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 538.62 \cdot 961.0 / (115 \cdot 1.0 - 0.6 \cdot 538.62)$
 $= 4.5141 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 538.62 \cdot 381.0 / (131 \cdot 1.0 + 0.4 \cdot 538.62)$
 $= 1.5690 \text{ mm}$

Required Nozzle thickness under External Pressure per UG-28 : 1.3071 mm

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	1480.0000 mm
Parallel to Vessel Wall, opening length	d	740.0000 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		47.5000 mm

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 130.6 / 115.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 130.6 / 115.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 115.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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Nozzle Calcs.: N2-Tube Outlet

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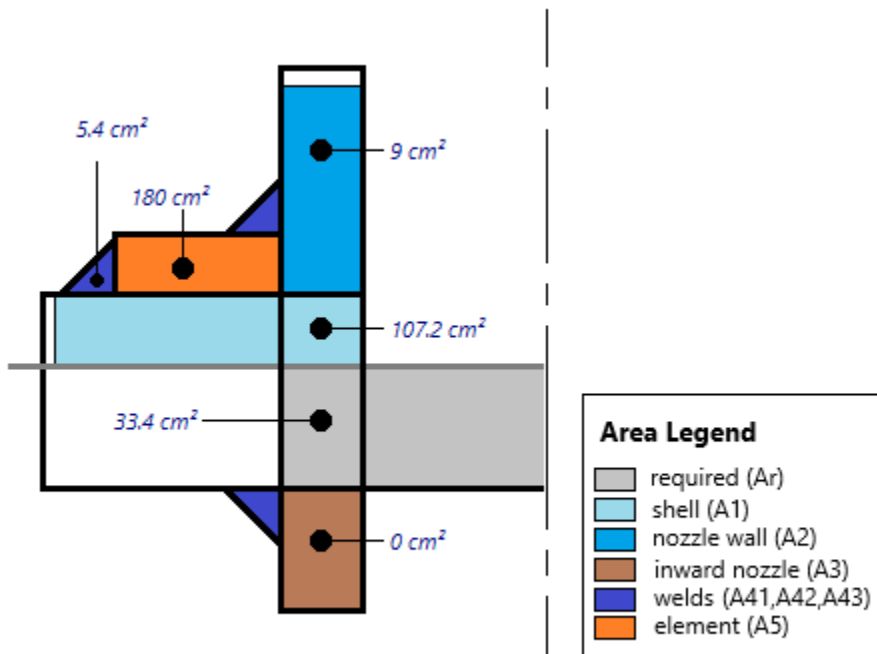


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	33.404	18.432	NA
Area in Shell	A1	107.196	103.736	NA
Area in Nozzle Wall	A2	8.959	9.208	NA
Area in Inward Nozzle	A3	0.000	0.000	NA
Area in Welds	A41+A42+A43	5.440	5.440	NA
Area in Element	A5	180.000	180.000	NA
TOTAL AREA AVAILABLE	Atot	301.596	298.385	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (740.0 \cdot 4.5141 \cdot 1.0 + 2 \cdot 11.0 \cdot 4.5141 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 33.404 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 740.0(1.0 \cdot 19.0 - 1.0 \cdot 4.514) - 2 \cdot 11.0 \\
 &\quad (1.0 \cdot 19.0 - 1.0 \cdot 4.5141) \cdot (1 - 1.0) \\
 &= 107.196 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 47.5) \cdot (11.0 - 1.57) \cdot 1.0
 \end{aligned}$$



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Nozzle Calcs.: N2-Tube Outlet

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$$= 8.959 \text{ cm}^2$$

Area Available in Welds [A41 + A42 + A43]:

$$= W_o^2 \cdot fr_3 + (W_i - can/0.707)^2 \cdot fr_2 + W_p^2 \cdot fr_4$$

$$= 12.0^2 \cdot 1.0 + (0.0)^2 \cdot 1.0 + 20.0^2 \cdot 1.0$$

$$= 5.440 \text{ cm}^2$$

Area Available in Element [A5]:

$$= (\min(D_p, DL) - (\text{Nozzle OD})) \cdot (\min(t_p, T_{lwp}, t_e)) \cdot fr_4$$

$$= (1362.0 - 762.0) \cdot 30.0 \cdot 1.0$$

$$= 180.000 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures $t_a = 4.5690 \text{ mm}$

Wall Thickness per UG16(b), $tr_{16b} = 4.5000 \text{ mm}$

Wall Thickness, shell/head, internal pressure $trb_1 = 5.5141 \text{ mm}$

Wall Thickness $tb_1 = \max(trb_1, tr_{16b}) = 5.5141 \text{ mm}$

Wall Thickness, shell/head, external pressure $trb_2 = 1.4322 \text{ mm}$

Wall Thickness $tb_2 = \max(trb_2, tr_{16b}) = 4.5000 \text{ mm}$

Wall Thickness per table UG-45 $tb_3 = 11.3312 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.331, \max(5.5141, 4.5)]$$

$$= 5.5141 \text{ mm}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(4.569, 5.5141)$$

$$= 5.5141 \text{ mm}$$

Available Nozzle Neck Thickness = 14.0000 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	30.1,	Allowable	:	130.6 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	305.2 N./mm ²	Passed
Occasional	:	8.9,	Allowable	:	173.7 N./mm ²	Passed
Shear	:	16.0,	Allowable	:	91.4 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Pad Weld for Nozzle skipped, as Nozzle is not a Carbon Steel material.

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, $t_g = 14.0$, $t_r = 1.489$, $c = 3.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.135$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -18 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $t_g = 20.0$, $t_r = 4.514$, $c = 1.0 \text{ mm}$, $E^* = 1.0$

Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.238$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -7 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C



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Nozzle-Shell Weld for Nozzle skipped as Nozzle is not a Carbon Steel material.

Gov. MDMT of the Reinforcement Pad : -104 °C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

UG-40 Modified by Appendix 1-7, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit) D1 1110.0000 mm

Parallel to Vessel Wall, opening length d 555.0000 mm

Normal to Vessel Wall (Thickness Limit), pad side Tlwp 47.5000 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Reinforcement Areas for Large Nozzle (Per App 1-7(a)): (cm²)

AREA AVAILABLE, A1 to A6		No Pad	With Pad
Area Required [Internal Pressure] AR		22.270	22.270
Area Available in Shell A1		53.598	53.598
Area Available in Nozzle Wall A2		5.187	8.959
Area Available in Inward Nozzle A3		0.000	0.000
Area Available in Welds A4		1.440	1.440
Area Available in Pad A5		0.000	104.400
Area Available in Hub A6		0.000	0.000
TOTAL AREA AVAILABLE Atot		60.225	168.398

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

The Nozzle does not fall within the scope of the diameter requirements of App. 1-7(b). Therefore, the Membrane and Bending stress checks are not required.

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 11.0000 mm

Intermediate Calc. for pad/shell Welds TminPad 19.0000 mm

Results Per UW-16.1:

Required Thickness Actual Thickness
 Nozzle Weld $7.7000 = 0.7 * t_{min}$ $8.4840 = 0.7 * W_o$ mm
 Pad Weld $9.5000 = 0.5 * T_{minPad}$ $14.1400 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A - A1 + 2 \cdot t_n \cdot f r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$$

$$= \max(0, (33.404 - 107.196 + 2 \cdot 11.0 \cdot 1.0 \cdot (1.0 \cdot 19.0 - 4.5141)) \cdot 115)$$

$$= \max(0, -82793.77) \text{ Kgf}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2 + A5 + A4 - (W_i - C_{an} / .707)^2 \cdot f r2) \cdot S_v$$

$$= (8.9595 + 180.0 + 5.44 - 0.0 \cdot 1.0) \cdot 115$$

$$= 227958.84 \text{ Kgf}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f r1)) \cdot S_v$$

$$= (8.9595 + 0.0 + 1.44 + (4.18)) \cdot 115$$

$$= 17096.34 \text{ Kgf}$$

Weld Load [W3]:

$$= (A2 + A3 + A4 + A5 + (2 \cdot t_n \cdot t \cdot f r1)) \cdot S$$



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$$= (8.9595 + 0.0 + 5.44 + 180.0 + (4.18)) \cdot 115$$

$$= 232860.44 \text{ Kgf}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$= (\pi/2) \cdot Dlo \cdot Wo \cdot 0.49 \cdot Snw$$

$$= (3.1416/2.0) \cdot 762.0 \cdot 12.0 \cdot 0.49 \cdot 131$$

$$= 93719. \text{ Kgf}$$

Shear, Pad Element Weld [Spew]:

$$= (\pi/2) \cdot DP \cdot WP \cdot 0.49 \cdot SEW$$

$$= (3.1416/2.0) \cdot 1362.0 \cdot 20.0 \cdot 0.49 \cdot 115$$

$$= 245858. \text{ Kgf}$$

Shear, Nozzle Wall [Snw]:

$$= (\pi \cdot (Dlr + Dlo)/4) \cdot (Thk - Can) \cdot 0.7 \cdot Sn$$

$$= (3.1416 \cdot 375.5) \cdot (14.0 - 3.0) \cdot 0.7 \cdot 131$$

$$= 120955. \text{ Kgf}$$

Tension, Pad Groove Weld [Tpgw]:

$$= (\pi/2) \cdot Dlo \cdot Wgpn \cdot 0.74 \cdot Seg$$

$$= (3.1416/2) \cdot 762.0 \cdot 25.0 \cdot 0.74 \cdot 138$$

$$= 311594. \text{ Kgf}$$

Tension, Shell Groove Weld [Tngw]:

$$= (\pi/2) \cdot Dlo \cdot Wgnvi \cdot 0.74 \cdot Sng$$

$$= (3.1416/2.0) \cdot 762.0 \cdot (14.0 - 0.0) \cdot 0.74 \cdot 131$$

$$= 165123. \text{ Kgf}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (SPEW + SNW) = (245858 + 120955) = 366813 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$= (Sonw + Tpgw + Tngw + Sinw)$$

$$= (93719 + 311594 + 165123 + 0) = 570436 \text{ Kgf}$$

Failure Path [Path 3-3]:

$$= (Spew + Tngw + Sinw)$$

$$= (245858 + 165123 + 0) = 410981 \text{ Kgf}$$

Summary of Failure Path Calculations:

Path 1-1 = 366813 Kgf, must exceed W = 0 Kgf	or W1 = 227958 Kgf
Path 2-2 = 570436 Kgf, must exceed W = 0 Kgf	or W2 = 17096 Kgf
Path 3-3 = 410981 Kgf, must exceed W = 0 Kgf	or W3 = 232860 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1891 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

Nozzle is O.K. for the External Pressure: 52 KPa.

The Drop for this Nozzle is : 78.8423 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 398.8423 mm

Warning:

Review Appendix 1-7 (b)(3) for notes regarding externally applied loads to large nozzles and U-2(g).

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:



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Diameter Basis for Cylindrical Shell	ID		
Shell Internal Corrosion Allowance	1.0000	mm	
Shell External Corrosion Allowance	0.0000	mm	
Shell Diameter	1920.000	mm	
Shell Thickness	20.0000	mm	
Shell Stress Concentration Factor	1.000		
Vessel Material	SA-240 304L		
Vessel Cold S.I. Allowable	Smc 115.00	N./mm ²	
Vessel Hot S.I. Allowable	Smh 115.00	N./mm ²	

Diameter Basis for Nozzle	OD		
Nozzle Corrosion Allowance	3.0000	mm	
Nozzle Diameter	762.000	mm	
Nozzle Thickness	14.0000	mm	
Nozzle Stress Concentration Factor	1.000		
Nozzle Material	SA-312 TP304		
Nozzle Cold S.I. Allowable	SNmc 137.64	N./mm ²	
Nozzle Hot S.I. Allowable	SNmh 130.59	N./mm ²	

Diameter of Reinforcing Pad	1362.000	mm	
Thickness of Reinforcing Pad	30.0000	mm	

Note:

External Forces and Moments in WRC 107/537 Convention:

These loads are assumed to be SUStained loads.

Design Internal Pressure	Dp	538.62	KPa.
Radial Load	P	-5353.89	Kgf
Circumferential Shear	Vc	3824.19	Kgf
Longitudinal Shear	Vl	3824.19	Kgf
Circumferential Moment	Mc	8027.71	Kg-m.
Longitudinal Moment	Ml	8027.71	Kg-m.
Torsional Moment	Mt	11468.14	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 762.0 + 2 \cdot 1.65 \cdot \sqrt{985.5(20.0 - 1.0)}$$

$$= 1213.564 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	1971.000 mm
Nozzle Outside Diameter	(d) =	762.000 mm
Vessel Thickness used	(T) =	49.000 mm
Nozzle Thickness used	(t) =	11.000 mm
T / t	=	4.455
d / t	=	69.273

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 2.452$$

Nr/P	=	0.027
Mr/P	=	0.033
M0/P	=	0.050
N0/P	=	0.226
MrD/Mc	=	0.086
NrDL/Mc	=	0.063
M0d/Mc	=	0.171



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NØDL/Mc = 0.432
MrD/Ml = 0.048
NrDL/Ml = 0.038
MØD/Ml = 0.076
NØDL/Ml = 0.516

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-22	-22	22	22
Outplane Bending (ML)	-19	19	19	-19
Normal Pressure Stress	10	10	10	10
Outplane Stress Summary	-19	6	63	11

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-12	12	12	-12
Inplane Pressure Stress	5	5	5	5
Inplane Stress Summary	-3	12	23	-9

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	2	2	2	2
Shear Stress Summary	3	3	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	20	13	63	21

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	4	4	4	4
Outplane Bending (P)	6	-6	6	-6
Outplane Membrane (Mc)	-18	-18	18	18
Outplane Bending (Mc)	-44	44	44	-44
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0



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Normal Pressure Stress	5	5	5	5
------------------------	---	---	---	---

Outplane Stress Summary	-46	29	79	-22
-------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	4	-4	4	-4
Inplane Membrane (Mc)	-2	-2	2	2
Inplane Bending (Mc)	-22	22	22	-22
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	10	10	10	10

Inplane Stress Summary	-9	26	40	-12
------------------------	----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	0	0	0	0
Torsional Shear (Mt)	2	2	2	2

Shear Stress Summary	1	1	3	3
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	46	30	79	22
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-22	-22	22	22
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	21	22	21	22

Hoop Stress Summary	4	4	48	49
---------------------	---	---	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	50	-50	50	-50
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-16	-16	16	16
Axial Bending (ML)	-146	146	146	-146
Axial Pressure Stress	10	10	10	10



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Nozzle Calcs.: N2-Tube Outlet

Nozl: 12 6:34pm

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Axial Stress Summary	-99	92	225	-167
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	2	2	-2	-2
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	11	11	11	11

Shear Stress Summary	14	14	8	8
----------------------	----	----	---	---

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	107	94	225	217
------------------------	-----	----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	4	4	4	4
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-18	-18	18	18
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	21	22	21	22

Hoop Stress Summary	7	8	44	45
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	2	2	2	2
Axial Bending (P)	50	-50	50	-50
Axial Membrane (Mc)	-16	-16	16	16
Axial Bending (Mc)	-278	278	278	-278
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10

Axial Stress Summary	-231	224	357	-299
----------------------	------	-----	-----	------

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-2	-2	2	2
Shear due to Torsion	11	11	11	11

Shear Stress Summary	8	8	14	14
----------------------	---	---	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	240	224	358	346
------------------------	-----	-----	-----	-----



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Nozzle Calcs.: N2-Tube Outlet

Nozl: 12

6:34pm

Sep 8, 2026

Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 762.0 + 2 \cdot 1.65 \cdot \sqrt{970.5(20.0 - 1.0)}$$

$$= 1210.114 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 1941.000 mm

Nozzle Outside Diameter (d) = 1362.000 mm

Vessel Thickness used (T) = 19.000 mm

Nozzle Thickness used (t) = 311.000 mm

T / t = 0.061

d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 7.092$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045

Mr/P = 0.071

M0/P = 0.019

N0/P = 0.024

MrD/Mc = 0.238

NrDL/Mc = 0.091

M0d/Mc = 0.066

N0DL/Mc = 0.044

MrD/Ml = 0.036

NrDL/Ml = 0.035

M0D/Ml = 0.012

N0DL/Ml = 0.020

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	3	3	3	3
Outplane Bending (P)	16	-16	16	-16
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (Ml)	-3	-3	3	3
Outplane Bending (Ml)	-11	11	11	-11
Normal Pressure Stress	26	27	26	27
Outplane Stress Summary	32	22	61	6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	6	6	6	6
Inplane Bending (P)	61	-61	61	-61
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (Ml)	-5	-5	5	5
Inplane Bending (Ml)	-34	34	34	-34
Inplane Pressure Stress	13	13	13	13
Inplane Stress Summary	41	-12	122	-70

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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Nozzle Calcs.: N2-Tube Outlet

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(Shear stress normal to longitudinal plane)			Top Outside	Top Inside	Bottom Outside	Bottom Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			2	2	2	2
Shear Stress Summary			2	2	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)		Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress		42	36	122	76

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)		3	3	3	3
Outplane Bending (P)		16	-16	16	-16
Outplane Membrane (Mc)		-7	-7	7	7
Outplane Bending (Mc)		-63	63	63	-63
Outplane Membrane (ML)		0	0	0	0
Outplane Bending (ML)		0	0	0	0
Normal Pressure Stress		13	13	13	13
Outplane Stress Summary		-37	56	104	-56

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)		6	6	6	6
Inplane Bending (P)		61	-61	61	-61
Inplane Membrane (Mc)		-14	-14	14	14
Inplane Bending (Mc)		-228	228	228	-228
Inplane Membrane (ML)		0	0	0	0
Inplane Bending (ML)		0	0	0	0
Inplane Pressure Stress		26	27	26	27
Inplane Stress Summary		-147	186	338	-241

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)		0	0	0	0
Outplane Shear (Vl)		0	0	0	0
Torsional Shear (Mt)		2	2	2	2
Shear Stress Summary		1	1	2	2

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)		Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress		147	186	338	241



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 Nozzle Calcs.: N2-Tube Outlet Nozl: 12 6:34pm Sep 8,2026

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	10	10	10	10	10	10	10	10
Circ. Pl (SUS)	-18	-18	26	26	-2	-2	2	2
Circ. Q (SUS)	-13	13	25	-25	-18	18	26	-26
Long. Pm (SUS)	5	5	5	5	5	5	5	5
Long. Pl (SUS)	-1	-1	1	1	-14	-14	22	22
Long. Q (SUS)	-8	8	16	-16	-38	38	50	-50
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	10.0	10.0	10.0	10.0	10.0	10.0	10.0	10.0
Pm+Pl (SUS)	12.0	12.0	36.0	36.0	17.0	17.0	27.0	27.0
Pm+Pl+Q (Total)	21.2	12.5	61.1	21.4	47.1	30.0	77.1	23.4

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	115.00	Passed
Pm+Pl (SUS)	36.00	172.50	Passed
Pm+Pl+Q (TOTAL)	77.10	344.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	26	27	26	27	26	27	26	27
Circ. Pl (SUS)	0	0	6	6	-8	-8	20	20
Circ. Q (SUS)	5	-5	27	-27	-167	167	289	-289
Long. Pm (SUS)	13	13	13	13	13	13	13	13
Long. Pl (SUS)	1	1	11	11	-4	-4	10	10
Long. Q (SUS)	27	-27	95	-95	-47	47	79	-79
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	0	0	0	0	0	0	0	0
Shear Q (SUS)	2	2	2	2	2	2	2	2
Pm (SUS)	26.0	27.0	26.0	27.0	26.0	27.0	26.0	27.0
Pm+Pl (SUS)	26.0	27.0	32.0	33.0	18.0	19.0	46.0	47.0
Pm+Pl+Q (Total)	41.4	35.2	119.1	77.1	149.0	186.0	335.0	242.0

Vessel Stress Summation Comparison (N./mm²):



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 Nozzle Calcs.: N2-Tube Outlet Nozl: 12 6:34pm Sep 8,2026

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	27.00	115.00	Passed
Pm+P1 (SUS)	47.00	172.50	Passed
Pm+P1+Q (TOTAL)	335.02	344.99	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	21	22	21	22	21	22	21	22
Circ. P1 (SUS)	-18	-18	26	26	-14	-14	22	22
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. P1 (SUS)	-14	-14	18	18	-14	-14	18	18
Long. Q (SUS)	-96	96	196	-196	-228	228	328	-328
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear P1 (SUS)	2	2	-2	-2	-2	-2	2	2
Shear Q (SUS)	11	11	11	11	11	11	11	11
Pm (SUS)	21.0	22.0	21.0	22.0	21.0	22.0	21.0	22.0
Pm+P1 (SUS)	8.1	8.9	47.2	48.2	11.7	12.6	43.3	44.2
Pm+P1+Q (Total)	106.2	93.9	224.5	216.7	239.7	224.4	356.5	345.0

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	22.00	130.59	Passed
Pm+P1 (SUS)	48.20	195.88	Passed
Pm+P1+Q (TOTAL)	356.54	391.76	Passed

Because only sustained loads were specified, the Pm+P1+Q allowable was 3 * Smh.

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ASME TS Calc:

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Input Echo, Tubesheet Number 1, Description: Fixed Tubesheet

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	1310.00 KPa.
Shell Maximum Operating Pressure	Psox,max	1310.00 KPa.
Shell Minimum Operating Pressure	Psox,min	51.70 KPa.
Shell Thickness	ts	16.0000 mm
Shell Internal Corrosion Allowance	cas	1.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	1926.000 mm
Shell Circumferential Joint Efficiency	Esw	0.850
Shell Temperature for Internal Pressure	Ts	160.00 °C
Shell Material		SA-516 70

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02700
Shell Allowable Stress at Temperature	Ss	138.00 N./mm ²
Shell Allowable Stress at Ambient		138.00 N./mm ²

Channel Data:

Channel Description: LHS ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	520.00 KPa.
Channel Maximum Operating Pressure	Ptox,max	520.00 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	14.0000 mm
Channel Internal Corrosion Allowance	cac	1.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	1920.000 mm
Channel Design Temperature	TEMPC	150.00 °C
Channel Flange Matl		SA-240 304L

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		S30403
Channel Allowable Stress at Temperature	Sc	115.00 N./mm ²
Channel Allowable Stress at Ambient		115.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	2895
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	25.4000 mm
Total Straight Tube Length	Lt	12800.00 mm
Straight Tube Length (bet. inner tubsht faces) L		12688.00 mm
Design Temperature of the Tubes		150.00 °C
Tube Material		SA-213 TP304L
Tube Material UNS Number		S30403
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	132.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	31.7500 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	2.1100 mm
Groove Weld Leg	ag	2.1100 mm
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	937.029 mm



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ASME TS Calc:

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Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	0.0000 mm
Tube-side pass partition groove depth	hg	0.0000 mm
Area of Tube Layout	Ap	25256.1 cm ²
Perimeter of Tube Layout	Cp	6023.4 mm

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf A

Tubesheet Design Metal Temperature	T	150.00 °C
Tubesheet Material		SA-765 II

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03047
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	56.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	10.0000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.0000 mm
Tubesheet Outside Diameter	A	1954.000 mm

Area of the Untubed Lanes	AL	305.8 cm ²
---------------------------	----	-----------------------

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	560.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Ignore Radial Thermal Exp. effects (4.18.8/4.18.9)		Yes
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	-52.0000 KPa.
Channel Side Min. Design Pressure	Ptd,min	-52.0000 KPa.
Mean Shell Metal Temp. along Shell len.	Tsm	0.00 °C
Mean Tube Metal Temp. along Tube length	Ttm	0.00 °C
Junction Stress Reduction option		Perform Plastic Calculation
Tubesheet Integral with		Both
Tubesheet Extended as Flange		No
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No



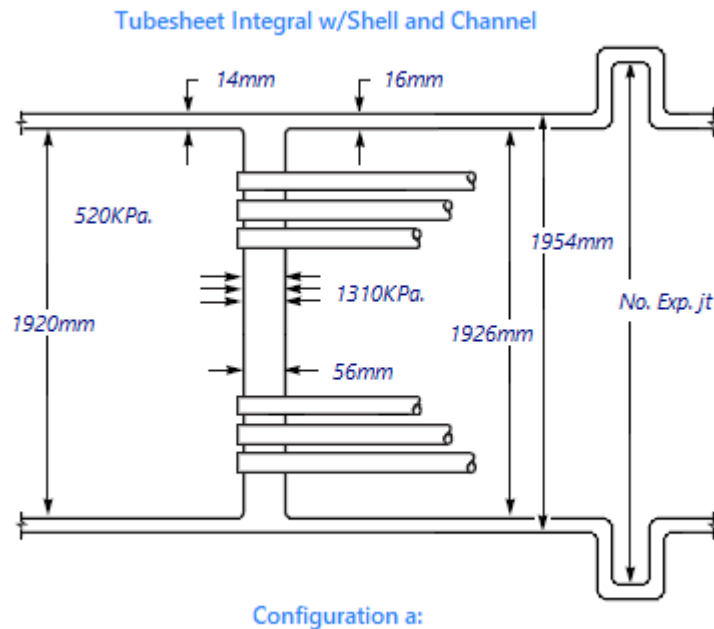
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ASME TS Calc:

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**ASME TubeSheet Results per Part UHX/4.18, 2025****Elasticity/Expansion Material Properties:**

Shell - TE-1 Carbon & Low Alloy Steels, Group 1
Shell - TM-1 Carbon Steels with C<= 0.3%

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000112153 /°C
Elastic Mod. at Design Temperature	160.0 °C	0.19457E+09 KPa.
Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000115190 /°C
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.20409E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TE-1 Austenitic Group 3 Steels Group 1
Channel - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. at Tubsht	21.1 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.18607E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

Tubes - TE-1 Austenitic Group 3 Steels Group 1
Tubes - TM-1 Material Group G

Th. Exp. Coeff. Metal Temp. along Len	0.0 °C	0.0000152988 /°C
Elastic Mod. at Design Temperature	150.0 °C	0.18607E+09 KPa.
Elastic Mod. at Metal Temp. along Len	0.0 °C	0.19650E+09 KPa.
Elastic Mod. at Tubsht. Design Temp.	150.0 °C	0.18607E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TE-1 Carbon & Low Alloy Steels, Group 1
TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	150.0 °C	0.19506E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Note:



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ASME TS Calc:

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The Elasticity and Alpha values are taken from Tables in ASME II D.

Please insure these properties are consistent with the type of Material for the tubes, shell, channel etc.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (572.0 \cdot (25.4/2 - 0)) / (115.0 \cdot 1.0 + 0.4 \cdot 572.0)$$

$$= 0.0630 + 0.0000 = 0.0630 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-3 at 150.00 °C
Elastic Modulus for Material 183300992.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	25.40	12688.00	12.04	50.0000	0.0075908	64.88
EMAWP = (4*B)/(3*(D/T)) = (4 *64.8769)/(3 *12.0379) = 7185.4194 KPa.						

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.6247	25.40	12688.00	40.66	50.0000	0.0006653	41.54
EMAWP = (4*B)/(3*(D/T)) = (4 *41.5429)/(3 *40.6625) = 1362.1243 KPa.						

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.6247 mm
Allowable Internal Pressure at Corroded thickness	20464.87 KPa.
Required Internal Design Pressure	572.00 KPa.
Allowable External Pressure at Corroded thickness	7185.42 KPa.
Required External Design Pressure	1362.00 KPa.
Required Thickness due to Shell Side pressure	0.6247 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME Fixed Tubesheet Calculations for Configuration a:

Results for Tubesheet Calculations Original Thickness:

ASME Section VIII-2 Part 4.18.8.4 Step 1:

Tube expansion depth ratio [p]:

$$= l_{tx} / h \text{ (modified for corrosion if present)}$$

$$= 0/56.0$$

$$= 0.0000 \text{ (must be } 0 \leq p \leq 1 \text{)}$$

Effective tube hole diameter [d*]:

$$= \max(dt - 2 \cdot tt(Et/E) \cdot (St/T/S) \cdot (\rho); dt - 2 \cdot tt)$$

$$= \max(25.4 - 2 \cdot 2.11 (0.18607E+09/1.9506E+09) \cdot (114/137) \cdot (0); 25.4 - 2 \cdot 2.11)$$

$$= 25.4000 \text{ mm}$$

Equivalent outer tube limit circle diameter [Do]:

$$= 2 \cdot r_o + dt$$

$$= 2 \cdot 937.029 + 25.4$$

$$= 1899.458 \text{ mm}$$

Determine the basic ligament efficiency for shear [μ]:

$$= (p - dt) / p$$

$$= (31.75 - 25.4) / 31.75$$

$$= 0.200$$



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ASME TS Calc:

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Equivalent outer tube limit radius [ao]:

$$\begin{aligned} &= D_o / 2 \\ &= 1899.458 / 2 \\ &= 949.7290 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{1 - 4 \cdot \min(AL \cdot CNVfactor; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2) } \\ &= 31.75 / \sqrt{1 - 4 \cdot \min(305.75 \cdot 100.0 ; 4 \cdot 1899.458 \cdot 31.75) / (\pi \cdot 1899.458^2) } \\ &= 31.9227 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (31.9227 - 25.4) / 31.9227 \\ &= 0.2043 \end{aligned}$$

Ratio [ps]:

$$\begin{aligned} &= a_s / a_o \\ &= 963.0 / 949.729 \\ &= 1.0140 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= a_c / a_o \\ &= 960.0 / 949.729 \\ &= 1.0108 \end{aligned}$$

Compute parameter [xt]:

$$\begin{aligned} &= 1 - Nt / ((dt - 2 \cdot tt) / (2 \cdot a_o))^2 \\ &= 1 - 2895 / ((25.4 - 2 \cdot 2.11) / (2 \cdot 949.729))^2 \\ &= 0.6401 \end{aligned}$$

Determine parameter [xs]:

$$\begin{aligned} &= 1 - Nt / (dt / (2 \cdot a_o))^2 \\ &= 1 - 2895 / (25.4 / (2 \cdot 949.729))^2 \\ &= 0.4823 \end{aligned}$$

Determine the value [h'g]:

$$\begin{aligned} &= \max((h_g - CATC); 0) \text{ (For pressure only cases)} \\ &= \max((0 - 0); 0) \\ &= 0.000 \text{ mm} \end{aligned}$$

ASME Section VIII-2 Part 4.18.8.4 Step 2:

Axial Shell Stiffness [Ks]:

$$\begin{aligned} &= \pi \cdot t_s (D_s + t_s) E_s / L \\ &= \pi \cdot 16.0 (1926.0 + 16.0) \cdot 19457E+09 / 12688.0 \\ &= 152647.6406 \text{ Kg/mm} \end{aligned}$$

Axial Tube Stiffness [Kt]:

$$\begin{aligned} &= \pi \cdot t_t (D_t - t_t) E_t / L \\ &= \pi \cdot 2.11 (25.4 - 2.11) \cdot 18607E+09 / 12688.0 \\ &= 230.8805 \text{ Kg/mm} \end{aligned}$$

Stiffness Factor [Ks,t]:

$$\begin{aligned} &= K_s / (Nt \cdot K_t) \\ &= 152647.641 / (2895 \cdot 230.881) \\ &= 0.22838 \end{aligned}$$

Rigidity Ratio [J]:

$$\begin{aligned} &= 1 / (1 + K_s / K_j) \\ &= 1 / (1 + 152647.641 / 0) \\ &= 1.0000 (= 1 \text{ if No Exp. Jt.}) \end{aligned}$$

Shell coefficient [β_s]:

$$\begin{aligned} &= ((12(1 - \nu_s^2))^{1/4}) / ((D_s + t_s) t_s) \\ &= ((12(1 - 0.3^2))^{1/4}) / ((1926.0 + 16.0) 16.0) \end{aligned}$$



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ASME TS Calc:

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$$= 0.0103 \text{ 1/mm}$$

Shell coefficient [ks]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2))$$

$$= 0.01 \cdot 0.19457\text{E}+09 \cdot 16.0^3 / (6(1 - 0.3^2))$$

$$= 1505240960.0000 \text{ KPa} \cdot \text{mm}^2$$

Shell coefficient [ls]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(1926.0 + 16.0) \cdot 0.15052\text{E}+1 / 56.0^3 \cdot (1 + 56.0 \cdot 0.01 + (56.0 \cdot 0.01)^2 / 2)$$

$$= 174203008.0000 \text{ KPa}$$

Shell coefficient [ds]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s)))$$

$$= 1926.0^2 / (4 \cdot 0.19457\text{E}+09 \cdot 16.0) \cdot (1 - 1926.0 \cdot 0.3 / (2(1926.0 + 16.0)))$$

$$= 0.25357\text{E}+00 \text{ mm/N./mm}^2$$

Channel coefficient [bc]:

$$= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{(D_c + t_c)t_c}$$

$$= ((12(1 - 0.31^2))^{1/4}) / \sqrt{(1920.0 + 14.0)14.0}$$

$$= 0.0110 \text{ 1/mm}$$

Channel coefficient [kc]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_c^2))$$

$$= 0.011 \cdot 0.18607\text{E}+09 \cdot 14.0^3 / (6(1 - 0.31^2))$$

$$= 1038311744.0000 \text{ KPa} \cdot \text{mm}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [lc]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (Cm\Delta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot Cq\Delta)$$

$$= 6(1920.0 + 14.0 / \cos(0)) \cdot 0.10383\text{E}+1 / 56.0^3 \cdot (1.0 + 56.0 \cdot 0.011 \cdot (1.0 + 1.0) / 2 + (56.0 \cdot 0.011)^2 / 2 \cdot 1.0)$$

$$= 124066088.0 \text{ KPa}$$

Channel coefficient [dc]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c)))$$

$$= 1920.0^2 / (4 \cdot 0.18607\text{E}+09 \cdot 14.0) \cdot (1 - 1920.0 \cdot 0.31 / (2(1920.0 + 14.0)))$$

$$= 0.29932\text{E}+00 \text{ mm/N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

ASME Section VIII-2 Part 4.18.8.4 Step 3:

$$h/p = 1.763780 ; \mu^* = 0.204328 ; E^* = 32717776 \text{ KPa}$$

$$E^*/E = 0.167730 ; \nu^* = 0.441119$$

Tube Bundle Stiffness Factor [Xa]:

$$= ((24(1 - \nu^*)Nt \cdot Et \cdot t(d_t - t_t)ao^2) / (E^* \cdot L \cdot H^3))^{1/4}$$

$$= ((24(1 - 0.441^2)2895 \cdot 0.18607\text{E}+09 \cdot 2.11(25.4 - 2.11)949.729^2) / (32717776 \cdot 12688.0 \cdot 56.0^3))^{1/4}$$

$$= 8.9201$$

Values from Table 4.18.3

$$Z_d = 0.002070 ; Z_v = 0.012558 ; Z_m = 0.159297$$

$$Z_a = 0.356288\text{E}+04 ; Z_w = 0.012558$$

ASME Section VIII-2 Part 4.18.8.4 Step 4:

Diameter ratio [K]:

$$= A / D_o$$

$$= 1954.0 / 1899.458$$



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ASME TS Calc:

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= 1.0287

Compute coefficient [F]:

$$= (1 - \nu) / E \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$

$$= (1 - 0.44) / 32717776 \cdot (0.17420E+09 + 0.12407E+09 + 0.19506E+09 \cdot \ln(1.03))$$

= 5.1893

Compute parameter [ϕ]:

$$= (1 + \nu) F$$

$$= (1 + 0.4411) 5.1893$$

= 7.4784

Compute parameter [Q1]:

$$= (p_s - 1 - \phi \cdot Z_v) / (1 + \phi \cdot Z_m)$$

$$= (1.014 - 1 - 7.4784 \cdot 0.0126) / (1 + 7.4784 \cdot 0.1593)$$

= -0.036481

Compute Parameter [Qz1]:

$$= (Z_d + Q1 \cdot Z_w) / 2 \cdot X_a^4$$

$$= (0.00207 + -0.03648 \cdot 0.01256) / 2 \cdot 8.92006^4$$

= 5.1018

Compute parameter [Qz2]:

$$= (Z_v + Q1 \cdot Z_m) / 2 \cdot X_a^4$$

$$= (0.01256 + -0.03648 \cdot 0.1593) / 2 \cdot 8.92006^4$$

= 21.3564

Compute parameter [U]:

$$= (Z_w + (p_s - 1) Z_m) X_a^4 / (1 + \phi \cdot Z_m)$$

$$= (0.0126 + (1.014 - 1) 0.1593) 8.92006^4 / (1 + 7.4784 \cdot 0.1593)$$

= 42.7129

ASME Section VIII-2 Part 4.18.8.4 Step 5:

Determine factor [γ_b]:

= 0 (config a)

Compute Parameter [γ]:

= 0.000 mm (For Pressure only cases)

Calculate parameter [ω_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta S (1 + h \cdot \beta_s)$$

$$= 1.014 \cdot 0.15052E+1 \cdot 0.0103 \cdot 0.253565 (1 + 56.0 \cdot 0.0103)$$

= 6296.3745 mm²

Calculate parameter [ω^*s]:

$$= a o^2 (p_s^2 - 1) \cdot (p_s - 1) / 4 - \omega_s$$

$$= 949.729^2 (1.014^2 - 1) \cdot (1.014 - 1) / 4 - 6296.375$$

= -6207.6997 mm²

Parameter [ω_c]:

$$= p_c \cdot k_c [\delta C (C_m \theta + \beta C + C_q \Delta \cdot h \cdot \beta C^2) - \zeta c (C_m \theta + h \cdot C_q \theta + h \cdot \beta C / 2)]$$

$$= 1.011 \cdot 0.10383E+1 [0.299323 (1.0 + 0.011 \cdot 1.0 \cdot 56.0) - 0 (1.0/56.0 + 1.0 \cdot 0.011/2)]$$

= 5604.9805 mm²

Parameter [ω^*c]:

$$= a o^2 [(p_c^2 + 1) \cdot (p_c - 1) / 4 - (p_s - 1) / 2] - \omega_c$$

$$= 949.729^2 [(1.01081^2 + 1) \cdot (1.01081 - 1) / 4 - (1.01397 - 1) / 2] - 5604.98047$$

= -6976.5479 mm²

Pressure [P^*S]:

= 0 For Pressure only cases or Configurations d,e,f,A,B,C,D



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ASME TS Calc:

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Pressure [P*C]:

= 0 For Pressure only cases or Configurations b,c,d,B,C,D

ASME Section VIII-2 Part 4.18.8.4 Step 6:

Pressure [P's]:

$$= Ps \{ xs + 2(1 - xs) vt + [2/Kst \cdot (Ds/Do)^2] vs - [(ps^2 - 1)/(J \cdot Kst)] - [(1 - J)/(2J \cdot Kst)] [(Dj^2 - (Ds)^2)/Do^2] \}$$

$$= 1310.0 \{ 0.482 + 2(1 - 0.482) 0.31 + [2/0.228 \cdot (1926.0/1899.458)^2] 0.3 - [(1.014^2 - 1)/(1.0 \cdot 0.228)] - [(1 - 1.0)/(2 \cdot 1.0 \cdot 0.228)] [(0^2 - (1926.0)^2)/1899.458^2] \}$$

$$= 4429.3945 \text{ KPa.}$$

Pressure [P't]:

$$= [xt + 2(1 - xt) vt + 1/(J \cdot Kst)] Pt$$

$$= [0.64 + 2(1 - 0.64) 0.31 + 1/(1 \cdot 0.228)] 520.0$$

$$= 2725.8013 \text{ KPa.}$$

Pressure [Py]:

$$= Nt \cdot Kt \cdot \gamma / (\pi \cdot ao^2)$$

$$= 2895 \cdot 230.881 \cdot 0 / (3.142 \cdot 949.729^2)$$

$$= 0.0000 \text{ KPa.}$$

Pressure [Pw]:

$$= -\gamma b \cdot U \cdot W* / (2 \cdot \pi \cdot ao^2)$$

$$= -0 \cdot 42.713 \cdot 0 / (2 \cdot \pi \cdot 949.729^2)$$

$$= -0.0000 \text{ KPa.}$$

Calculate the pressure [Prim]:

$$= -(U/ao^2) \cdot (\omega*s \cdot Ps - \omega*c \cdot Pt)$$

$$= -(42.713/949.729^2) \cdot (-6207.7 \cdot 1310.0 - -6976.548 \cdot 520.0)$$

$$= 213.2972 \text{ KPa.}$$

Calculate the pressure [Pw]:

$$= (U/ao^2) \cdot (\omega*s \cdot P*s - \omega*c \cdot P*c)$$

$$= (42.713/949.729^2) \cdot (6296.3745 \cdot 0 - 5604.9805 \cdot 0)$$

$$= 0.0000 \text{ KPa.}$$

Determine the effective pressure [Pe]:

$$= J \cdot Kst / (1 + J \cdot Kst(Qz1 + (ps - 1)Qz2)) \cdot (P's - P't + Py + Pw + Pw + Prim)$$

$$= 0.1000E+01 \cdot 0.228 / (1 + 1.0 \cdot 0.228 (5.102 + (1.014 - 1)21.356)) \cdot (4429.395 - 2725.801 + 0 + 0 + 0 + 213.297)$$

$$= 196.0216 \text{ KPa.}$$

ASME Section VIII-2 Part 4.18.8.4 Step 7:

Determine factor [Q2]:

$$= [((\omega*s \cdot Ps - \omega*c \cdot Pt) - (\omega*s \cdot P*s - \omega*c \cdot P*c)) + W* \cdot \gamma b / (2 \cdot \pi)] / (1 + \phi \cdot Zm)$$

$$= [((-6207.7 \cdot 1310.0 - -6976.548 \cdot 520.0) - (6296.375 \cdot 0 - 5604.98 \cdot 0)) \cdot 0 + 0 \cdot 0 / (2 \cdot \pi)] / (1 + 7.47843 \cdot 0.1593)$$

$$= -209.6151 \text{ Kgf}$$

Calculate factor [Q3]:

$$= Q1 + 2 \cdot Q2 / (Pe \cdot ao^2)$$

$$= -0.036 + 2 \cdot -209.615 / (196.022 \cdot 949.729^2)$$

$$= -0.05973$$

Fm Value from Table 4.18.3 = 0.029866

Tube sheet Bending Stress - original thickness [σ]:

$$= (1.5 \cdot Fm / \mu*) \cdot (2 \cdot ao / (H - h'g))^2 \cdot Pe$$



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ASME TS Calc: Case: 1 6:34p Sep 8, 2026

$$= (1.5 \cdot 0.0299 / 0.2043) \cdot (2 \cdot 949.729 / (56.0 - 0))^2 \cdot 196.02$$

$$= 49.4489 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ allowed]:

$$= 1.5 \cdot S = 1.5 \cdot 138.0 = 207.0 \text{ N./mm}^2$$

Tubesheet Bending Stress - final thickness [σ_f]:

$$= (1.5 \cdot F_m / \mu \cdot) \cdot (2 \cdot a_o / (h - h'g))^2 \cdot P_e$$

$$= (1.5 \cdot 0.0173 / 0.2043) \cdot (2 \cdot 949.729 / (17.852 - 0))^2 \cdot 143.86$$

$$= 206.9916 \text{ N./mm}^2$$

Reqd Tubesheet Thickness, for Bending Stress (including CA) [HReqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 17.852 + 0 + 0$$

$$= 17.8520 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 8:

Shear Stress check [τ limit]:

$$= (2 \cdot \mu \cdot h / a_o) \cdot \min(0.8S; 0.533S_y)$$

$$= (2 \cdot 0.2 \cdot 56.0 / 949.729) \cdot 110.398$$

$$= 2.6038 \text{ N./mm}^2$$

Shear Stress is not required to be computed; [P_e] <= τ ulimit

Could not converge on a minimum required tubesheet thk. to satisfy the Shear Stress criteria.



Use the best Estimated Required Tubesheet Thickness value as design guidance & Compare the Actual Stresses versus Allowable Stresses.

$$\text{Tubesheet thickness} = 12.1370 \text{ mm}$$

$$\text{Tubesheet Shear Stress} = 21.3555 \text{ N./mm}^2$$

Required Tubesheet Thickness for Given Loadings (Including CA) [H reqd]:

$$= \max(H_{reqB}, H_{reqS})$$

$$= \max(17.852, 12.137)$$

$$= 17.8520 \text{ mm}$$

ASME Section VIII-2 Part 4.18.8.4 Step 9:

The F_{tmin} and F_{tmax} Coefficients from Table 4.18.4:

$$F_{tmin} = -0.2669, F_{tmax} = 4.1775$$

First Extreme Tube Axial Stress from among all the tubes [σ_{t1}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmin}) / (X_t - X_s)$$

$$= ((1310.0 \cdot 0.4823 - 520.0 \cdot 0.6401) - (196.022) \cdot -0.267) /$$

$$(0.6401 - 0.4823)$$

$$= 2.2276 \text{ N./mm}^2$$

Second Extreme value of Tube Axial Stress from among all the tubes [σ_{t2}]:

$$= ((P_s \cdot x_s - P_t \cdot x_t) - P_e \cdot F_{tmax}) / (X_t - X_s)$$

$$= ((1310.0 \cdot 0.4823 - 520.0 \cdot 0.6401) - (196.022) \cdot 4.178) /$$

$$(0.6401 - 0.4823)$$

$$= -3.2962 \text{ N./mm}^2$$

Maximum Tube Axial Stress [$\sigma_{t,max}$]:

$$= \max(|\sigma_{t,1}|; |\sigma_{t,2}|) = 3.296 \text{ N./mm}^2$$

Allowable Tube Stress [$\sigma_{t,A}$]:



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ASME TS Calc:

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$$= Sot = 114.9979 \text{ N./mm}^2$$

Check for Buckling as some of the tubes are in compression.

Determine the Factor of Safety [Fs]:

$$= \max((3.25 - 0.25(Z_d + Q3 \cdot Z_w)Xa^4); 1.25)$$

$$= \max((3.25 - 0.25(0.002 + -0.06 \cdot 0.013)8.92^4); 1.25)$$

$$= 1.2500 \text{ (Should be } \leq 2 \text{)}$$

Determine the factor [rt]:

$$= (\sqrt{ (dt^2 + (dt - 2 \cdot tt)^2) }) / 4$$

$$= (\sqrt{ (25.4^2 + (25.4 - 2 \cdot 2.11)^2) }) / 4$$

$$= 8.2680 \text{ mm}$$

Determine the factor [Ct]:

$$= \sqrt{ (2 \cdot \pi^2 \cdot Et / Syt)}$$

$$= \sqrt{ (2 \cdot \pi^2 \cdot 0.18607E+09 / 132)}$$

$$= 166.8131$$

Determine the factor [Ft]:

$$= k \cdot L / r$$

$$= 1.0 \cdot 560.0 / 8.268$$

$$= 67.7312$$

Buckling Allowable Stress [Stb]:

$$= (Sy_t / Fs) \cdot (1 - Ft / (2 \cdot Ct))$$

$$= (132 / 1.25) \cdot (1 - 67.7312 / (2 \cdot 166.813))$$

$$= 84.162 \text{ N./mm}^2 \text{ (Never greater than Sot)}$$

Note: The Axial Compressive stress in tubes is within limits.

Largest tube-to-tubesheet Joint Load [Wt]:

$$= \sigma_{t,max} \cdot \text{Tube Area}$$

$$= 3.3 \cdot 1.5438$$

$$= 51.89 \text{ Kgf}$$

Tube Metal Area [At]:

$$= \pi (do - t) t$$

$$= \pi (25.4 - 2.11) 2.11$$

$$= 1.544 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (do - t) S$$

$$= \pi \cdot 2.11 (25.4 - 2.11) 114.998$$

$$= 1810.353 \text{ Kgf}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot ag (do + 0.67 \cdot ag) Sw \text{ (but not } > Ft \text{)}$$

$$= 0.85 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 115.0$$

$$= 1771.6150 \text{ Kgf}$$

Weld Strength Factor [fw]:

$$= S / Sw \text{ where } Sw = \min(S \text{ and } St)$$

$$= 114.998 / 114.998$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot af (do + 0.67 \cdot af) Sw \text{ (but not } > Ft \text{)}$$

$$= 0.55 \cdot \pi \cdot 2.11 (25.4 + 0.67 \cdot 2.11) 115.0$$

$$= 1146.3392 \text{ Kgf}$$



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ASME TS Calc:

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Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{(0.75 \cdot do)^2 + 1.07 \cdot t \cdot (do - t) \cdot fw \cdot fd} - 0.75 \cdot do \right)$$

$$= 2 \left(\sqrt{(0.75 \cdot 25.4)^2 + 1.07 \cdot 2.11 \cdot (25.4 - 2.11) \cdot 1.0 \cdot 1.0} - 0.75 \cdot 25.4 \right)$$

$$= 2.667 \text{ mm}$$

Minimum Required Fillet Weld Leg [afr]: 1.3334 mm

Minimum Required Groove Weld Leg [agr]: 1.3334 mm

Tube-Tubesheet Jt allowable, 1810.35 is >= tube strength 1810.35 Kgf

Note: This tube-tubesheet joint is a Full Strength joint

ASME Section VIII-2 Part 4.18.8.4 Step 10:

Shell Axial Membrane Allowable Stress:

$$= S_s \cdot E_{sw} = 138.0 \cdot 0.85 = 117.3 \text{ N./mm}^2$$

Axial Membrane Stress in Shell [σ_s, m]:

$$= a o^2 / ((D_s + t_s) t_s) \cdot [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 949.729^2 / ((1926.0 + 16.0) 16.0) \cdot [196.02 + (1.014^2 - 1)(1310.0 - 520.0)] + 963.0^2 \cdot 520.0 / ((1926.0 + 16.0) 16.0)$$

$$= 21.8568 \text{ N./mm}^2$$

ASME Section VIII-2 Part 4.18.8.4 Step 11:

Note:

For a given Shell thickness of 16.0 mm, the minimum Shell length adjacent to the tubesheet should be 315.981 mm

Shell Membrane Stress due to Joint Interaction [σ_s, m]:

$$= a o^2 / ((D_s + t_s) t_s) [P_e + (p_s^2 - 1)(P_s - P_t)] + a s^2 \cdot P_t / ((D_s + t_s) t_s)$$

$$= 949.729^2 / ((1926.0 + 16.0) 16.0) [196.02 + (1.014^2 - 1)(1310.0 - 520.0)] + 963.0^2 \cdot 520.0 / ((1926.0 + 16.0) 16.0)$$

$$= 21.8568 \text{ N./mm}^2$$

Shell Bending Stress due to Joint Interaction [σ_s, b]:

$$= 6 \cdot k_s / t_s^2 \{ \beta_s [\delta_s \cdot P_s + a s^2 \cdot P \cdot s / (E \cdot t_s)] + 6(1 - \nu^2) / (E \cdot h) (a o / h)^3 (1 + h \cdot \beta_s / 2) [P_e (Z_v + Z_m \cdot Q_1) + 2 a o^2 \cdot Z_m \cdot Q_2] \}$$

$$= 6 \cdot 0.15052 E + 1 / 16.0^2 \{ 0.01 [0.253565 \cdot 1310.0 + 963.0^2 \cdot 0 / (0.31131 E + 1)] + 6(1 - 0.44^2) / (32717776) (949.73 / 56.0)^3 (1 + 56.0 \cdot 0.01 / 2) [196.0 (0.013 + 0.159 \cdot -0.036) + 2 / 949.73^2 \cdot 0.159 \cdot -209.615] \}$$

$$= 140.4039 \text{ N./mm}^2$$

Shell Stress Summation vs. Allowable

$$\therefore |\sigma_s, m| + |\sigma_s, b| \leq 1.5 \cdot S_s$$

$$\therefore |21.86| + |140.4| \leq 207.0 \text{ N./mm}^2$$

$$\therefore 162.26, \text{ must be } \leq 207.0 \text{ N./mm}^2$$

Note:

For a given Channel thickness of 14.0 mm, the minimum Channel length adjacent to the tubesheet should be 295.112 mm

Channel Membrane Stress due to Joint Interaction [σ_c, m]:

$$= a c^2 \cdot P_t / ((D_c + t_c) t_c \cdot \cos(\theta))$$

$$= 960.0^2 \cdot 520.0 / ((1920.0 + 14.0) 14.0 \cdot \cos(0))$$

$$= 17.701 \text{ N./mm}^2$$

Channel Bending Stress due to Joint Interaction [σ_c, b]:

$$= 6 \cdot k_c / t_c^2 \cdot \{ (\delta_c \cdot C_m \Delta + \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6(1 - \nu^2) / (E \cdot h) \cdot (a o / h)^3 \cdot (C_m \theta +$$



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ASME TS Calc:

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$$\begin{aligned}
 & h \cdot \beta c / 2 \cdot C m \Delta \cdot [P e (Z v + Z m \cdot Q 1) + 2 / a o^2 \cdot Z m \cdot Q 2] \} \\
 & = 6 \cdot 0.10383 E + 1 / 14.0^2 \cdot \{ (0.299323 \cdot 1.0 \cdot 0.011 - 0 \cdot 1.0) \cdot 520.0 \\
 & \quad - 6 ((1 - 0.441^2) / 32717776) \cdot \\
 & \quad (949.729 / 56.0)^3 \cdot (1.0 + 56.0 \cdot 0.011 / 2 \cdot 1.0) \cdot [196.0 (0.01256 + 0.1593 \\
 & \quad \cdot -0.036) + 2 / 949.729^2 \cdot 0.1593 \cdot -209.615] \} \\
 & = 36.6916 \text{ N./mm}^2
 \end{aligned}$$

Channel Stress Summation vs. Allowable

$$:: | \sigma_{c,m} | + | \sigma_{c,b} | \leq 1.5 \cdot S_c$$

$$:: | 17.7 | + | 36.69 | \leq 172.5 \text{ N./mm}^2$$

$$:: 54.39 \text{ must be } \leq 172.5 \text{ N./mm}^2$$

Computations completed for ASME tubesheet configuration: a

Stress/Force Summary for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet Bend. Stress	49.4 <=	207.0 N./mm ²	Pass
Tubesheet Shear Stress	7.3 <=	110.4 N./mm ²	Pass
Maximum Tube Stress	3.3 <=	115.0 N./mm ²	Pass
Minimum Tube Stress (Buckling)	-3.3 <=	-84.2 N./mm ²	Pass
Maximum Force on any one Tube	51.9 <=	1810.4 Kgf	Pass
Axial Membrane Stress in Shell	21.9 <=	117.3 N./mm ²	Pass
Shell Stress (jt. inter.)	162.3 <=	207.0 N./mm ²	Pass
Channel Stress (jt. inter.)	54.4 <=	172.5 N./mm ²	Pass

Thickness Results for Loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	17.852	56.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.333	2.110	Ok
Tube-Tubesheet Groove Weld Leg :	1.333	2.110	Ok

Fixed Tubesheet results per ASME Div. 2 Part 4.18.8 2025

Results for 16 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	25.925 ...	66	207	12 110	Ps+Pt-Th	D1 Ok
D2uc	38.508 ...	117	207	20 110	Ps+Pt-Th	D2 Ok
D3uc	17.852 ...	49	207	7 110	Ps+Pt-Th	D3 Ok
D4uc	5.112 ...	2	207	... 110	Ps+Pt-Th	D4 Ok
O1uc	21.318 ...	101	438	21 110	Ps+Pt+Th	O1 Ok
O2uc	10.054 ...	71	438	9 110	Ps+Pt+Th	O2 Ok
O3uc	4.154 ...	21	438	2 110	Ps+Pt+Th	O3 Ok
O4uc	10.160 ...	40	438	10 110	Ps+Pt+Th	O4 Ok
D1c	38.400 ...	95	207	15 110	Ps+Pt-Th-c	D1 Ok
D2c	50.169 ...	166	207	24 110	Ps+Pt-Th-c	D2 Ok
D3c	30.648 ...	68	207	9 110	Ps+Pt-Th-c	D3 Ok
D4c	17.835 ...	3	207	... 110	Ps+Pt-Th-c	D4 Ok
O1c	33.582 ...	145	438	25 110	Ps+Pt+Th-c	O1 Ok
O2c	23.959 ...	100	438	11 110	Ps+Pt+Th-c	O2 Ok
O3c	17.021 ...	24	438	2 110	Ps+Pt+Th-c	O3 Ok
O4c	22.723 ...	56	438	11 110	Ps+Pt+Th-c	O4 Ok
Max:	50.1686 ... mm	0.801	0.226	(Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:



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ASME TS Calc:

Case: 1 6:34p Sep 8,2026

Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[(+)-Th]:
With or Without Thermal Expansion, Tt,mx & Ts,mx

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

[O1, O2, O3, O4]:
Operating Load Cases using the Maximum and Minimum Operating Pressures and
Operating Temperatures

Shell Axial Membrane Stress Summary:

Case#	---- Shell Stresses ----				:	---- Shell Band Stress ----				: Pass Fail
	Ten	Allwd	Cmp	Allwd		Ten	Allwd	Cmp	Allwd	
D1uc	6	117	...	...		...	...	...	...	Ok
D2uc	15	117	...	...		...	...	...	...	Ok
D3uc	22	117	...	...		...	...	...	...	Ok
D4uc	1	117	-1	-101		...	...	...	...	Ok
O1uc	1	460	-1	-101		...	...	...	...	Ok
O2uc	8	460	...	...		...	...	...	...	Ok
O3uc	14	460	...	...		...	...	...	...	Ok
O4uc	8	460	-8	-101		...	...	...	...	Ok
D1c	7	117	...	...		...	...	...	...	Ok
D2c	15	117	...	...		...	...	...	...	Ok
D3c	23	117	...	...		...	...	...	...	Ok
D4c	1	117	-1	-100		...	...	...	...	Ok
O1c	...	460	...	...		...	...	...	...	Ok
O2c	8	460	...	...		...	...	...	...	Ok
O3c	16	460	...	...		...	...	...	...	Ok
O4c	7	460	-7	-100		...	...	...	...	Ok
Max RATIO	0.196			0.074		...			...	

Tube, Shell and Channel Stress Summary:

Case#	---- Tube Stresses ----				Tube Loads		Shell Stress		Channel Stress		Pass Fail
	Ten	Allwd	Cmp	Allwd	Ld	Allwd	Stress	Allwd	Stress	Allwd	
D1uc	7	114	-3	-84	109	1810	60	206	117	172	Ok
D2uc	11	114	-11	-84	165	1810	206	206	72	172	Ok
D3uc	3	114	-3	-84	52	1810	162	206	54	172	Ok
D4uc	...	114	...	-84	4	1810	4	206	9	172	Ok
O1uc	15	229	-3	-84	235	3620	105	460	167	344	Ok
O2uc	4	229	-1	-75	68	3620	137	460	8	344	Ok
O3uc	5	229	...	...	80	3620	98	460	107	344	Ok
O4uc	8	229	-1	-84	132	3620	65	460	52	344	Ok
D1c	8	114	-3	-84	122	1810	66	206	127	172	Ok
D2c	12	114	-12	-84	185	1810	221	460	78	172	Ok
D3c	4	114	-4	-84	58	1810	173	206	59	172	Ok
D4c	...	114	...	-84	5	1810	4	206	10	172	Ok
O1c	16	229	-3	-84	251	3620	113	460	180	344	Ok
O2c	4	229	-2	-84	69	3620	148	460	10	344	Ok
O3c	5	229	...	...	77	3620	104	460	115	344	Ok
O4c	9	229	-1	-84	135	3620	68	460	56	344	Ok



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ASME TS Calc:

Case: 1 6:34p Sep 8, 2026

Max RATIO 0.102 0.140 0.102 0.996 0.736

Summary of Thickness Comparisons for 16 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	50.169	56.000	Ok
Tube thickness :	0.625	2.110	Ok
Tube-tubesheet fillet weld leg :	1.333	2.110	Ok
Tube-tubesheet groove weld leg :	1.333	2.110	Ok

Min Shell length of thk, (16.000) adj. to tubesheet: 315.981 mm

Min Channel length of thk, (14.000) adj. to tubesheet: 295.112 mm

Note: This is a full strength Tube to Tubesheet Joint.

Summary of Axial Differential Expansion between Shell and Tubes :

Due to Thermal Expansion Shell Compresses by : -1.096 mm

Due to Pressure Shell Compresses by : -0.134 mm

Due to Pressure + Thermal Shell Compresses by : -1.228 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1295.94	0.998	1748.28	1.000
Tubesheet Shear Stress	3771.07	1.000	6401.00	1.000
Tube Tensile Stress	8169.12	1.000	13656.34	1.000
Tube Compressive Stress	2243.72	0.144	9994.50	1.000
Tube-Tubesheet Joint load	8169.12	1.000	13656.34	1.000
Shell Stress (Axial, Junction)	1295.94	0.998	1748.28	1.000
Tubesheet-Channel Junction	1209.87	1.000	1748.28	1.000
Tube Pressure Stress	20464.86	1.000	7185.41	1.000
Minimum MAWP	1209.87		1748.28	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	1760.03	1.000	2515.60	0.997
Tubesheet Shear Stress	5167.57	1.000	7838.81	1.000
Tube Tensile Stress	9092.35	1.000	15244.81	1.000
Tube Compressive Stress	20362.57	1.000	13972.47	1.000
Tube-Tubesheet Joint load	9092.35	1.000	15244.81	1.000
Shell Stress (Axial, Junction)	1760.03	0.343	2515.60	0.997
Tubesheet-Channel Junction	1561.95	1.000	2515.60	0.997
Tube Pressure Stress	20464.86	1.000	9707.49	1.000
Minimum MAPnc	1561.95		2515.60	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Summary of Exchanger Pressure and Thermal Displacements: (mm)

Load Case#	Due to Pressure		Due to Thermal Exp.		Computed Delta Ms	
	UnCorroded	Corroded	UnCorroded	Corroded	UnCorroded	Corroded
1	0.0274	0.0275	...	...	...	...
2	-0.1341	-0.1343	...	...	...	...



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ASME TS Calc:

Case: 1 6:34p Sep 8, 2026

3	-0.1097	-0.1098	...	...	...	...
4	0.0030	0.0030	...	...	...	...
5	0.0222	0.0222	-1.0940	-1.0957	...	...
6	-0.1319	-0.1321	-1.0940	-1.0957	...	...
7	-0.1097	-0.1098	-1.0940	-1.0957	...	...
8	...	...	-1.0940	-1.0957	...	...

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(56.0/4, \min(56.0, 16.0))$$

$$= 16.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(1310.0/2515.6, 1)$$

$$= 0.521$$

Thickness Ratio = 0.521, Temperature Reduction per Fig. UCS 66.1 = 30 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-14 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-44 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \text{tubesheet thickness}/4$$

$$= 56.0/4$$

$$= 14.000 \text{ mm}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{pt}/\text{Tubeside MAPnc}, 1)$$

$$= \min(520.0/1561.95, 1)$$

$$= 0.333$$

Thickness Ratio = 0.333, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-18 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

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Problems/Failures Summary:

Step: 34

6:34pm

Sep 8, 2026

Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: A standard flange seems to be governing the Shellside MAWP!**

Please review all reports carefully!

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PV Elite® Pressure Vessel Design Notes

Project Information	
<i>Project Number</i>	
<i>Client</i>	
<i>Date</i>	
<i>Designer</i>	
<i>Design Code</i>	
<i>Additional Standards</i>	

Design Conditions		
<i>Internal Design Pressure:</i>		
<i>External Design Pressure:</i>		
<i>Design Int. Temperature:</i>		
<i>Design Ext. Temperature:</i>		
<i>Corrosion Allowance:</i>		

Vessel Geometry		
<i>Vessel Type</i>		
<i>Diameter</i>		
<i>Tangent-to-Tangent Length</i>		
<i>Head Type</i>		
<i>Number of Segments</i>		

Material Specifications	
<i>Shell:</i>	
<i>Head:</i>	
<i>Nozzle:</i>	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
<i>Support Type</i>		
<i>Number of Supports</i>		
<i>Support Material</i>		
<i>Seismic Code</i>		
<i>Wind Code</i>		
<i>Insulation</i>		
<i>Pressure Test</i>		
<i>Radiography</i>		
<i>Fatigue Analysis</i>		

Analysis Notes: